



**EXPERIMENTAL VALIDATION OF
CLASSIFICATION METHODOLOGIES FOR
MULTIFREQUENCY AND POLSAR IMAGES**

VALIDAZIONE SPERIMENTALE DI
METODOLOGIE DI CLASSIFICAZIONE PER
IMMAGINI SAR MULTIFREQUENZA E POLARIMETRICHE

**DEGREE THESIS IN
COMPUTER SCIENCE ENGINEERING**

UNIVERSITY of GENOA

in collaboration with

DLR (GERMAN AEROSPACE CENTER)

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*"Non quia difficilia
sunt non audemus,
sed quia non audemus
difficilia sunt."*

Seneca

ABSTRACT

Image classification is an important aspect in different applications. Specifically, image classification of remote sensing data is important for several applications like topography, oceanography, forestry, agriculture, urban planning, environmental and prediction and evaluation of natural disasters. This thesis is focused on image classification of *SAR (Synthetic Aperture Radar)* Images. A theoretical study is presented which summarises the most important aspects of radar and SAR and the theory about stochastic processes.

We analysed SAR images with two different types of algorithms: one which is based on *MRFs (Markov Random Fields)* with *ICM (Iterated Conditional Mode)* that is a supervised classifier; the other based on *K-Means* algorithm. We combined these algorithms with different feature transformations and filters to obtain a better classification. To compare the results we used a statistical analysis based on “*Confusion Matrices*”.

Particularly, our attention was dedicated to develop and test the MRF-ICM method for classification of SAR images. In the past, these methods are not used for classification of only SAR images but also with optical images. In addition, we studied the image properties in order to obtain the better data representation. Among these properties, we consider feature transformations in order to study the image from a different points of view. In addition, we tested the accuracy of the Log-normal probability density function to describe the image and a relative classification. Instead, for the unsupervised classification, we used, as described above, K-Means algorithm which was used to compare different classifications. Actually, we studied the feature extraction of *PolSAR (Polarized SAR)* images which consists in parameters *alpha* and *entropy* (α and H). Thus, the results consist in classifications with these parameters and, then, we compare the results of supervised and unsupervised classification separately.

SOMMARIO

La classificazione di immagini è un importante aspetto in diverse applicazioni. In particolare, la classificazione di immagini di dati telerilevati è importante per i campi quali: topografia, oceanografia, studio di foreste, agricoltura, cartografica, studio, analisi e previsione di disastri naturali come, per esempio, alluvioni, uragani, ecc... Questa tesi è concentrata sulla classificazione di immagini *SAR* (*Synthetic Aperture Radar* – Radar ad Apertura Sintetica). È presentata una parte teorica che riassume le proprietà più importanti del radar e, in particolare, del SAR ed è descritta la teoria riguardante i processi stocastici.

Abbiamo analizzato le immagini SAR con due differenti tipi di algoritmi: uno basato sui *MRFs* (*Markov Random Fields* – Campi Markoviani) con *ICM* (*Iterated Conditional Mode* – Modo Iterativo Condizionale) che consiste in un classificatore supervisionato (*Supervised*); l'altro basato sull'algoritmo *K-Means* (Classificatore non supervisionato – *Unsupervised Classifier*). Abbiamo, inoltre, combinato questi metodi di classificazione con estrazioni di *feature* (“caratteristiche”) e diversi filtri per ottenere migliori risultati. Per confrontare questi risultati abbiamo usufruito dell'analisi statistica basata sulle “*Confusion Matrices*”(Matrici di Confusione).

In particolare, la nostra attenzione si è concentrata nello sviluppo e nel test del metodo MRF-ICM per la classificazione delle immagini SAR. Fino ad oggi, non state effettuate, con tale metodo, classificazioni di pure immagini SAR, ma di immagini SAR in combinazione con dati ottici. Inoltre, sono state studiate le proprietà dell'immagine tra le quali quella usata per i test al fine di ottenere buoni risultati. Tra le proprietà, ricordiamo le trasformazioni di *feature* che consistono in una trasformazione di spazio e permettono, con alcune condizioni, di ottenere alcuni vantaggi. Fra le altre proprietà studiate abbiamo usato come modello probabilistico per l'immagine SAR la densità di probabilità Log-normal che è la una distribuzione empirica che molto bene si adatta a questo tipo di immagini. Per la classificazione non supervisionata, invece, è stato usato l'algoritmo di K-Means per studiare due diversi metodi di classificazione. Infatti, per questo studio sono state usate immagini *PolSAR* (*SAR Polarimetrico*) da cui sono stati ricavati i parametri *alpha* ed *entropia* (α and H) che descrivono l'immagine da un altro punto di vista. In definitiva, sono state studiati e sperimentati diverse metodologie per la classificazione supervisionata e non-supervisionata di cui sono stati analizzati i risultati separatamente.

ACKNOWLEDGMENTS

We would like to gratefully acknowledge the supervision of Prof. *S.B. Serpico* of *DIBE (of Genoa University)* during this work. We thank *DLR (German Aerospace Center)* for the organization and the possibility to improve our knowledge. Particularly, we like to thank Prof. *Mihai Datcu* for the important supervision and for the theoretical and technical discussions about image classification in remote sensing applications. In addition, it is necessary to remember the help in every kind of problem of Dr. *Gabriele Moser* of *DIBE*. We feel indebted to him for the numerous interventions and clarifications.

We are grateful to all our colleagues of *DLR (German Aerospace Center)* for their continued support and their advice how to organise the work. There is necessity to thank the *DLR* which permitted to do this work in two months and it supplied all the components in order to developed our job peaceful.

Finally, we are forever indebted to *our parents* and *our colleague* of *Genoa University Giuseppe De Fazio*, for their help which permitted this experience and this thesis.

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1 INTRODUCTION

Environmental monitoring, earth-resource mapping, and defence (security systems) may involve broad-area imaging at high and medium resolution. The imagery must often be acquired in inclement weather or during night as well as day. Synthetic Aperture Radar (SAR), both on airborne and spaceborne platforms, provides such a capability. SAR systems take advantage of the long-range propagation characteristics of radar signals and of the complex information processing capability of modern digital electronics to provide high and median resolution imagery. Synthetic Aperture Radar complements photographic and other optical imaging capabilities thanks to the minimum constraints on time-of-day and atmospheric conditions and thanks to the unique responses of terrain and cultural targets to radar frequencies. ERS-1 was the first instrument in a series of orbital SAR planned to have long lifetimes and semi-operational capabilities. Consequently ERS-2, JERS-2 and RADARSAT satellite systems were created. ENVISAT was launched in 2001. These satellites allow dynamic processes to be observed over most of the Earth's surface by providing a long series of accurate measurements of the backscatter coefficient¹. This has a significant impact in many scientific domains: vegetation mapping and monitoring, hydrology, sea-ice mapping and geology. Particularly, Synthetic Aperture Radar technology has provided terrain structural information to geologists for mineral exploration, oil spill boundaries on water to environmentalists, sea state and ice hazard maps to navigators, and reconnaissance and targeting information to military operations. There are many other applications or potential applications. Some of these, particularly civilian, have not yet been adequately explored because lower cost electronics are just beginning to make SAR technology economical for smaller scale uses. Spaceborne radar systems use single frequencies and polarization with modest resolution, because of the constraints imposed by their deployment in space. On the contrary, more complex airborne systems have demonstrated the advantages of multiple frequencies and polarizations. These advantages

¹ The backscattering or backward scattering coefficient, in unit of m^{-1} , indicates the attenuation (reduction in light intensity) caused by scattering at angles from 90° to 180°.

were and will be further proved from space by SIR-C/X-SAR, the SRTM missions of the Space Shuttle, and the TerraSAR and Cosmo/SkyMed missions.

1.1 BACKGROUND

Data acquired by radar systems must be interpreted to yield information about the imaged area. However, the task of information extraction and the detection of significant features are very time-consuming if performed semi automatically by an image analyst, especially when a large amount of remote sensing data must be examined. Thus there has been and there is a development of automatic algorithms to extract relevant information quickly and in reproducible way. When polarimetric, multi-frequency and multi-temporal images are used the need of automatic or semi-automatic methods is even larger and all available information must be fused. To be valuable, an automatic data interpretation process has to be based on a solid understanding. Actually, these images extracted from satellites need several kinds of classification for different purposes. The image classification is an important and difficult task which may also require taking into account the process of image generation. Particularly, for the SAR images this problem is bigger than in other cases because these images present a lot of noise, thus, they need particular processing steps. In the last part of this paragraph, we shall explain shortly which range of electromagnetic spectrum is used for remote sensing. Actually, there are several radars which work in different ranges. Particularly, the Synthetic Aperture Radar works in the range of the microwaves shown in Figure 1.1.

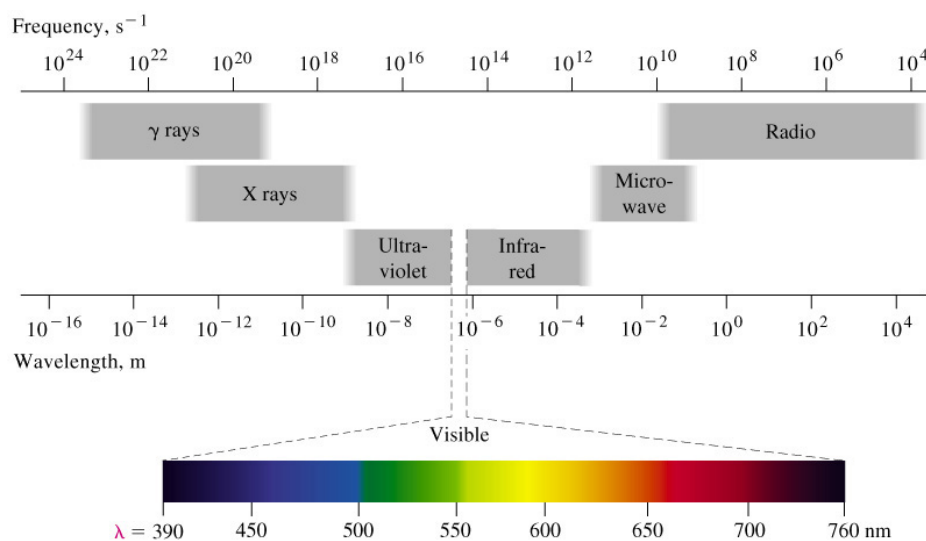


Figure 1.1: Electromagnetic Spectrum.

1.2 GOAL AND MOTIVATIONS

The goal of this thesis is to present several approaches to classification of multimodal SAR images. Due to its advantages, as compared to optical sensors, and to its steadily increasing resolution, SAR imagery is getting more and more important as a source of remote sensing information. Basic features of potential interest, which can be observed in SAR data are homogeneous regions of different medium backscatter, (i.e., of different cross-section or backscatter coefficient), textured regions such as forests, edges giving information for data classification and strong scatterers (which are mostly related to man-made targets). However, appropriate tools are still needed for automated interpretation and extraction of such features from SAR data.

In this thesis we explain several classification approaches from different points of view. We used several methods among which the main one is MRF (Markov Random Field) applied with ICM (Iterated Conditional Mode) estimation. Through this technique, a lot of experimental tests have been performed, together with different transformations of the image data. Other important experiments which are described in this thesis regard the K-Means unsupervised classification. The above mentioned transformations were done in order to obtain a better classification. Actually, we did many classifications which provide different results. We performed an evaluation and assessment of the processing chain to determine which approach achieved the best classification performance.

1.3 OUTLINE OF THE THESIS

The aim of this thesis is to apply different classification schemes to the analysis of multi-feature SAR images and to describe the algorithms which were implemented to this end. In our tests we used two types of images (one multifrequency–SAR image and the other one originated with L frequency and with 4 polarizations–POLSAR image) and we focused on Markov Random Fields for supervised classification and on the K-Means algorithm for unsupervised classification. In order to explain these parts in a better way we summarized in

the first chapter the theory about radar and, particularly, the description of Synthetic Aperture Radar (SAR). In addition, the properties of this category of radar and the related physical operations are described. To discuss image classification it is necessary to explain the theory about stochastic processes in order to model the image pixel like a random variable and to define probability density functions (PDFs) which better model the statistics of such variables. The other chapters before the discussion of our tests are concentrated about the description of image transformations and the adopted approaches. The developed program is based on Markov Random Fields with Iterated Conditional Mode estimation and thus, there is a short chapter about this development and the theory about this kind of classification and what it is possible to do with it. The other approach which we used for the test is based, as described above, on K-Means, applied to POLSAR images. To do this experiment we used, therefore, a commercial software package, called ENVI, which provides functionalities to manipulate and show images. Instead, the image transformations were developed with a specific programming language, namely: IDL (Interactive Data Language) which is a good interpretative language to compute operations with matrices and vectors through which we represent images.

Chapter 11 presents the description and the discussion of our tests. In this part several tests with different transformations are described. Particularly, it is possible to divide this chapter in two principal parts: one for supervised classification with MRF and the other one for unsupervised classification with K-Means. These two parts are described from different points of view: for example, there are classification results which are good with respect to the quality of classification but which require long processing time, and for some kinds of applications this may not be acceptable. So, in this way, a comparison taking into account these issues and the different transformations is presented. For the first part it was possible to do an evaluation of the classification because we had the test image to compute the accuracy of classification. In the second part, instead, this was not possible because we did not have the test image for these experiments. Hence, a description of results and a qualitative comparison between classification results are provided.

2 PRINCIPLES OF SYNTHETIC APERTURE RADAR (SAR)

2.1 SIDE-LOOKING GEOMETRY OF SAR SYSTEMS

The geometry of a SAR is shown in Figure 2.1

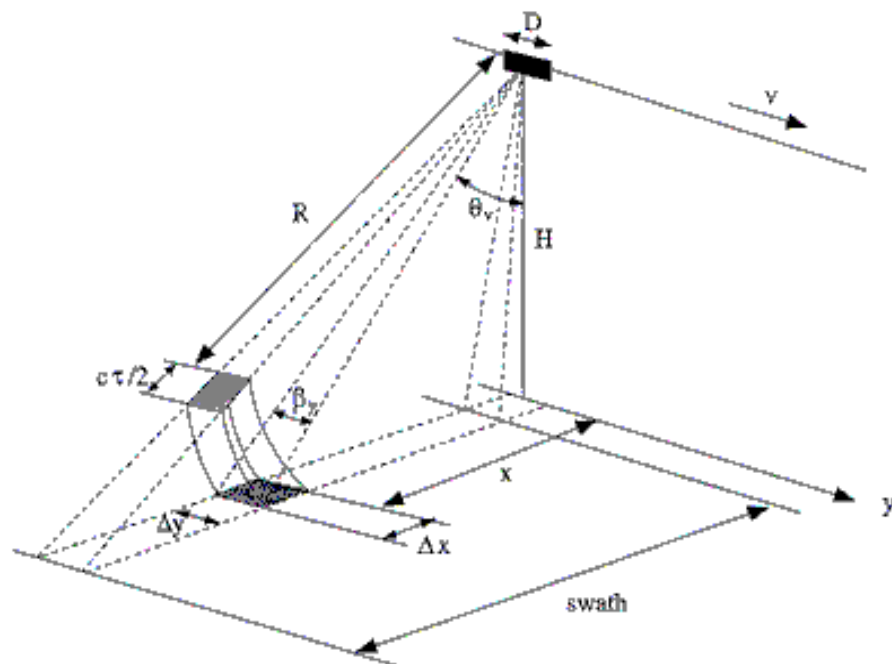


Figure 2.1: Geometry of a side-looking radar system used for surface imaging. The radar with an antenna of width D is flying at altitude H with speed v . The incidence angle for the considered resolution cell is θ_v . The resolution in range and azimuth are denoted by r_x and r_y respectively.

We can see a platform moving with velocity v at altitude H carrying a side-looking radar antenna that illuminates the Earth's surface with pulses of electromagnetic radiation. The direction of travel of the platform is known as the azimuth direction and is denoted by y . The distance of the radar track is measured in the range direction x . For simplicity, we only deal with the case where the antenna is oriented along to the flight path, not squinted.

2.1.1 BASIC PROPERTIES OF SIDE-LOOKING RADARS

Considering a radar system mounted on a platform, either airborne or spaceborne, as depicted in Figure 2.1, an emitted pulse is received after $\Delta t = 2R/c$, where c is the speed of light ($c \approx 3 \cdot 10^8$ m/s) and R is the path of the emitted wave. The emitted waves can be described by rectangular pulses of duration τ that are repeatedly sent with a pulse repetition frequency f_T with $T^1 \gg \tau$. The sensor resolution that is the minimum distance between distinguishable different objects can then be calculated to be $\Delta R = c \tau / 2$ [2.3]. For the ground resolution this results in [2.3]

$$r_x = \frac{c \tau}{2 \sin \theta_v} \quad (2.1)$$

where θ_v is the look angle, as shown in Figure 2.2. Thus, resolution is improved with growing range distance x . For instance, typical look angle varies from 20 to 60 for RADARSAT depending on the used mode. The pulse length τ is the main parameter to increase resolution because θ_v is constrained to a certain range due to the signal-to-noise ratio. It's important that the emitted pulses contains enough energy to guarantee a sufficient amount of reflected intensity at the receiver, but, at the same time, they must be as short as possible. The best theoretical signal, a Dirac pulse, is technically not realizable.

There are other considerations to take: the resolution in azimuth r_y depends on the antenna aperture in azimuth β_y . The width of the main antenna lobe at -3 dB is given by

$$\beta_y = \frac{\lambda}{D} \quad (2.2)$$

¹ where $T = \frac{1}{f_T}$ and it represents the period of the rectangular pulses.

where λ is the wavelength of the emitted signal and D denotes the antenna length in azimuth. The resolution in the azimuth direction is

$$r_y = \beta_y R = \frac{\lambda R}{D} = \frac{\lambda H}{D \cos \theta_v} \quad (2.3)$$

It depends on the wavelength λ , the altitude H and the antenna size D . As it is difficult to have a great variation of H and λ for physical reasons, the resolution can be only increased by the antenna size D .

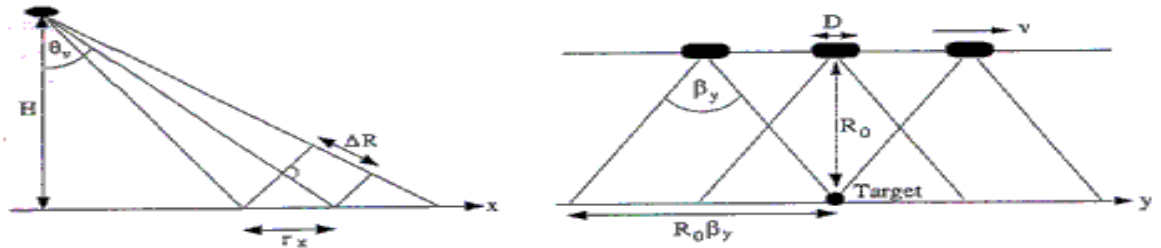


Figure 2.2: Resolution of SAR, in range and azimuth. Left: Imaging geometry in range direction. Right: target illumination by the moving sensor used to generate the synthetic aperture.

2.2 RADIOMETRICAL AND GEOMETRICAL EFFECTS

SAR systems detect the distance between the sensor emitting a microwave pulse and the target reflecting the energy back to the receiving antenna. This range measurement principle leads to specific geometric distortions in the processed SAR image, which makes SAR images more difficult to interpret than optical images, especially for inexperienced users. This kind of degradation does not occur in imaging of flat, e.g. agricultural, areas, but in imaging of rough areas, e.g. mountainous regions.

The most relevant SAR specific topographically induced distortions are illustrated in Figure 2.3 and can be summarized as follows [2.4]:

- **Foreshortening:** For slopes facing the sensor, the area on the ground that is mapped onto one SAR resolution cell [2.4] is larger than for the case of flat terrain. This so called foreshortening situation has two consequences: 1) Due to the change in ground resolution, foreshortening areas appear compressed in the SAR image, i.e. their extension in range direction is reduced; 2) As a consequence of the energy concentration, foreshortening areas are characterized by brighter image gray values. The received energy within a resolution cell is higher due to the larger imaged area.
- **Layover:** Layover is an extreme case of foreshortening and occurs where, due to steep terrain slopes, the top of mountain is closer to the sensor than its bottom. Layover areas appear as particularly bright regions in the image with an inverted original geometrical order. In Figure 2.3 this is depicted by points three and four: we can note the reverse position of these points in the slant range image compared to their actual location in ground coordinates.

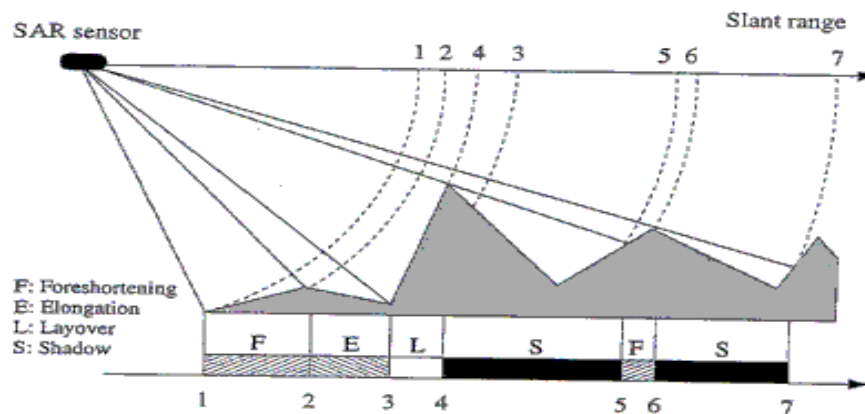


Figure 2.3: The most relevant SAR specific topographically induced distortions.

- **Elongation:** In contrast to the foreshortening effect, slopes facing away from the sensor lead to rather dark, elongated regions in the SAR image. The explanation for this behaviour is dual to the one of foreshortening.
- **Shadows:** Similarly to optical images, areas which are not illuminated by the radar beam are called radar shadows. In the image, shadow areas appear as dark regions only corrupted by thermal noise.

We can draw the following conclusions: the topography of the sensed area not only affects the radiometry of the data (bright areas for layover and foreshortening, dark areas for shadow), but also their geometry by changing the image resolution, called slant range resolution, in comparison with the ground range resolution, as a function of the terrain slopes. The resolution of the slant range data is not constant but varies over the image.

2.2.1 RADIOMETRIC CORRELATION

In order to correct SAR layover areas, the accumulated energy in one image pixel has to be redistributed among those ground resolution cells, which are mapped onto that particular pixel. This problem of energy redistribution is also encountered in foreshortening areas, where each pixel contains the energy from a larger area, comprising generally more than one ground resolution cell. Concerned terrain facets on a digital elevation model (DEM) are adjacent to each other and can be approximated to have identical local incidence angles and reflectivity. Therefore the radiometric correlation can be performed by a division by the size of the imaged area, which, as a first approximation, is a function of the local incidence angle, which, in general, is calculated by using DEMs of similar (or, if available, higher) resolution.

2.2.2 GEOMETRIC CORRELATION

In case of geometrically induced effects, the slant range image must be resampled to its nominal ground resolution. Geocoding is the name of this procedure, which consists in minimizing geometrical distortions and resampling the image to a homogenous, predefined map grid. To this end, it's necessary to know the exact elevation for each pixel. The purpose of geocoding is to generate a map-like representation of the satellite image, where the SAR image is aligned with a Cartesian map projection grid. Otherwise, the uncorrected data can hardly be interpreted due to effects in slant range images, such as mountains which appear to be leaning towards the sensor.

2.3 BASIC SCATTERING MECHANISM

Radar systems emit electromagnetic waves with wavelengths ranging from a few centimetres up to one decimetre (around 10GHz) and receive the backscattered reflection from the imaged surface. The measured time between emission and reception of the reflected wave is used to localize the scatterer or target (the term target denotes both single objects and distributed scatterers). In Figure 2.4 different backscattering mechanisms depending on the micro and macroscopic properties of the scatterer are showed in Figure 2.4.

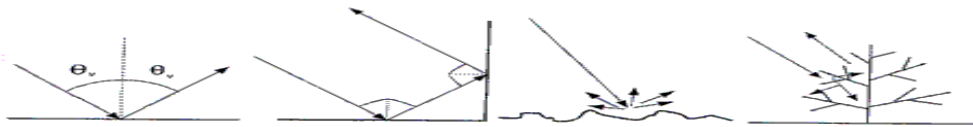


Figure 2.4: Different backscattering mechanisms. From left to right: Scattering from a smooth surface, double bouncing reflection, diffuse scattering from rough surfaces, volume scattering.

- **Reflection from smooth surfaces and double bouncing:** a smooth surface reflects very few of the incoming energy back to the emitting antenna. There are only two cases in which we have a greater reflection of energy, i.e. when the incidence angle θ_v between sensor and surface is zero or when there is “double bouncing”. For this reason, highways and calm lakes appear rather dark in SAR images. Buildings usually appear bright due to the double bouncing effect.
- **Reflection from rough surfaces and volume scattering:** rough surfaces, i.e. non-specular reflectors, reflect the wavefront in multiple directions. The sensor only receives a part of the emitted energy. This effect is called diffuse scattering. The same effect occurs for volume scattering, where the wavefront partly penetrates the scatterers (in the forest for example). The wavelength of the system depends on the volume scattering and on the depth of penetration.

An interesting example is represented by the case of the surface of the sea. When the sea is calm, the incident angle θ_v between sensor and surface is not zero and we have a situation in which it's not possible to receive the reflected energy. For this reason the receiver maps the

input in a black image. Otherwise, if the sea is slightly rough, a partial reflection of the emitted signal is obtained, so the signal produced by the sensor is a gray image. Considering the last case, i.e., when the sea is very rough, the sensor on the platform can receive a large amount of energy because the surface reflects in all directions including the same direction in which the radar signal was emitted.

SAR images can be computed by measuring the reflected energy and the time delay between emission and reflection and by doing an appropriate processing, based on the electromagnetic properties of the imaged surface.

2.4 SAR FOCUSING

2.4.1 RANGE FOCUSING - PULSE COMPRESSION

Pulse compression is a technique which allows increasing dramatically the range resolution. In this case, the radar does not emit a rectangular signal modulated by a carrier frequency f_c , but a signal of duration τ with a linearly modulated frequency called chirp [2.1]. The phase of the emitted signal which is limited in time by $\Delta t = \tau$ is given by

$$\phi_x(t) = 2\pi \left(f_c t + \frac{Kt^2}{2} \right) \quad (2.4)$$

with the signal bandwidth described by $B = K\tau$. The received signal is passed through a matched filter [2.2] which is equivalent to a convolution with an ideal chirp. It can be demonstrated that the range resolution is then determined by (2.5)

$$r_x = \frac{c}{2B \sin \theta_v} \quad (2.5)$$

This is equivalent to emit a rectangular pulse of duration $\tau' = \frac{1}{B}$.

2.4.2 AZIMUTH FOCUSING - SYNTHETIC APERTURE

In order to improve the resolution we can use a frequency modulation, similar to the pulse compression in range, obtained by using the Doppler Effect in the azimuth direction. Specifically, flying at the speed v , the sensor travels the distance $\Delta y = v \Delta t$ within Δt . We assume that at time $t = 0$ the platform is at $y = 0$, where the radial distance R_0 to a given target is minimal.

Since the travelled distance Δy is small compared to R_0 , the distance $R(t)$ can be approximated to the target as a function of time $t = y/v$ by

$$R(t) \approx R_0 + \frac{v^2 t^2}{2R_0} \quad (2.6)$$

with a phase shift between emitted and received signal of

$$\phi_y(t) = 2\pi f_c \frac{2R(t)}{c} = \frac{2\pi}{\lambda} \left(2R_0 + \frac{v^2 t^2}{R_0} \right) \quad (2.7)$$

This linear modulation of the frequency, seen as a time function, can be processed by employing a matched filter. As a given target stays within the antenna beam of width $R_0 \beta_y$ on ground for the time $\Delta t = R_0 \beta_y / v$ the platform movement simulates a large antenna aperture, also called synthetic aperture, with increased resolution of

$$r_y = \frac{D}{2} \quad (2.8)$$

Thus, the synthetic aperture neither depends on the wavelength, nor on the distance to the target. Several advanced signal processing are needed to reach this theoretical resolution. In addition, especially for SAR systems, it's necessary to use methods to compensate effects like range migration of targets [2.5].

2.5 SPECKLE NOISE AND MULTILOOKING REDUCTION TECHNIQUES

One of the applications of SAR systems consists in generating a map of the NRCS (Normalized Radar Cross Section) $\sigma^\circ(x,y)$, which is related to backscattering properties. Actually, each pixel of the image is obtained through the vectorial sum of the contributions of a large number of scatterers; consequently, the reflection of one zone to a constant σ° fluctuates due to the effect of the random combination of the contributions present in each singular cell. For this reason the image pixels have different gray values and this yields a granular effect that can be attributed to a multiplicative noise known with the name of speckle [2.6]. (Figure 2.6).

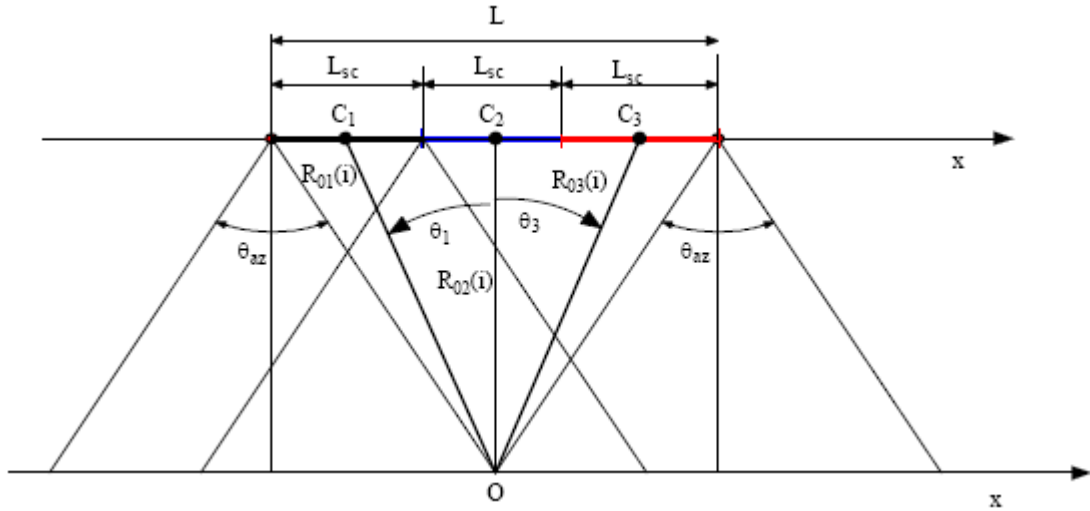


Figure 2.5: Multilook technique.

Typical speckle reduction technique consists in averaging pixels of several independent images of the same zone; this technique is called *multilooking*. Actually, it is possible to obtain images of the same zone with a lower resolution in azimuth subdividing the synthetic aperture L in K subapertures of length $L_{sc}=L/K$ ($K=3$ in Figure 2.5).

We denote by $R_{01}, R_{02}, \dots, R_{0K}$ the distances between point O and the centre of the synthetic aperture and by x_i position of the centres C_i . Processing the obtained data in the K -th subaperture, focusing on the point O with the term

$$H_K(x) = e^{-j \frac{2\pi(x-x_K)^2}{R_{0K}\lambda}} \quad (2.9)$$

we obtain images of the same point 0 (related to the same zone) from K different angles of azimuth θ_i and with resolution, $\delta_x = \frac{D_{az}}{2} K$ reduced of a K factor. The diversity of azimuth of the images allows the images to be independent. Averaging the K images we have a great reduction of speckle, how we can see in Figure 2.7, obtained by a of 3-look multilooking of the image in the Figure 2.6.

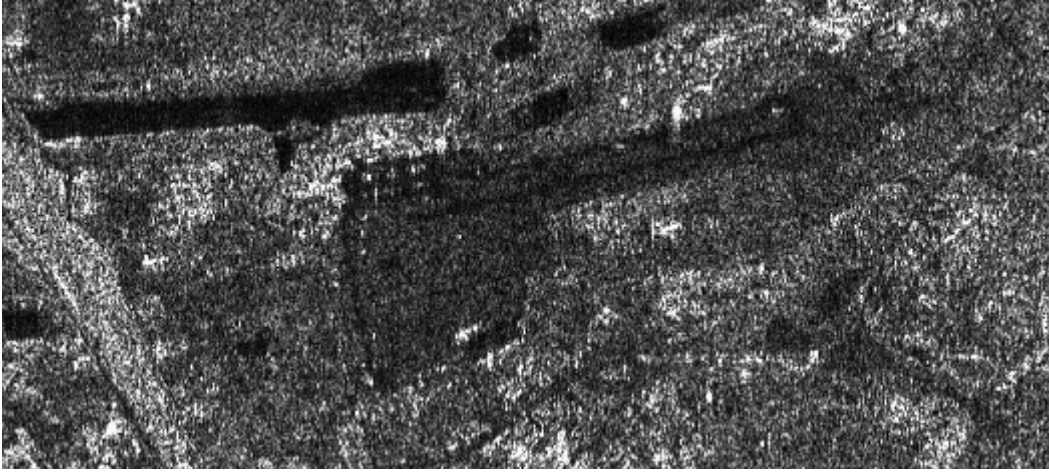


Figure 2.6: 1 look SAR images of an urban area corrupted by speckle.



Figure 2.7: 3 look SAR image generated from Figure 2.6 (the S/N ratio is enhanced).

3 OVERVIEW OF SAR SENSORS

3.1 SAR REMOTE SENSING

Imaging radars are airborne or spaceborne radars, which generate a reflectivity map of an illuminated area through transmission and reception of electromagnetic energy. Among other types of microwave sensors, special attention has been devoted to the synthetic aperture radar (SAR) because of its high spatial and multivariuous information content.

The development of the synthetic array radar originated in 1951 with Carl Wiley, who postulated the use of Doppler information for increasing the azimuth resolution of the conventional side-looking aperture radar (SLAR) [3.1] . Based on this idea, the first SAR image was produced by researches at the University of Michigan in 1958, using an optical processing method [3.2].

Precision radar optical processors and hologram were developed and fine resolution strip maps were obtained by the mid 1960's. Later, in the early 1970's, digital signal processing methods were introduced to obtain off-line or non-realtime SAR images of high quality [3.3], [3.4]. Since these early days, SAR systems evolved to an essential and power tool in geosciences and remote sensing. SAR data are applicable in many scientific fields. Besides traditional applications in geography and for topographic and thematic mapping, nowadays SAR sensors are also utilised in areas such as oceanography, forestry, agriculture, urban planning, environmental and prediction evaluation of natural disasters.

SAR sensors operate in the microwave region of the electromagnetic spectrum with typical wavelengths between 1cm and several metres. As an active system, a SAR emits by itself microwave radiation to the ground and measures the electric field backscattered by the

illuminated ground patch. These measurements are transformed into a high and medium resolution image. As SAR systems are operated actively, they can perform equally well during day and night.

Looking at the transmission spectrum, it can be observed that the Earth's atmosphere is nearly transparent in the microwave region. Electromagnetic waves with wavelengths longer than 1cm even permeate almost undisturbed small water drops. Thus, the operation of a SAR is possible even in the presence of clouds, fog and rain, which are limiting factors in optical remote sensing. This is of great importance for areas which are regularly covered by haze or clouds, like for example rain forests in the tropics. Weather independence combined with day and night operation capabilities makes SAR an operational monitoring device for the entire Earth's surface, a task which cannot be achieved with optical sensors. Radar images contain quite different information as compared with images obtained from optical or infrared sensors. While in the optical range mainly molecular properties on the object surfaces are responsible for the characteristics object reflectivity, in the microwave region dielectric and geometrical properties become relevant for the backscattering. Radar images, therefore, emphasise the relief and morphological structure of the observed terrain as well as changes in the ground conductivity, for example caused by differences in soil moisture. Because of the sensitivity to dielectric properties, SAR images, in principle, can also provide information about the condition of vegetation, an important fact for agricultural and forestry applications.

Another important feature of SAR data results from the propagation characteristics of microwaves. Due to their long wavelength, microwaves are capable to penetrate vegetation and even the ground up to a certain depth [3.5], [3.6]. The penetration capabilities depend on the wavelength as well as on the complex dielectric constants, conductivities and densities of the observed targets. Shorter wavelengths, like the X-bands, show typically a high attenuation and are mainly backscattered on the surface or on the top of the vegetation. Consequently, at these wavelengths mainly information about this layer is collected. Longer wavelengths, like L- and P-band, normally penetrate deep into vegetation and often also into the ground. The backscattering then contains contributions from the entire volume.

One main problem in the analysis of SAR data is to determine exactly this superposition of different scattering contributions. Although certain typologies of desired information are contained in the data, they are not accessible because only the total backscattering can be measured. An inversion of the measured data to retrieve parameters of interest is often

ambiguous and cannot be solved without “a priori” information. Another problem is that the exact origin of the backscattering is, in principle, unknown as the entire SAR geometry has no evaluation symmetry. Thus the elevation angle and topographic height of the observed scatterer remains unknown. Two main extensions of conventional SAR have been pursued in the past to resolve these limitations: SAR interferometry and SAR polarimetry.

3.2 SAR INTERFEROMETRY

SAR interferometry (INSAR) is a technique, which analyses the phase difference between two SAR images acquired from slightly different positions. This phase difference is related to the terrain topography of the scene and can be used to generate high resolution digital elevation models (DEMs). The first basic experiments were made in 1974 by L.C. Graham demonstrating the capability of interferometric SAR for topographic mapping [3.7]. In the following years only little attention was paid to SAR interferometry. Experimental efforts were made by R.M. Goldstein and H.A. Zebker in the '80s, using a simple two-antenna modification of the NASA/JPL airborne AIRSAR system to demonstrate singlepass SAR interferometry [3.8]. In 1988 A. Gabriel and R.M. Goldstein showed the possibility of repeat-pass interferometry using spaceborne L-band data acquired by the SEASAT sensor [3.9]. Finally, first results from repeat-pass interferometry using the Canadian airborne CCRS sensor, in X- and C-band, were published in 1992 by A.L. Gray and P.J. Farris-Manning [3.10].

In 1991, the European Space Agency ESA launched the ERS-1 remote sensing satellite, carrying (beside a number of other sensors) a C-band SAR system. This was the starting point for numerous advances in SAR interferometry [3.11]-[3.12]. ERS-1, originally designed mainly for oceanographic applications, quickly turned out to possess great potential for the generation of large scale high precision DEMs from areas all over the world. The main reasons for this have been the continuous and nearly global mapping performed by ERS-1, combined with a very good orbit resolution necessary for operational DEM generation. ERS-2 followed ERS-1 in 1995 to continue its operation; and, for a while during the 'TANDEM' mission period, it was possible to operate both in parallel. In 1994, two missions with the

Shuttle. Imaging Radar SIR-C/X-SAR were created, for the first time providing spaceborne multi-frequency (X-, C- and L- band) interferometric data in a repeat-pass mode. In the second mission over some areas even fully polarimetric were acquired [3.13], [3.14]. Parts of the same instrument, augmented by a second antenna mounted on a 60m boom, were used again in February 2000 for the Shuttle Radar Topography Mission (SRTM) [3.15].

In addition, to these and some other spaceborne sensors, also several airborne single-pass interferometric systems are in use. Some significant examples currently are the NASA/JPL TOPSAR system (C- and L- band) in the US; the Japanese NASDA/CDR airborne PAL-SAR (X- and L-band); the EMISAR system (C-band) of the Danish Center for Remote Sensing; the commercial German AeS-1 (X-band) of the company AeroSensing; and the E-SAR system of German Aerospace Center (DLR) which operates in X-band in a single pass-mode and in a L- and P-band fully polarimetrically in a repeat pass mode [3.16], [3.17], [3.18], [3.19].

Besides topographic mapping, an extended version of SAR interferometry, called differential interferometry, can be used for precise mapping of elevation changes. This technique allows the detection of surface deformation on a scale smaller than the radar wavelength, usually in the millimetre range. This high precision enables a large scale detection and monitoring of ecological stress-change processes like sudden co-seismic displacements and long-term tectonic movements with spaceborne sensors. Also volcanic in mountainous areas as well as ice deformations and glacier dynamics can be detected with this method [3.20], [3.21], [3.22], [3.23], [3.24], [3.25].

Interferometric SAR data additionally possess content of a different nature as compared with SAR images alone. The correlation or coherence between two SAR images is very sensitive to changes in the arrangement of the scatterers inside the resolution cells (particularly, the coherence of multi-temporal, multi-frequency or changing processes, for example taking place in vegetation layers [3.26], [3.27], or on natural surfaces [3.28]). Finally, multi-baseline approaches take into account influences of the imaging geometry on the interferometric coherence and can even resolve to a limited extent for the spatial distribution of scatterers in a volume [3.29], [3.30].

3.3 SAR POLARIMETRY

SAR polarimetry (POLSAR) is another major extension of conventional single channel SAR imaging. Like all electromagnetic waves also microwaves have a vectorial nature, and a complete description of the scattering problem in radar science requires a matrix formulation. This is the task of radar polarimetry, a technique which was initiated by the introduction of the concept of the 'scattering matrix' by G.W. Sinclair in 1948 [3.31], [3.32]. Since radar polarimetry requires advanced hardware devices, which have not been available in the late 1940's and the 1950's, radar polarimetry only remained a theoretical concept and its practical use for civil applications was not really recognised.

This situation changed in the early 1980's with the availability of polarimetric SAR data from the NASA/JPL airborne AIRSAR system, which allowed in practice the implementation of more recent works by E.M. Kennaugh, J.R. Huynen [3.33] and W.-M. Boerner [3.34]. Since then, SAR polarimetry has become an established technique in remote sensing. This was supported by the growing number of polarimetric airborne sensors like DLR's E-SAR or NASA/JPL's AIRSAR systems, providing high resolution polarimetric data in several frequency bands. Additionally, in 1994, two SIR-C/X-SAR Shuttle missions took place, recording for the first time spaceborne fully polarimetric data in C- and L-bands. During the second mission this was also combined with repeat-pass interferometric data acquisition.

One special characteristic of SAR polarimetry is that it allows a discrimination of different types of scattering mechanisms. This becomes possible because the observed polarimetric signatures depend strongly on the actual scattering process. In comparison to conventional single-channel SAR, the inclusion of SAR polarimetry can lead to a significant improvement in the quality of classification and segmentation results [3.35], [3.36], [3.37]. Certain polarimetric scattering models [3.38] even provide direct physical ground parameters like soil moisture and surface roughness [3.39], as well as unsupervised classification methods with automatic identification of different scatterer characteristics and target types [3.40], [3.41].

SAR polarimetry additionally offers some limited capabilities for separating multiple scattering mechanisms occurring inside the same resolution cell and can be deemed as a first

step in resolving the ambiguous scattering problem in SAR, as mentioned above. With polarimetric decomposition techniques a received signal can be split into a sum of three¹ scattering contributions with orthogonal polarimetric signatures [3.42]. This can be used for extracting the corresponding target types in the image, even in the case that they are occurring as superimposed. Also, if a signal is disturbed by undesired orthogonal contributions, in this way the relevant components can be extracted, improving results for several applications [3.43].

Finally, polarimetric SAR interferometry (POLINSAR) combines the capability of interferometry to extract height information with polarimetric decomposition techniques [3.44]. With POLINSAR the topographic height of the phase centre of each extracted scattering mechanism can be estimated independently. This means that a limited volumetric imaging can be achieved with this technique, allowing for example an estimation of tree heights or underlying ground topography [3.45]. In recent times also increased attention was paid to model-based parameter estimation based on polarimetric interferometric data. This technique aims to invert scattering models in order to determine from the measured signals the physical parameters of interest [3.46], [3.47].

The concept of SAR polarimetry nowadays is applied in many scientific fields. Significant areas of application are particularly found in agriculture and forestry for crop monitoring, species identification and biomass estimation. Also in geology and hydrology the possibility of a characterisation for of ground roughness and of soil and snow moisture content is of great interest. Finally, in oceanography, SAR polarimetry is used for monitoring of wave systems, thermal and current fronts, and for estimating ice-age and thickness in polar regions [3.21], [3.48], [3.49], [3.50], [3.51], [3.52].

¹ Intrinsically four in the case of SAR, but under the assumption of reciprocal symmetric backscattering, the two cross-polar components are equal.

4 BASICS OF RADAR POLARIMETRY

4.1 DESCRIPTION OF ELECTROMAGNETIC WAVES

The purpose of this section is to illustrate the main features of the electromagnetic waves that are the base of the polarimetry. All aspects of macroscopic electromagnetic phenomena can be described in terms of the set of Maxwell equations that have the form [4.1], [4.2], [4.3]:

$$\vec{\nabla} \times \vec{E}(\vec{r}) = -j\omega\vec{B}(\vec{r}) \quad (4.1)$$

$$\vec{\nabla} \times \vec{H}(\vec{r}) = \vec{J}(\vec{r}) + j\omega\vec{D}(\vec{r}) \quad (4.2)$$

$$\vec{\nabla} \cdot \vec{B}(\vec{r}) = 0 \quad (4.3)$$

$$\vec{\nabla} \cdot \vec{D}(\vec{r}) = \rho(\vec{r}) \quad (4.4)$$

where:

- $\vec{E}$ is the electric field intensity vector [*Volt/meter*],
- $\vec{B}$ is the magnetic flux density vector [*Tesla*],
- $\vec{H}$ is the magnetic field intensity vector [*Ampere/meter*],
- $\vec{D}$ is the current displacement vector [*Coulomb/meter²*],
- $\vec{J}$ is the electric current density vector [*Ampere/meter*],
- ρ is the electric charge density [*Coulomb/meter³*].
- ω is the angular frequency of the fields (assumed to be monochromatic) [*rad/s*]

$\vec{E}, \vec{B}, \vec{H}, \vec{D}, \vec{J}$ and ρ are all real-valued functions of time t and spatial location $\vec{r}$, $\vec{r}$ being a position vector defined with respect to a specified coordinate system. Generation, propagation

and interactions of EM (electromagnetic) waves with matter are governed by these laws. Consider now a simple physical system such as the one in Figure 4.1 composed of a transmitting antenna, a target and a receiving antenna; define a global Cartesian coordinate system, with basis vectors $\hat{x}$, $\hat{y}$ and $\hat{z}$, with its origins within the target. The plane containing the directions of propagation of the transmitted and scattered waves is the *scattering plane*; referring to it, the complex transverse components of the electric field illuminating the scatterer are expressed in terms of a local right-handed coordinate system $(\hat{h}_1, \hat{v}_1 \text{ and } \hat{n}_1)$, coincident with the transmit antenna, defined so that the n_1 -axis is directed towards target. In a similar way a local coordinate system $(\hat{h}_2, \hat{v}_2, \hat{n}_2)$ is defined with its origin in the receiving system.

In the far field and for small targets, waves can be treated as planar; thus, the electric fields of three monochromatic (i.e. completely polarized) waves may be expressed as follows [4.3]:

$$\vec{E}_T = (E_{T_{h_1}}^2 + E_{T_{v_1}}^2)^{1/2} [\cos \alpha_T \hat{h}_1 + \sin \alpha_T e^{j\delta_T} \hat{v}_1] \cdot \exp j(\omega t - \vec{k} \cdot \vec{r}_1 + \phi_T) \quad (4.5)$$

$$\vec{E}_R = (E_{R_{h_2}}^2 + E_{R_{v_2}}^2)^{1/2} [\cos \alpha_R \hat{h}_2 + \sin \alpha_R e^{j\delta_R} \hat{v}_2] \cdot \exp j(\omega t + \vec{k} \cdot \vec{r}_2 + \phi_R) \quad (4.6)$$

$$\vec{A} = (A_{h_2}^2 + A_{v_2}^2)^{1/2} [\cos \alpha_A \hat{h}_2 + \sin \alpha_A e^{j\delta_A} \hat{v}_2] \cdot \exp j(\omega t - \vec{k} \cdot \vec{r}_2 + \phi_A) \quad (4.7)$$

where, in all the equations, $\alpha = \arctan(E_v / E_h)$ (E_h and E_v being the absolute values of the complex components of the electric field), δ and ϕ are the relative and absolute phases respectively, subscripts T and R stand for transmitter and receiver, subscripts 1 and 2 specify the coordinate system and $\vec{A}$, the *antenna height*, is the wave that the receiving antenna would radiate in the $+\hat{n}_2$ direction if it acted as a transmitter.

The three waves propagate in the $+\hat{n}_1$, $-\hat{n}_2$ and $+\hat{n}_2$ directions respectively. In the case of backscattering, $\vec{E}_T \approx \vec{A}$ only if the same antenna is used for transmitting and receiving. When this condition is held to be true, one can then write [4.4]

$$\vec{E}_T = \frac{Z_0 I_t}{2\lambda r} \vec{A} \quad (4.8)$$

where Z_0 is the impedance of the medium, I_t the terminal antenna current, λ the wavelength and $r = r_1 = r_2$.

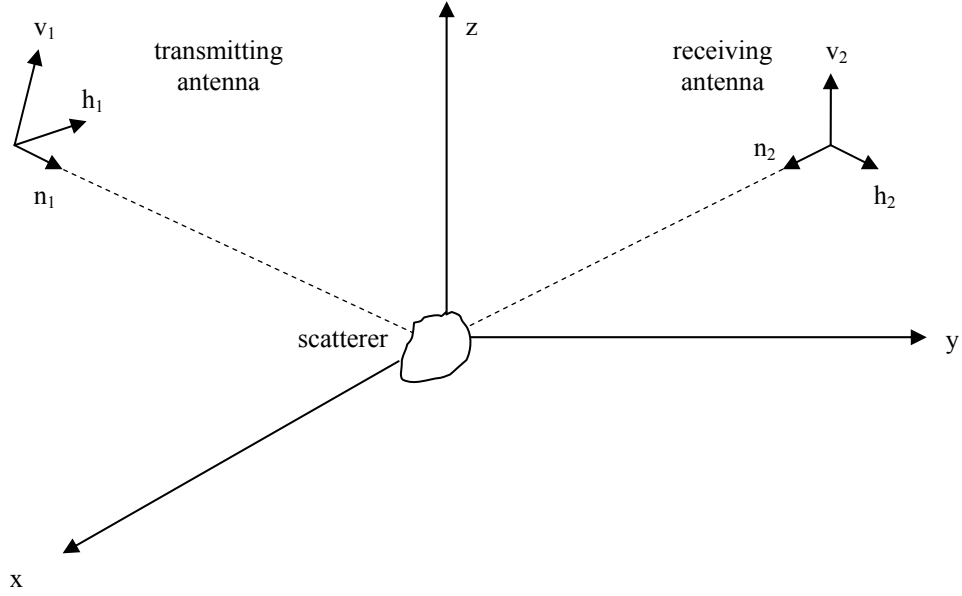


Figure 4.1: General bistatic scattering geometry and local coordinate systems.

For every monochromatic EM wave, the tip of the vector representing the electric field describes an ellipse on each generic fixed plane normal to the direction of propagations [4.5], the polarization of a wave receding from an observer is denoted *right-handed* if the electric field vector appears to be rotating clockwise in this plane and *left-handed* if it appears to be rotating counterclockwise on the chosen perpendicular plane. Therefore $\vec{A}$ may be defined as a *left-handed* circularly polarized wave (because of its $+n_2$ propagation) and $\vec{E}_R$ as a *right-handed* one (as it propagates along the $-\hat{n}_2$ direction) assuming $\alpha = \pi/4$ and $\delta = \pi/2$.

4.2 POLARIZED WAVES

Considering a plane electromagnetic (*EM*) wave, polarization refers to the locus of the electric field vector in the plane perpendicular to the direction of propagation. While the length of the vector represents the *amplitude* of the wave, and the rotation rate of the vector represents the *frequency* of the wave, polarization refers to the *orientation* and *shape* of the pattern traced by the tip of the vector. The waveform of the electric field strength (voltage) of an *EM* wave can be predictable (the wave is polarized) or random (the wave is unpolarized), or a combination

of both. In the latter case, the degree of polarization describes the ratio of polarized power to total power of the wave. An example of a fully polarized wave would be a *monochromatic* sine wave, with a single, constant frequency and stable amplitude.

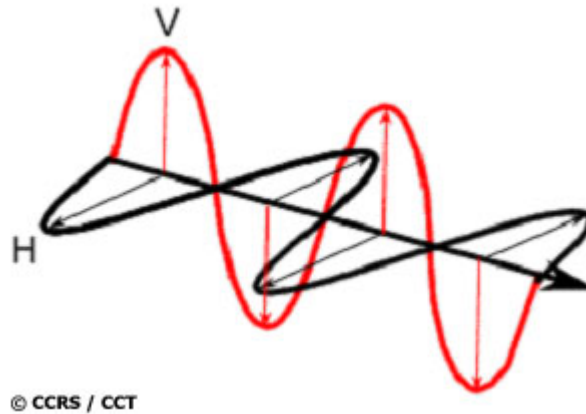


Figure 4.2: Examples of horizontal (*black*) and vertical (*red*) polarizations of a plane electromagnetic wave.

Many radars are designed to transmit microwave radiation that is either horizontally polarized (*H*) or vertically polarized (*V*). A transmitted wave of either polarization can generate a backscattered wave with a variety of polarizations. It is the analysis of these transmitted and received polarization combinations that constitutes the science of radar polarimetry.

Any polarization on either transmission or reception can be synthesized by using *H* and *V* components with a well-defined relationship between them. For this reason, systems that transmit and receive both of these linear polarizations are commonly used. With these radars, there can be four combinations of transmit and receive polarizations:

- *HH* - for horizontal transmit and horizontal receive;
- *VV* - for vertical transmit and vertical receive;
- *HV* - for horizontal transmit and vertical receive;
- *VH* - for vertical transmit and horizontal receive.

The first two polarization combinations are referred to as "*like-polarized*" because the transmit and receive polarizations are the same. The last two combinations are referred to as "*cross-polarized*" because the transmit and receive polarizations are orthogonal to each

another. Radar systems can have one, two or all four of these transmit/receive polarization combinations. Examples include the following types of radar systems:

SINGLE POLARIZED	• HH or VV (or possibly HV or VH)
DUAL POLARIZED	• HH and HV, VV and VH, or HH and VV
ALTERNATING POLARIZATION	• HH and HV, alternating with VV and VH
POLARIMETRIC	• HH, VV, HV, and VH

Table 4.1: Different types of radar systems.

The *relative phase* between channels is measured in a polarimetric radar, and is a very important component of the measurement. In the other radar types, relative phase may or may not be measured. The alternating polarization mode has been introduced on ENVISAT. As both wavelength and polarization affect how a radar system "sees" the elements in the scene radar imagery collected using different polarization and wavelength combinations may provide different and complementary information. Furthermore, when three polarizations are combined in a colour composite, the information is presented such that an image interpreter can infer more information about the surface characteristics.

4.3 SCATTERING MATRIX

The primary description of how a radar target or surface feature scatters *EM* energy is given by the scattering matrix. We can consider an electromagnetic wave transmitted with a particular polarization which is reflected by the target in the receiver direction with a polarization which is generally different from the incident one. This polarization change is modelled by a 2×2 complex matrix S called scattering matrix. Particularly, the scattering matrix S relates the incident electric field E_i to the scattered field E_s by

$$\begin{bmatrix} E_{s1} \\ E_{s2} \end{bmatrix} = \frac{e^{jkr}}{r} \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \begin{bmatrix} E_{i1} \\ E_{i2} \end{bmatrix} \quad (4.9)$$

where subscripts 1 and 2 usually refer to horizontal (H) and vertical (V) polarizations, respectively, k (i.e. $k = 2\pi/\lambda$) is the wavenumber, and r is the range. The scattering matrix, thus, defines the interaction completely. Its elements are, in general complicated functions of frequency, target orientation and shape, relative orientation of the polarization planes, etc. The diagonal elements S_{11} , S_{22} and the off-diagonal ones S_{12} , S_{21} are called respectively *co-* and *cross-polarized* components. In the most general case, there are seven independent parameters in the scattering matrix: four amplitudes and three phases. Indeed, an overall absolute phase may be neglected, since the power received from the scatterer is independent of this phase. For a reciprocal scattering medium, $S_{12} = S_{21}$ [4.8] and there are only five independent parameters in the scattering matrix.

Now we define a matrix M , which is composed of the complex value of channel for each pixel of the uncalibrated polarimetric image. Because of path length differences, gain differences, coupling between channels and noise, the measured (complex valued) matrix is not equal to the scattering matrix. So the scattering matrix S has been related to the measured matrix M in this way [4.6], [4.7]:

$$M = R^T S T + N$$

$$M = \begin{bmatrix} R_{11} & R_{21} \\ R_{12} & R_{22} \end{bmatrix} \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \begin{bmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{bmatrix} + \begin{bmatrix} N_{11} & N_{12} \\ N_{21} & N_{22} \end{bmatrix} \quad (4.10)$$

where R and T are the receive and transmit distortion matrices, respectively, superscript t denotes transpose and the matrix N represents the noise present in each channel. For an ideal radar system, R and T are both identity matrices multiplied by complex constants. As the off-diagonal terms of R and T represent leakage or coupling between channels, they ideally would be very small.

5 POLSAR DATA REPRESENTATION

5.1 REPRESENTATION THROUGH 4 PARAMETERS

The scattering matrix is a fundamental tool in remote sensing problems and it is represented as a 2 by 2 *complex matrix*:

$$S = \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix} \quad (5.1)$$

Usually, this matrix is symmetric because the value of the polarizations HV and VH are equal. With this model the POLSAR image (polarimetric SAR image) is represented by 4 parameters (S_{HH} , S_{VV} , S_{HV} and S_{VH}). So the image has four complex channels (for four polarizations: HH , VV , HV and VH). It is also useful to describe an important matrix called coherency matrix which is computed locally for every pixel. This means that this matrix do not consider the spatial context (like, for instance, the covariance matrix does). To compute the coherence matrix it is useful to represent the pixel as a vector with, in this case, four components (like the number of channels):

$$\underline{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} S_{HH} \\ S_{HV} \\ S_{VH} \\ S_{VV} \end{bmatrix} \quad (5.2)$$

Actually, the coherence matrix is computed by the formula:

$$[T] = \underline{x} \cdot \underline{x}^{*^T} \quad (5.3)$$

where $*$ indicates the conjugate complex and T is for the row vector. The image is so represented with this feature vector and with the coherence matrix [which is indicated with $[T]$ in (5.3)].

5.2 REPRESENTATION THROUGH 3 PARAMETERS

If we make the assumption that $S_{HV} = S_{VH}$ then it is possible to represent the feature vector in an another way.

$$\underline{x} = \begin{bmatrix} S_{HH} + S_{VV} \\ S_{HH} - S_{VV} \\ 2S_{HV} \end{bmatrix} \quad (5.4)$$

So the coherence matrix changes from 4 by 4 to 3 by 3 sized. With this representation it is possible to obtain other important parameters. Obviously the representation in (5.4) is not acceptable if the value S_{HV} and S_{VH} are not equal. However in more cases these values are very similar. With (5.4) the new coherence matrix is the following one [5.1]:

$$[T] = \frac{1}{2} \begin{bmatrix} (S_{HH} + S_{VV})(S_{HH} + S_{VV})^* & (S_{HH} + S_{VV})(S_{HH} - S_{VV})^* & 2(S_{HH} + S_{VV})S_{HV}^* \\ (S_{HH} - S_{VV})(S_{HH} + S_{VV})^* & (S_{HH} - S_{VV})(S_{HH} - S_{VV})^* & 2(S_{HH} - S_{VV})S_{HV}^* \\ 2S_{HV}(S_{HH} + S_{VV})^* & 2S_{HV}(S_{HH} - S_{VV})^* & 4S_{HV}S_{HV}^* \end{bmatrix} \quad (5.5)$$

There are different transformations which permit to obtain a diagonal matrix from $[T]$. Each of these ones exhibit the same structure which is useful to extract some important parameters [5.1].

$$[T] = [U_3] \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} [U_3]^* \quad (5.6)$$

The matrix $[U_3]$ is a rotation matrix which permits to obtain the diagonal matrix in (5.6). In this way the eigenvalues of the diagonal matrix are easy to compute; they are the diagonal elements.

$$[U_3] = \begin{bmatrix} \cos \alpha_1 & \cos \alpha_2 & \cos \alpha_3 \\ \sin \alpha_1 \cos \beta_1 e^{i\delta_1} & \sin \alpha_2 \cos \beta_2 e^{i\delta_2} & \sin \alpha_3 \cos \beta_3 e^{i\delta_3} \\ \sin \alpha_1 \cos \beta_1 e^{i\gamma_1} & \sin \alpha_2 \cos \beta_2 e^{i\gamma_2} & \sin \alpha_3 \cos \beta_3 e^{i\delta_3} \end{bmatrix} \quad (5.7)$$

5.3 THE PARAMETER ENTROPY H

The dominant matrix (of the eigenvalues) is determined from the maximum eigenvector of $[T]$ and has a degree of randomness or entropy defined from the eigenvalues¹ as

$$H = \sum_{i=1}^n -P_i \log_n P_i \quad (5.8)$$

with

$$P_i = \frac{\lambda_i}{\sum_{j=1}^n \lambda_j} \quad (5.9)$$

where $n = 3$ for backscattering problems (so that, by construction, $0 \leq H \leq 1$).

There are two limitations to this simplified model:

- 1) It is limited to reflection symmetric media, where by definition the co- and cross-linear polarization scattering coefficients are uncorrelated i.e., $E(S_{HH} S_{HV}^*) = E(S_{VV} S_{HV}^*) = 0$ [5.1], [5.2]. Even if common, such media are not the only ones of interest in remote sensing applications.

¹ Suppose to have a matrix A ($\mathcal{V}^n \rightarrow \mathcal{V}^n$), a vector $x \in \mathcal{V}^n$ ($x \neq 0$) and a scalar $\lambda \in \mathbb{C}$. In the situation in which $Ax = \lambda x$, λ is known as eigenvalue and x is its associated eigenvector.

- 2) If the lowest eigenvalues λ_2 and λ_3 are not equal, then the entropy is not an unique measure of the randomness of the scatterer. In this case the degree of polarization of the scattered waves depends on the polarization of the incident wave.

If the scattering medium does not have reflection symmetry, then a preferred orientation angle β for the scatterer exists and we can accommodate this one by introducing a new parameterization of the eigenvectors of $[T]$ as showed in (5.10):

$$\underline{e} = [\cos \alpha \quad \sin \alpha \cos \beta e^{i\delta} \quad \sin \alpha \cos \beta e^{i\gamma}]^T \quad (5.10)$$

The parameterization of a 3 by 3 unitary matrix is in terms of column vectors with different parameters $\alpha_1 \beta_1$ etc. In general, the 3 by 3 matrix has only eight independent parameters since the columns are not only unitary but mutually orthogonal. This means that in practice α_1 , α_2 and α_3 are not independent. By including the angle β we have allowed the possibility that the cross and co-polar channels may be correlated. The second modification is to allow the possibility of a nondegenerate eigenvalue spectrum ($\lambda_2 \neq \lambda_3$). In this case we can consider the following statistical model of the scatterer.

5.4 THE PARAMETER α

We model the target as a three symbol Bernoulli process i.e., the target is represented by three $[S]$ matrices [the columns of $[U_3]$ in (5.7)] which occur with probabilities P_i (defined in (5.8) by the normalized eigenvalues so that $P_1 + P_2 + P_3 = 1$). In this way, for example, the parameters α can be associated to a random sequence:

$$x(n) = \alpha_1 \alpha_2 \alpha_1 \alpha_3 \alpha_2 \alpha_1 \alpha_1 \alpha_3 \dots \quad (5.11)$$

and our best estimate (in the sense of maximum likelihood) of the parameters is given by the mean of this sequence which is easily evaluated as follows:

$$\bar{\alpha} = P_1\alpha_1 + P_2\alpha_2 + P_3\alpha_3 \quad (5.12)$$

In this way, we extract the mean parameters of the dominant scattering mechanism from the coherence matrix, as a vector $\underline{f}$ such that

$$\underline{f} = [\bar{\alpha} \quad \bar{\beta} \quad \bar{\delta} \quad \bar{\gamma}] \quad (5.13)$$

where the parameters α , β , δ and γ are defined in (5.7). In this way, we can concentrate our attention on two parameters: entropy (H) and alpha (α) which are useful to represent the image from different points of view. Actually, the output image derived from the extraction of these parameters has two channels. So the feature vector which represents the pixel is a vector with two elements. Through these parameters it is possible to classify with different algorithms or it is possible to classify the image in the plane H/α which is divided in 9 parts according to the values of H and α . This issues will be addressed in the next chapters about the unsupervised classifications (Chapter 8).

6 BASICS OF STOCHASTIC PROCESS

6.1 INTRODUCTION

We introduce some concepts about stochastic processes. The definition of stochastic process is the following one:

"A stochastic process is a family of random variables which depend on a parameter t ".

A stochastic process is indicated with the notation: $\{X_t, t \in T\}$ where t it is a parameter and T is the range of the possible values of t . Usually, t is interpreted as a time variable. The domain of variability of t can be a subset of real numbers (coinciding, eventually, with the entire axis of the times); it can assume discrete values, as well. The random variable X_t takes values in X , named *space of the states*, that can be a continuous set (in such case we are speaking about *continuous stochastic process*), or discrete (and in this case we are speaking about *discrete stochastic process*). Elements $x_i \in X$, that are the values which the random variable X_t can take, are said *states of the system* and they represent the possible results of an experiment. The variable X_t of a given phenomenon is drawn among them according to a given probability distribution. A random variable is not only the values which can take, but also if the distribution of probability is not, therefore in order to know a stochastic process, it is necessary to know the values which X_t can take and the joint distribution and distributions of probability among them. It is possible to consider the simplest stochastic processes in which the range of variability of t is made of discrete values. There are processes for which "knowing the past and the present does not allow determining the future"; other ones, instead, for which the future is determined from the present, without considering the past. In order to understand this concept in a better way, we consider the following experiment: we suppose to toss a coin an infinite number of times and that each test happens every

t seconds. In a Cartesian diagram, for every toss, we represent a step towards the high if the result is head (H), towards the bottom if the result is cross (C). After some t seconds from the beginning of the experiment, we will have a particular situation indicated with $x(t)$ which exclusively depends on the fact that we have head or cross. If the results of the launches are, for example, the ones in Table 6.1, $x(t)$ is given by the plot in Figure 6.1.

1	2	3	4	5	6	7	8
H	H	H	C	H	C	C	H

Table 6.1: A test experiment to introduce the stochastic process.

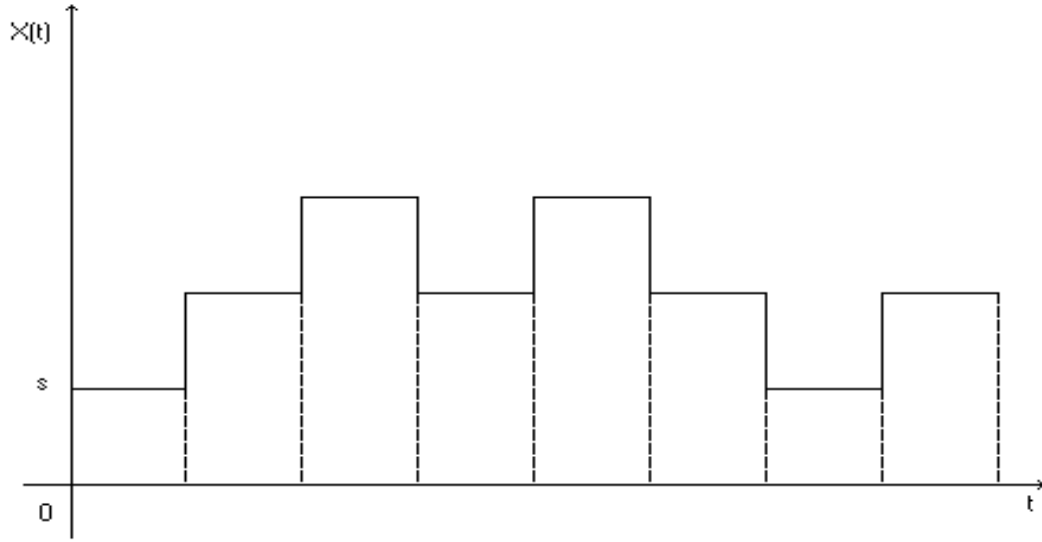


Figure 6.1: Graph of the results of the experiment.

The realization (or history) of a stochastic process constitutes a visible trace of the process. Considering the case of n tosses in which, for k times head is obtained; we will have:

$$t = n\tau \quad (6.1)$$

with τ the interval between every launch; exactly k steps towards the high and $n-k$ steps towards the bottom

$$x(n\tau) = ks - (n-k)s = (2k-n)s \quad (6.2)$$

At k , it is possible to assign the value $0, 1 \dots n$ thus $X(n\tau)$ is a random variable which assumes the values¹:

$$P_k = \binom{n}{k} \left(\frac{1}{2}\right)^{n-k} \left(\frac{1}{2}\right)^k = \binom{n}{k} \left(\frac{1}{2}\right)^n \quad (6.3)$$

With this simple example, it is possible to show what a stochastic process is and what this one represents. It is now the moment to emphasize stochastic process about images, in the field of remote sensing. In this field, the process does not depend on the time but it depends on spatial coordinates.

6.2 FEATURE VECTOR

Let us consider a vector x . We can view it like a random variable X and, therefore, as a sample of a stochastic process. The stochastic process is, therefore, one statistical model which allows studying and processing the second aspects of image processing in a better way. Specifically, the process under analysis regards the possible series of values that the vector of the features can take. As the feature represents the properties of a given pixel, the random variable x can take all the admissible values in the range defined by the properties that it represents. For this reason, it is usual to associate a PDF (probability density function) with the random variable X .

¹ This problem follows the Binomial Distribution for which it is possible to prove the formula in Equation (6.3).

6.3 DATA CHARACTERIZATION

Historically, SAR Data models have been largely directed at describing single-point statistics through the PDF. Initially, therefore, it is important to consider whether typical SAR images are actually capable of representation in such sample terms. There are fundamental problems in terms of nature of the scene being imaged, the physical properties of the imaging process, and finally the analysis of measured data. Image formation also introduces variability through physical differences in the imaging processes. SAR images depend on the system wavelength due to differing foliage penetration, for example. Geometrical effects due to the incidence angle will influence the scene through processes such as shadowing. Finally, the resolution of the imaging system will have a strong effect on the appearance of the clutter because it will average out features on a smaller scale than the resolution while preserving larger ones. In view of the number of processes that contribute to differences between images, it is doubtful whether it is ever worth attempting to define data models with great precision. We need only be concerned with general properties that can be exploited in image interpretation.

There are different types of PDF to represent the modelling for features vector. It is necessary a compromise between choosing a simple model (characterized by few degrees of freedom) that fails to reproduce the observed PDF accurately and a more sophisticated model (with many degrees of freedom) that fits the data.

A variety of PDFs having two degrees of freedom have been proposed, such as Weibull, lognormal and K distributions. The additional degree of freedom allows them to represent different contrasts in the data. However, there are occasions where even this two-parameter PDF fails to provide a reasonable match to the data and it is necessary to introduce more parameters.

6.3.1 EMPIRICAL DATA DISTRIBUTIONS

Early measurements of the distributions on radar data were carried out with a low resolution so that objects, such as trees and houses were much smaller than a resolution cell. In order to describe higher resolution data, it was necessary to introduce further parameters to characterize the image contrast. Log normal and Weibull are two such distributions that were widely applied. For example, the amplitude of low-resolution sea clutter has been fitted to the Rayleigh distribution², corresponding to pure speckle, while the log normal was applied at a higher resolution. These distributions can be fitted to other amplitude or intensity data.

The log-normal distribution is given by:

$$P(x) = \frac{1}{x\sqrt{2\pi V}} \exp\left[-\frac{(\ln x - \beta)^2}{2V}\right] \quad (6.4)$$

where x is a generic observable, and β and V are the mean and variance of $\ln x$.

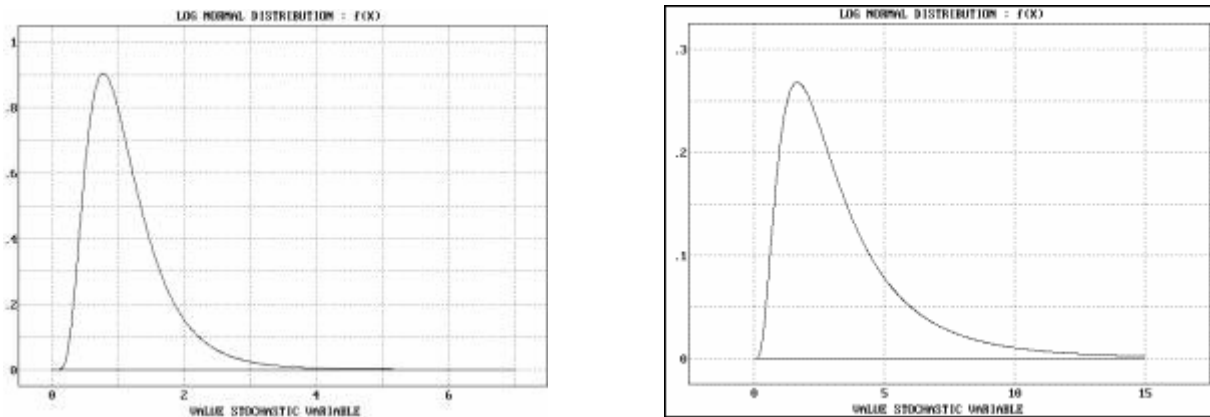


Figure 6.2: Plot of the lognormal probability density function with different values of the parameters (left: $\beta=0$, $V=0.5$; right: $\beta=1$, $V=0.7$).

The mean value and n-th normalized moment of the distribution are

² Rayleigh distribution with probability function: $P(r) = \frac{r}{s^2} e^{-\frac{r^2}{2s^2}}$ where $r \in [0, \infty)$

$$E(x) = \exp\left[\beta + \frac{V}{2}\right] \quad \text{and} \quad \frac{E(x^n)}{E(x)^n} = \exp\left[n(n-1)\frac{V}{2}\right] \quad (6.5)$$

The log-normal distribution has a problem with the observable scene having a zero value. Actually it is necessary to adopt a different strategy. Thus, when the observable scene has the value zero the probability of log-normal distribution is zero. In addition it is a very poor representation of single-look intensity speckle.

The Weibull Distribution is given by

$$P(x) = \frac{cx^{c-1}}{b^c} \exp\left[-\left(\frac{x}{b}\right)^c\right] \quad (6.6)$$

where b is a scaling parameter, and c controls the shape.

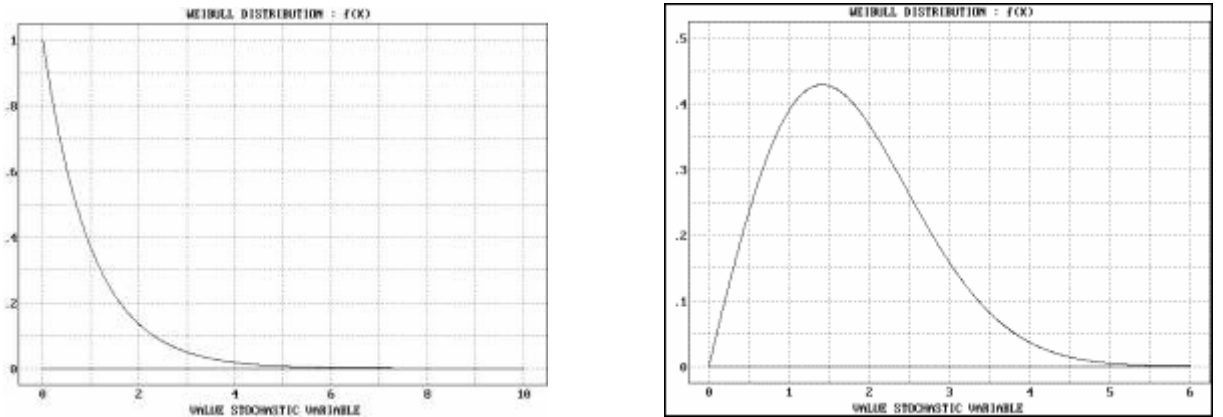


Figure 6.3: Plot of the Weibull probability density function with different parameters (left: $b=1$, $c=1$; right: $b=2$, $c=2$).

The mean and n -th normalized moments of this distribution are

$$E(x) = b\Gamma\left(1 + \frac{1}{c}\right) \quad \text{and} \quad \frac{E(x)^n}{E(x)^n} = \frac{\Gamma(1 + n/c)}{\Gamma^n(1 + 1/c)} \quad (6.7)$$

The Weibull distribution is identical to a Rayleigh PDF when $c=2$ and to a negative exponential when $c=1$. Thus, it can describe single-look speckle precisely for either amplitude or intensity. Unfortunately, it cannot represent multilook speckle exactly.

The general formula for the probability density function of the gamma distribution is:

$$P(x) = \frac{x^{\alpha-1} e^{-x\lambda}}{\Gamma(\alpha)\lambda^{-\alpha}} \quad (6.8)$$

where λ is the shape parameter, α is the location parameter and Γ is the gamma function which is defined as follows:

$$\Gamma(\alpha) \equiv \int_0^{\infty} t^{\alpha-1} e^{-t} dt \quad (6.9)$$

Since the general form of probability functions can be expressed in terms of the standard distribution, all subsequent formulas in this section are given for the standard form of the function.

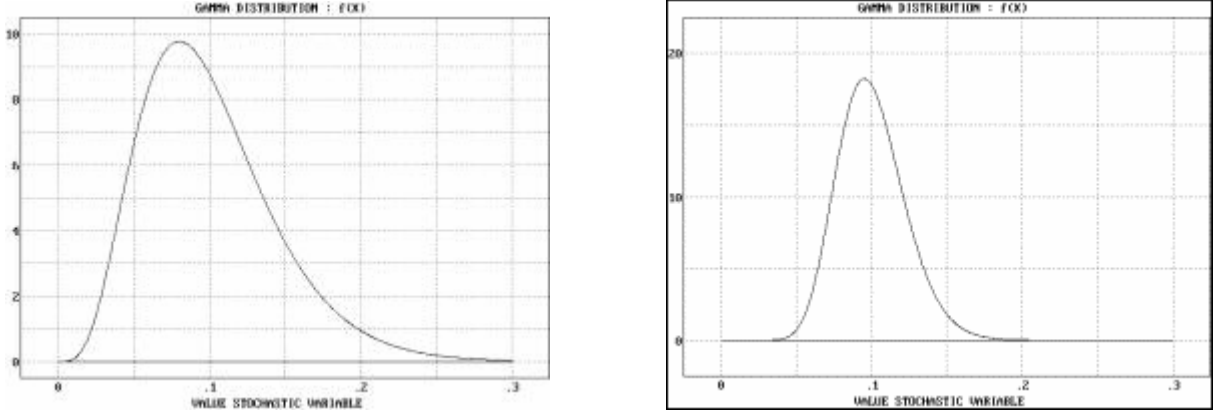


Figure 6.4: Plot of the Gamma probability density function with different parameters (left: $\alpha=5$, $\lambda=50$; right: $\alpha=20$, $\lambda=200$).

6.3.2 COVARIANCE ANALYSIS OF DATA

We have defined the different PDFs which are useful to represent the probability of a stochastic variable x . Actually, this variable is associated to every pixel which is considered independently. For this reason, it is necessary to explain some data property. The first one of these properties is the Covariance Matrix. We consider a random vector $\underline{X}$ whose components are random variables X_i . Given any pair of components, X_i and X_j , we denote their covariance as either $\text{cov}(X_i, X_j)$ or $\sum_{i,j}$. The covariance is defined by the expectation:

$$\text{cov}(X_i, X_j) = E[(X_i - \mu_i)(X_j - \mu_j)] \quad (6.10)$$

where μ_i and μ_j are the means of X_i and X_j . By definition, covariance is symmetric, with $\Sigma_{i,j} = \Sigma_{j,i}$. Also, the covariance of any component X_i with itself is the variance of that component:

$$\text{cov}(X_i, X_i) = E[(X_i - \mu_i)^2] = \text{var}(X_i) \quad (6.11)$$

We collect all the covariances of a random vector $\underline{X}$ in a “covariance matrix”:

$$\mathbf{\Sigma} = \begin{pmatrix} \Sigma_{1,1} & \Sigma_{1,2} & \Sigma_{1,3} & \cdots & \Sigma_{1,n} \\ \Sigma_{2,1} & \Sigma_{2,2} & \Sigma_{2,3} & & \vdots \\ \Sigma_{3,1} & \Sigma_{3,2} & \Sigma_{3,3} & & \vdots \\ \vdots & & & \ddots & \\ \Sigma_{n,1} & & \cdots & & \Sigma_{n,n} \end{pmatrix} \quad (6.12)$$

Thanks to the symmetry property of covariances, this is necessarily a symmetric matrix. It can be shown that covariance matrices are positive definite or positive semi definite. The magnitude of a covariance depends upon the standard deviations of the two components X_i and X_j . To obtain a more direct indication of how two components co-vary, we scale covariances to obtain correlation. Given any pair of components, X_i and X_j we denote their correlation as either $\text{cor}(X_i, X_j)$ or $\rho_{i,j}$.

The correlation is defined as

$$\rho_{i,j} = \frac{\text{cov}(X_i, X_j)}{\sigma_i \sigma_j} \quad (6.13)$$

where σ_i and σ_j are the standard deviations of X_i and X_j . By construction, a correlation is always a number between -1 and 1 . Correlation inherits the symmetry property of covariance: $\rho_{i,j} = \rho_{j,i}$. If X_i and X_j are independent, their correlation is 0 . The converse is not true.

As for the covariances, we can collect all the correlations of a random vector $\underline{X}$ in a symmetric “correlation matrix”:

$$\mathbf{\rho} = \begin{pmatrix} \rho_{1,1} & \rho_{1,2} & \rho_{1,3} & \cdots & \rho_{1,n} \\ \rho_{2,1} & \rho_{2,2} & \rho_{2,3} & & \vdots \\ \rho_{3,1} & \rho_{3,2} & \rho_{3,3} & & \\ \vdots & & & \ddots & \\ \rho_{n,1} & & \cdots & & \rho_{n,n} \end{pmatrix} \quad (6.14)$$

7 ELEMENTS OF PARAMETER ESTIMATION

7.1 INTRODUCTION OF ELEMENT PARAMETER ESTIMATION

The parameter estimation theory together with signal estimation and decision theory, provides powerful tools for many typologies of image processing like restoration, segmentation, feature extraction, and clustering. Image information extraction requires recognizing which structures can be identified. Indeed, a hypothesis testing rule is after introduced as a partitioning of the measurement space into disjoint sets. Each set explains a certain scene structure. We can derive the decision rule learning probability distribution from the observed data, i.e. estimate the likelihood over the space partition in the assumed hypotheses by measuring frequencies of measurements under known assumptions. The Bayesian approach consists in interpreting probabilities based on a system of axioms which describe the incomplete information [7.1] and allow an interference¹ process that assigns probabilities for a given data set and a hypothesis space. Thus, the Bayesian approach enables a self-consistent interference process, permitting the specifications of information retrieval problems directly using models in the form of probability distributions [7.2]. Particularly, the Bayesian information retrieval from

¹ Interference process: the result of this process depends on all the possible histories of the same process. It is based on a paradox of quantum mechanics [7.2].

remote-sensing (RS) images is essentially an inverse problem consisting of searching for the scene that best explains the observed data.

7.2 PARAMETER ESTIMATION

We consider now the following model for the observation y :

$$y = f\{x\} + n \quad (7.1)$$

where f represents a deterministic model of the image formation process (the forward model), x is the physical parameter vector we want to extract from the observations and n is the measurement noise. The physical parameter vector x is a realization of a stochastic process. In order to obtain classical parameter estimation it is necessary to minimize the Bayes risk R defined over the signal space. Specifically, the risk R is the expectation of a cost function defined over the joint space of observations (Y) and model parameters (X).

$$R = E[c(x, \hat{x})] = \int \int c(x, \hat{x}) p(x, y) dx dy \quad (7.2)$$

The cost function is a measure of the quality of the estimated parameter $\hat{x}$, being defined as a “distance” between the true and the desired (but unknown) value of x . There are different kinds of cost functions, but the most frequently used ones are the quadratic and uniform (Figure 7.1).

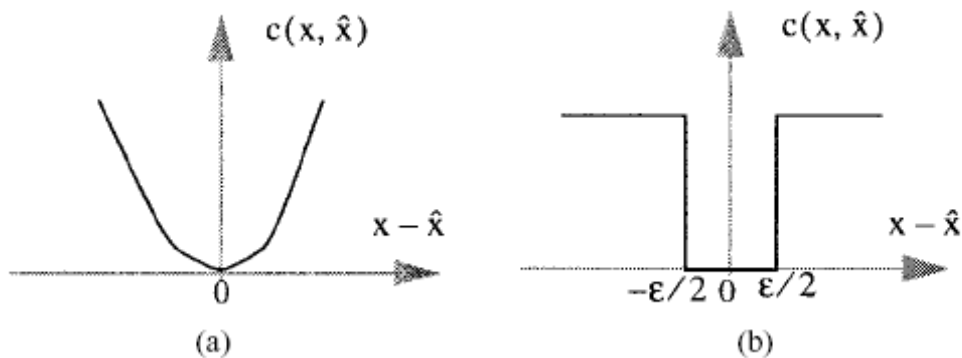


Figure 7.1: (a) Quadratic cost function that leads to the MMSE estimator and (b) uniform cost function leading to the MAP estimator.

Minimizing the risk using a quadratic cost function, we obtain the minimum mean square estimator (MMSE) which is given by [7.5]:

$$\hat{x}_{MMSE} = \int_X x p(x|y) dx \quad (7.3)$$

which is the conditional mean of the desired parameter. Minimizing the risk and using a uniform cost function, at the limit $\varepsilon \rightarrow 0$, we obtain the Maximum “*A Posteriori*” (MAP) estimator [7.5]:

$$\hat{x}_{MAP} = \arg \max_x \{ \log p(y|x) + \log p(x) \} \quad (7.4)$$

A special case is the estimation of a deterministic parameter a , whose probability distribution function is a δ distribution

$$p(x) = \delta(x - a) \quad (7.5)$$

In this case, MMSE degenerates into the originally assumed value and cannot extract any information about x from the observation y :

$$\hat{x}_{MMSE} = \int_X x \delta(x - a) dy = a \quad (7.6)$$

The estimation problem is reformulated defining the likelihood function $p(x|y)$ and searching for its maximum. The maximum likelihood (ML) estimator is defined as:

$$\hat{x}_{ML} = \arg \max_x \log p(y|x) \quad (7.7)$$

Using a cost function, therefore, we obtain a type of prior information. This fact is observed in the role played by posterior distributions:

$$p(x|y) = \frac{p(y|x)p(x)}{p(y)} \quad (7.8)$$

Assuming x and n to be independent for the observation model in (7.1) the likelihood factor can be rewritten as:

$$p(x|y) = \frac{p_n(y - f(x))p(x)}{p(y)} \quad (7.9)$$

where p_n is the probability distribution function (PDF) of the noise. So the expression for the posterior encapsulates the deterministic prior knowledge represented by the forward model and, in addition, the knowledge about the observation noise and the prior information about the desired parameter are also included. So we can consider MAP to be a complete frame for model-based approaches in information extraction. In general there are several parameter estimation methods and some of those ones are presented in Table 7.1

LIKELIHOOD	PRIOR	log - POSTERIOR	REMARKS
$p(y x)$	$p(x)$	$\log[p(x y)]$	
A known PDF	A known PDF	$\log[p(y x)] + \log[p(x)]$	Full MAP estimation
A known PDF	Uniform PDF	$\log[p(y x)]$	ML estimation
	Unknown a) $MENT$ $e^{H(x) - C(x)}$	$\log[p(y x)] + \log[p(x)]$	MAPMENT
	Unknown b) Jeffrey's $ I(x) ^{1/2}$	$\log[p(y x)] + \log[p(x)]$	MAP a special case
A symmetric PDF	A known PDF		MMSE estimation
Gaussian	Surface smoothness condition	$-\left(\ y - f(x)\ + \lambda C(x)\right)$	CLS estimation
Gaussian	Unspecified	$-\ y - f(x)\ $	LS estimation

Table 7.1: Summary of classical parameter estimation method as particular cases of the MAP approach.

Indeed, Table 7.1 summarizes the basic parameter estimation methods from the point of view of the capability to model a problem. MAP is a complete approach, encapsulating both stochastic models and deterministic prior knowledge. The least-squares (LS) estimation is a weak approach, handling no prior knowledge about the information retrieval problem. Using

different noise and prior models MAP itself is, therefore, sufficient to define the other estimators. ML is a least-committed estimator, in the sense that it chooses the parameters from the observation space where the noise is zero with maximum probability. Due to this property, ML is in many situations preferred to other methods. The standard regularization of the solution of ill-posed problems leads to objective functional similar to the conditional least squares (CLS) [7.5]. The difference lies in the physical meaning of the regularization operator $C(*)$, which in the deterministic approach is understood as the degree of smoothness of a reconstructed signal [7.5].

7.3 QUALITY OF THE ESTIMATED PARAMETERS

As the estimated parameters are functions of the observations that are random variables so they are themselves random variables. The quality of the estimated parameters [7.3] is evaluated by their conditional mean relative to $p(y|x)$

$$E[x] = E[\hat{x}(y)] \quad (7.10)$$

If the conditional mean equals the true value of the parameter, the estimator is unbiased. A stronger quality criterion for the estimated parameters is the conditional variance relative to $p(y|x)$

$$\sigma_x^2 = E[\hat{x}(y) - E[\hat{x}(y)]]^2 \quad (7.11)$$

A good estimator must have a variance as small as possible. The limit of the quality of an estimator is given by the Cramer-Rao bound

$$\sigma_x^2 \geq \frac{1}{E\left[\frac{\partial}{\partial x} \log p(y|x)\right]^2} \quad (7.12)$$

The Cramer-Rao bound can be expressed using the Kullback-Leibler ($K-L$) divergence

$$\sigma_x^2 \geq \lim_{x \rightarrow x_0} \left[\frac{2D(p_x \| p_{x_0})}{(x - x_0)} \right]^{-1} \quad (7.13)$$

where D is the $K-L$ divergence

$$D(p_x \| p_{x_0}) = \int_{-\infty}^{+\infty} p(y|x) \log \frac{p(y|x)}{p(y|x_0)} dy \quad (7.14)$$

If the parameter is a vector, the quality of the estimator is measured by its covariance matrix

$$Cov_x = E\left[\left[\hat{x}(y) - E[\hat{x}(y)]\right]\left[\hat{x}(y) - E[\hat{x}(y)]\right]^T\right] \quad (7.15)$$

The lower bound of the quality is given by

$$Cov_x \geq \frac{1}{I(x)} \quad (7.16)$$

where $I(x)$ is the *Fisher information matrix*² whose (K, p) -entry is given by:

$$I_{k,l}(x) = \int_{-\infty}^{+\infty} p(y|x) \left[\frac{\partial}{\partial x_k} \log p(y|x_k) \right] \cdot \left[\frac{\partial}{\partial x_l} \log p(y|x_l) \right] dy \quad (7.17)$$

² Let $X(n) = X(x_1, x_2, \dots, x_n)$ be a random vector in $\mathcal{R}^n$ and let $f_X(x)$ be a probability distribution on X with continuous first and second order partial derivatives. The Fisher information matrix of X is $n \times n$ the matrix I_X whose (i, j) th the entry is given by

$$(I_X)_{i,j} = E\left(\frac{\partial \ln f_X(x)}{\partial x_i} \frac{\partial \ln f_X(x)}{\partial x_j}\right) = \int_{\mathcal{R}^n} \frac{\partial \ln f_X(x)}{\partial x_i} \frac{\partial \ln f_X(x)}{\partial x_j} f_X(x) \partial^{(n)} x$$

7.4 BAYESIAN PARADIGM

The probability theory provides the following basic product and summation rules:

$$p(A, B|C) = p(A|B, C)p(B|C) \quad p(A|B) + p(\bar{A}|B) = 1 \quad (7.18)$$

where A, B, C are events, with $P(C) > 0$ and $P(B) > 0$. For example, if AB and BA represent the same event, we can change A and B on the right-hand side of (7.18) and if $p(B|C) > 0$, we obtain the *Bayes* theorem:

$$p(A|B, C) = \frac{p(B|A, C)p(A|C)}{p(B|C)} \quad (7.19)$$

So, the fundamental equations of probability theory are uniquely determined as rules of inference. If the probability theory is a set of normative rules for conducting interference, the Bayes theorem is to be seen as a method of translating information into a probability assignment. Thus, the Bayesian approach consists of interpreting probabilities based on a system of axioms describing the incomplete information rather than randomness and enabling the interference process, which for a given data set and a hypothesis space assigns probabilities to the hypothesis. The classical decision and estimation theory relies on the availability of some knowledge expressed as pdf's and a definition of a cost function. As already observed from Table 7.1, the specification of the cost seems to be discretionary. It is the Bayesian approach that enables the self-consistent interference process, thus making possible the specification of information retrieval problems directly using models in the form of probability distributions.

7.4.1 BAYESIAN MODEL INFERENCE

According to the Bayesian paradigm, a statistical model is defined as a parametric stochastic model and a prior distribution defined on the parameter space $\{p(y|x), p(x)\}$ [7.4]. The *Bayes* theorem (7.19) shows how the information on x is updated by the information contained in the data y . Models are decomposed into simpler hierarchical layers thanks to the complexity of the observed scenes. A *hierarchical Bayes* model is a Bayesian stochastic model $\{p(y|x), p(x)\}$ in which the prior $p(x)$ is decomposed into conditional distributions: $\{p(x|\theta_1), p(\theta_1|\theta_2) \dots p(\theta_m)\}$, where $p(\theta_m)$ is the marginal distribution.

7.4.2 BAYESIAN INFORMATION RETRIEVAL

One of the main difficulties in the interpretation of RS images is their intrinsic high complexity (both radiometric and structural). In real applications, very seldom sufficient training data sets and ground truth are available to allow models for scattering and structures to be extracted from the observed scenes. Therefore, the interpretation of RS images using computer vision methods is often an ill-posed approach. Alternatively, we can consider the scene understanding approach, which aims at extracting information about the physical characterization, structure and geometry of three-dimensional (3-D) scenes from two-dimensional (2-D) RS images. The task of the scene understanding process is to find the scene that best explains the observed data. Scene understanding is an ill-posed inverse problem. Inspired by (7.9), we present an algorithm for the solution of the scene understanding problem (Figure 7.2). Trying to interpret an RS image, we assume to know some elements in advance. The assumed known information (*prior models*) can be used to simulate the scene, e.g., a landscape, and observe it virtually with a simulated sensor (forward model). The resulting

simulated image can now be compared to the actual RS image. The error will depend on the quality of the guess of the parameters describing the scene.

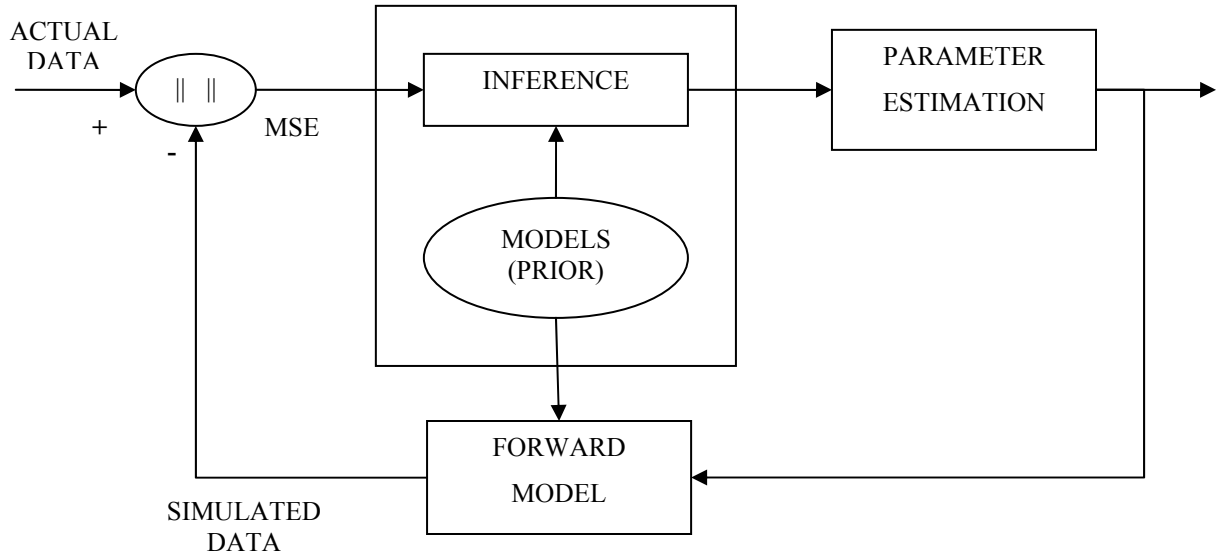


Figure 7.2: Parameter estimation paradigm. The MAP formalism or the model fitting in the *Bayesian* approach inspire the idea of an iterative algorithm for information extraction from observation (7.9). We have to compare the observations y with the simulated data based on a guess of the parameters. The parameter x is restricted to take values only according to a certain prior model.

A set of fixed and variable parameters controlling the image synthesis characterizes the scene model. Thus, one model could describe all aspects of a real scene. A pragmatic approach is to consider a library of rather simple models, each one describing a part of reality. The next task is to select, based on the analysis of the observed data, the models that can best explain different parts of the scene. For the derivation of goodness measure depending on the information we have the data and the parameters that describe it. Through this goodness measure (evidence) appropriate models can be selected. The diagram in Figure 7.3 illustrates the model selection process.

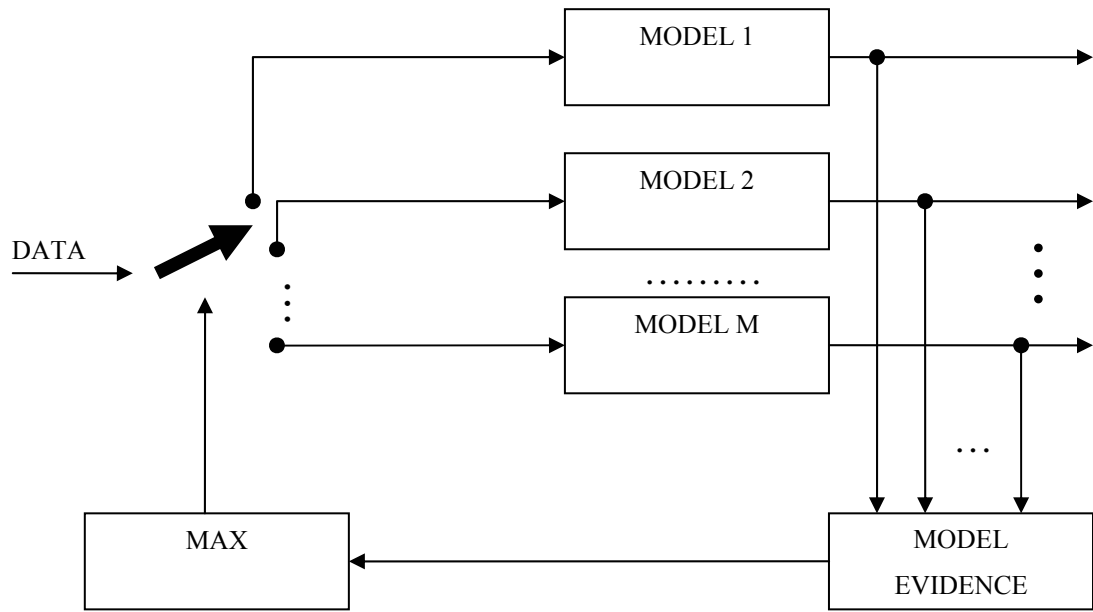


Figure 7.3: Model selection paradigm. The diagram refers to the shaded area in Figure 7.2. Structures in RS images are explained based on *a priori* knowledge stored in a library of models. We extract information about the identity of the scene by selecting the model that best explains it.

Once a model has been selected as being evident for explaining certain aspects of the scene, parameter estimation follows as the next step. It is necessary to search for the values of the parameters in such a way that the error between the observed and the simulated image is minimized. These values represent a guess of the scene parameters. The Bayesian inference method consists of interpreting probabilities as a measure of incomplete information rather than randomness and gives an alternative understanding of the information hidden in the measured data. The solution of ill-posed inverse problems is generally very sensitive to the assumed model.

This is the critically of inverse problem methods: they can introduce unrealistic behaviour in the results. The importance of the Bayesian inference consists in the possibility to select the best result that corresponds to the most plausible model for the available data.

8 METHODS FOR IMAGE CLASSIFICATION

8.1 INTRODUCTION TO CLASSIFICATION PROBLEMS

In general, the classification is a processing technique which applies quantitative methods to the values in a digital yield or remotely sensed scene to group pixels with similar digital number values into feature classes or categories [8.1]. Usually, this is accomplished using general mathematical pattern recognition and pattern classification techniques, although more specific methods have been developed especially suited for various sensors (as in the case of the H/α parameters for polarimetric SAR [8.2]). Then, the patterns are the pixels themselves and it is customary to represent them as column vectors like:

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad (8.1)$$

where $x_1, x_2, \dots, x_n$ are the spectral values of the pixel in bands $1, 2, \dots, n$, respectively. Therefore, classification involves labelling the pixels as belonging to particular classes using the characteristics of the available data. During classification, each pixel of an image is no longer identified by its position in the chosen spatial reference system but by means of the vector belonging to the feature space relative to the spectral values. One, then, establishes a

set of rules to label the pixels according to their particular properties in this space. To derive these rules, it is also necessary to define boundaries among the different classes in the n -dimensional feature space.

Two main types of classification procedures may be identified: supervised and unsupervised classification methods. Each type finds applications in the analysis of remote sensing image data. These may be used as alternative approaches but are also often combined into hybrid methodologies.

In the supervised case, a specialist identifies terrain classes in a scene, and the class boundaries are identified in the parameter space that serve to separate the classes. This is called "training", and the training data can be chosen from the scene itself, or from previously acquired scenes that possess similar characteristics, or can be produced by specific "in-situ" campaigns. After the training, the algorithm automatically assigns classes to each pixel based on the predetermined class means or boundaries.

In a basic unsupervised classifier, the algorithm has no prior information about the scene content or about the terrain classes. The algorithm examines the parameter space for each scene, and assigns classes and boundaries based on the clustering of pixels. Sometimes, the classes and boundaries can be based upon physical models. In either case, the operator must identify each class manually after the class assignment.

Supervised classifiers have the drawback of requiring operator input, and the obtained classes tend to be scene specific. The unsupervised classifiers often yield classes whose physical meaning is uncertain.

8.1.1 PIXEL LABELLING

Recognition that image data exists in sets of spectral classes, and identification of those classes as corresponding to specific ground cover types, is carried out using the techniques of pattern recognition or pattern classification and their neural network counterparts. The patterns are the pixels themselves, or strictly the mathematical pixel vectors that collect the sets of brightness values.

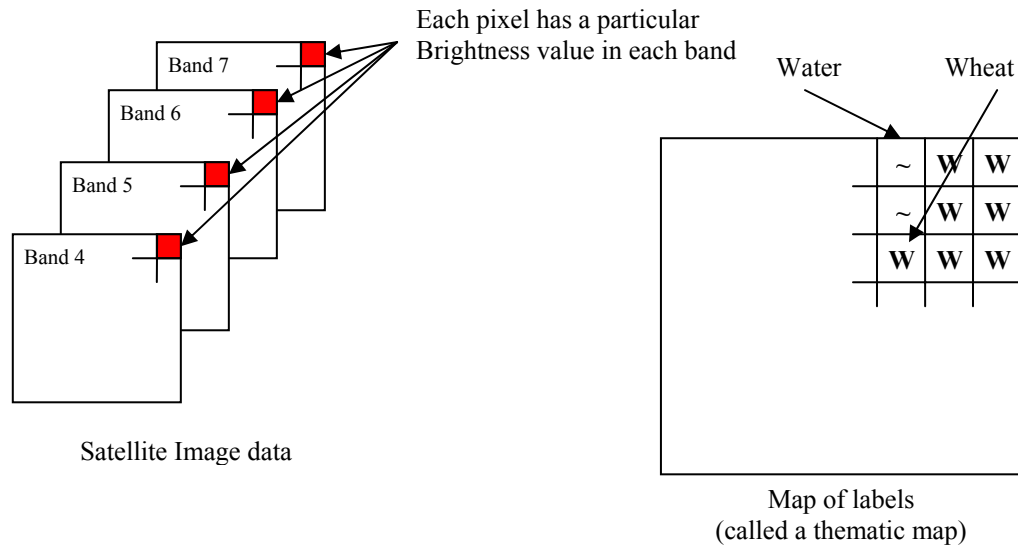


Figure 8.1: The role of classification in labelling pixels in remote sensing image data. Here an illustration from Landsat multispectral scanner data is depicted.

This is depicted as a mapping in Figure 8.1. In the terminology of statistics this is more properly referred to as allocation rather than classification. However throughout this thesis, classification, categorization, allocation and labelling are generally used synonymously.

8.2 SUPERVISED CLASSIFICATION

Supervised classification is the procedure most often used for automated quantitative analysis of remote sensing image data. It is usually based on an important assumption, namely that each spectral class can be described by a probability distribution in the multispectral space; this will be a multivariate distribution with as many variables as the dimensions of the space. Such a distribution describes the chance of finding a pixel belonging to that class at any given location in the multispectral space. Almost all the algorithms used for this purpose consist of the same sequences of basic steps:

1. Decide the set of ground coverages into which the image is to be segmented, for example “water”, “grass”, “field”...
2. A set of prototype pixels, the *training data*, is then selected for each class of the desired set. This choice may be done using information from ground surveys, aerial photograph, topographic maps or any other sources of reference data.
3. The following step consists of describing each class by means of some particular parameters, mainly statistical ones such as the first and second order moments, depending on the chosen classifier. These parameters characterize the adopted probability model or directly define the partition of the multispectral space. The set of parameters for a given class is sometimes called the signature of the class.
4. Using the trained classifier labels, every pixel of the image is assigned to one of the desired ground cover types (information classes). Operatively, the complete image is semi-automatically classified after the user has directly identified just a portion of it in the second step when defining the training data.
5. Produce tabular summaries or thematic maps summarizing the classification results.

There are a lot of algorithms that perform supervised classification: *maximum likelihood* [8.6], *minimum distance* [8.6], *parallelepiped* [8.6], etc.

8.2.1 MAXIMUM LIKELIHOOD CLASSIFICATION

This kind of classification is based on the estimation of the statistical distributions of the spectral components [8.1]. This allows evaluating the chance of finding a pixel belonging to a certain class at any given location in the multispectral space. A working hypothesis often adopted is that the probability distribution for each class is a multivariate normal, or Gaussian, distribution. Hence, the probability density function of finding a pixel belonging to a given class ω_i in a position x of the multispectral space is described by [8.6]:

$$p(\underline{x}|\omega_i) = \frac{1}{(2\pi)^{\frac{n}{2}}(|\Sigma_i|)^{\frac{1}{2}}} \exp\left\{-\frac{1}{2}(\underline{x}-\underline{m}_i)^t |\Sigma_i|^{-1} (\underline{x}-\underline{m}_i)\right\} \quad (8.2)$$

where $i = 1, 2, \dots, l$ is the index identifying the l ground cover classes, $\underline{m}_i$ the mean vector of the data in class ω_i and Σ_i the *covariance matrix* for that class. For each i , the Σ_i matrix is defined and estimated (in the second relation of (8.3)) as [8.6]:

$$\Sigma_i = E[(\underline{x} - \underline{m}_i)(\underline{x} - \underline{m}_i)^T] \quad (8.3)$$

The class parameters m_i and Σ_i ($i = 1, 2, \dots, l$) can be estimated by using the ML approach described in Section 7.2, as follows [8.6]:

$$\begin{aligned} \hat{\underline{m}}_i &= \frac{1}{r_i} \sum_{j=1}^{r_i} \underline{x}_{ji} \\ \hat{\Sigma}_i &= \frac{1}{r_i} \sum_{j=1}^{r_i} (\underline{x}_{ji} - \hat{\underline{m}}_i)(\underline{x}_{ji} - \hat{\underline{m}}_i)^T \end{aligned} \quad (8.4)$$

where r_i is the number of training samples for ω_i and $\underline{x}_{ji}$ is the j -th training sample for ω_i ($j = 1, 2, \dots, r_i$). In particular, it is possible to prove that $\hat{\underline{m}}_i$ is an unbiased estimate of $\underline{m}_i$, while $\hat{\Sigma}_i$ is slightly biased; this bias can be removed by redefining $\hat{\Sigma}_i$ as:

$$\hat{\Sigma}_i = \frac{1}{r_i - 1} \sum_{j=1}^{r_i} (\underline{x}_{ji} - \hat{\underline{m}}_i)(\underline{x}_{ji} - \hat{\underline{m}}_i)^T \quad (8.5)$$

The multidimensional normal distribution is completely specified by its mean vector and its covariance matrix. The covariance matrix introduced here is different from the one associated to polarimetric data. It allows, for instance, to see if a correlation exists between the responses of a pair of spectral bands: if they are correlated, the corresponding off-diagonal elements of the covariance matrix are large as compared to the diagonal terms (whereas little correlation yields off-diagonal terms close to zero). In order to determine the class or category to which a pixel characterized by a vector $\underline{x}$ belongs, the conditional probabilities:

$$p(\omega_i | \underline{x}) \quad \text{for } i = 1, 2, \dots, l \quad (8.6)$$

play a relevant role. In particular, the vector $\underline{x}$ describes the pixel as a point in the multispectral space. The probability $p(\omega_i | \underline{x})$ gives the likelihood that the correct class is ω_i for the pixel at the position $\underline{x}$. Hence, the classification is performed assigning a pixel vector $\underline{x}$ to class ω_i when the condition:

$$p(\omega_i|\underline{x}) > p(\omega_j|\underline{x}) \quad \forall j \neq i \quad (8.7)$$

is satisfied. The problem is then to estimate the probabilities $p(\omega_i|\underline{x})$ that are unknown. If sufficient training data is available for each class (each ground cover type defined by the user), the probability distribution, which describes the chance of finding a pixel from ω_i at the position $\underline{x}$, can be calculated. This new probability distribution is indicated with $p(\underline{x}|\omega_i)$. The desired $p(\omega_i|\underline{x})$ and the estimated $p(\underline{x}|\omega_i)$ are related by the Bayes theorem [8.3]:

$$p(\omega_i|\underline{x}) = \frac{p(\underline{x}|\omega_i)p(\omega_i)}{p(\underline{x})} \quad (8.8)$$

Where $p(\omega_i)$ is the probability that class ω_i occurs in the image and $p(\underline{x})$ is the probability of finding a pixel from any class at the location $\underline{x}$. The $p(\omega_i)$ are called *a priori* or *prior* probabilities, since they express the probabilities with which the class membership of principle, the $p(\omega_i|\underline{x})$ are *posterior* probabilities. Using (8.8), the classification rule of (8.7) may be written as:

$$\underline{x} \in \omega_i \quad \text{if} \quad p(\underline{x}|\omega_i)p(\omega_i) > p(\underline{x}|\omega_j)p(\omega_j) \quad \forall j \neq i \quad (8.9)$$

Mathematical convenience results if discriminant functions like:

$$g_i(\underline{x}) = \ln\{p(\underline{x}|\omega_i)p(\omega_i)\} = \ln\{p(\underline{x}|\omega_i)\} + \ln\{p(\omega_i)\} \quad (8.10)$$

are used, so that (8.9) is expressed as:

$$\underline{x} \in \omega_i \quad \text{if} \quad g_i(\underline{x}) > g_j(\underline{x}) \quad \forall j \neq i \quad (8.11)$$

At this stage it is assumed that the probability distribution for each class is of the form of the multivariate normal model (see (8.2)). This is an assumption, rather than a demonstrable property for natural spectral or information classes; however it is chosen according to the well-known properties of the multivariate normal density. The final form for the discriminant function, based upon the assumption of normal statistics, is then:

$$g_i(\underline{x}) = \ln p(\omega_i) - \frac{1}{2} \ln(\det|\Sigma_i|) - \frac{1}{2} (\underline{x} - \underline{m}_i)^t |\Sigma_i|^{-1} (\underline{x} - \underline{m}_i) \quad (8.12)$$

The boundaries among different classes, thanks to the second order equation that characterizes the discriminant function, are quadratic curves such as parabolas, circles and ellipses. As it will be shown later, the boundaries in the minimum distance approach are expressed by linear equations; hence, the higher order decision rules make the maximum likelihood classification more efficient for partitioning multispectral space than the minimum distance one. It is also possible to introduce a threshold in the case that insufficient training data are available to estimate the parameters of the class distributions; the threshold of each class is calculated from the knowledge of its “*a priori*” probability and its covariance matrix [8.6].

8.2.2 MINIMUM DISTANCE CLASSIFICATION

The effectiveness of maximum likelihood classification is related to the accuracy in the estimation of the mean vector $\underline{m}_i$ and the covariance matrix Σ_i for each spectral class ω_i ($i=1, 2, \dots, l$) [8.1]. This, in turn, depends upon having a sufficient number of training pixels for each of those classes. If this condition is not satisfied, an inaccurate estimation of Σ_i results, leading to poor classification results. When the number of training samples per class is limited, it is worth trying to use a classification approach based only on the use of $\underline{m}$, whose evaluation, even for small quantities of samples, can be more accurate than that of the covariance matrix elements. A classifier which only needs the mean vector of the various classes to be estimated is the minimum distance, which should be more precisely indicated as “minimum distance to class means” classifier. Indeed, this algorithm operates by placing a pixel in the class of the nearest mean in the multispectral measurement space. The discriminant function for the minimum distance classifier is derived from the expression of the squared Euclidean distance between the position of the generic sample $\underline{x}$ to be classified and the mean value $\underline{m}_i$ of i -th class:

$$d^2(\underline{x}, \underline{m}_i) = (\underline{x} - \underline{m}_i)^t (\underline{x} - \underline{m}_i) \quad (8.13)$$

Expanding the product gives:

$$d^2(\underline{x}, \underline{m}_i) = \underline{x}^t \underline{x} - 2\underline{m}_i^t \underline{x} + \underline{m}_i^t \underline{m}_i \quad (8.14)$$

Classification is performed on the basis of:

$$\underline{x} \in \omega_i \quad \text{if} \quad d(\underline{x}, \underline{m}_i) < d(\underline{x}, \underline{m}_j) \quad \forall j \neq i \quad (8.15)$$

As $\underline{x}^t \underline{x}$ is common to all $d^2(\underline{x}, \underline{m}_i)$, it can be neglected. By reversing the signs of the expressions above, the assignment to a class is again expressed by the condition (8.11) with:

$$g_i(\underline{x}) = 2\underline{m}_i^t \underline{x} - \underline{m}_i^t \underline{m}_i \quad (8.16)$$

As a consequence of the linearity of the discriminant functions, the decision surfaces for this classifier are planes in the n -dimensional spectral space. The minimum distance classification is proven to be of value when the number of training samples is limited .

8.2.3 PARALLELEPIPED CLASSIFICATION

This kind of supervised classification is very simple; in principle, the classifier is trained by inspecting histograms of the individual spectral components of the selected training data. The upper and lower significant bounds of the histograms are identified as the edges of a multidimensional parallelepiped for each class. If, on classification, pixels are found to lie in such a parallelepiped, they are labelled as belonging to that class. Relevant drawbacks of these criteria are that pixels in regions between parallelepipeds can not be assigned to any class and pixels in overlapping areas can not be separated: in both cases, these pixels are not classified. No statistical constraint in the definition of the class membership are taken into account here.

8.2.4 NUMBER OF TRAINING PIXELS REQUIRED FOR EACH CLASS

As mentioned above, supervised classification is based on training set. Indeed, it is very important to have sufficient training samples for each spectral class in order to allow reasonable estimates of the elements of the mean vector and covariance matrix to be computed. For an n dimensional multispectral space at least $n+1$ samples are required to avoid the covariance matrix being singular. For this reason it is necessary to have as many training pixels as possible, particularly, as the dimensionality of the feature space increases, since in higher dimensional spaces there is an increased chance of having some individual dimensions poorly represented.

8.3 UNSUPERVISED CLASSIFICATION

In this kind of classification pixels are assigned to spectral classes without prior knowledge of the existence and property of classes. Most often it is performed using clustering methods. These procedures can be used to determine at first the number and location of the spectral classes from which the data are drawn and then the spectral class of each pixel. The user finally identifies those classes “*a posteriori*”, by associating a sample in each class with available data, for instance maps and information from ground truth.

8.3.1 K-MEANS CLASSIFICATION

This first method of classification is comprised into generic classification called “*Clustering classification*”. This part of classification is obviously unsupervised and, generally, the clustering is the formal study of algorithms and methods to group objects with similar characteristics.

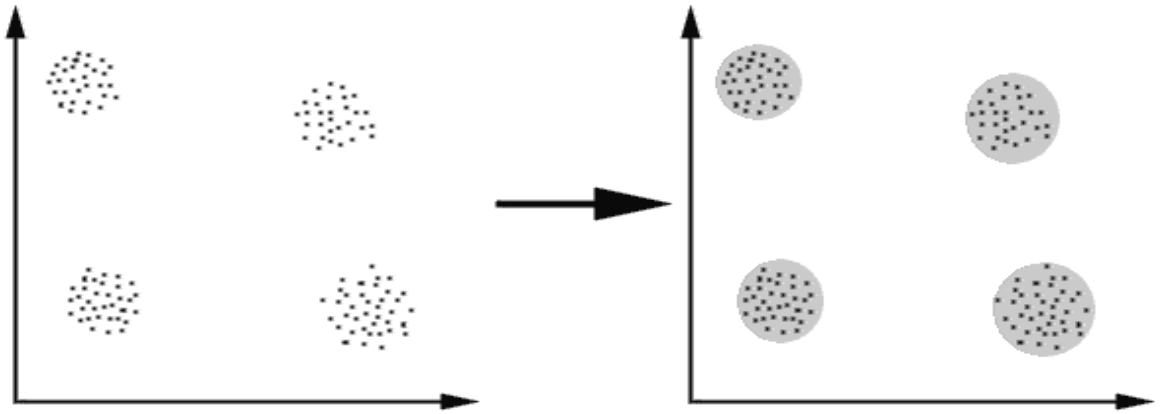


Figure 8.2: Clustering of image.

The clustering can be considered the most important unsupervised deepening problem. Among all the algorithms K-Means clustering represents one of the best common approaches and, for this reason, it is wide used. The purpose is to divide a set of objects in K clusters in order to minimize a known objective function. This function is defined preliminarily (it is possible to choose several functions which are different with respect to the ways in which they represent objects ...to distinct natural classes). The most commonly used function is the mean square error:

$$(\underline{x} - \underline{\mu}_i)(\underline{x} - \underline{\mu}_i)^T \quad (8.17)$$

where i represent the cluster index, $\underline{x}$ the pixels (it is a vector because it is normal to work with images with several bands) and $\underline{\mu}_i$ represents the mean value of the i -th cluster ($i=1, 2,$

..., k). *K-Means algorithm* is an iterative method which requires several steps in order to obtain the classification. These steps are summerized in the following table:

STEP	OPERATION
1	Choose the first K mean values
2	Assign every pixel to the nearest mean values
3	Compute again the K mean value in order to minimize the objective function
4	Repeat steps 2 and 3 until the changes are below a prefixed tolerance

Table 8.1: Steps of the *K-MEANS algorithm*.

It is possible to prove that the procedure always terminates in a finite number of iterations, However the procedure could stop itself when it finds the local minima. The result is, therefore, affected by the choice of the first mean values. In addition, it does not guarantee a relation between the cluster labels of neighbouring pixels .

8.3.2 UNSUPERVISED CLASSIFICATION BASED ON $H/A/\alpha$ PARAMETERS

In any classifier, the choice of parameters is important, and in the case of polarimetric radar data, content-independent scattering models can be used to get parameters that provide a reasonable class separation. An example is the $H/A/\alpha$ set of parameters derived from an eigenvalue decomposition of the coherency matrix. The $H/A/\alpha$ algorithm was developed by Cloude and Pottier, who showed that terrain classes sometimes produced distinct clustering in the H/α plane.

The H/α plane is drawn in Figure 8.3. The observable alpha values for a given entropy are bounded between curves *I* and *II* (i.e., the shaded areas are not valid) because the averaging of

the different scattering mechanisms (i.e., averaging of the different eigenvectors) restricts the range of the possible values as the entropy increases. As H and α are both invariant to the type of polarization basis used, the H/α plane provides a useful representation of the information in the coherency matrix.

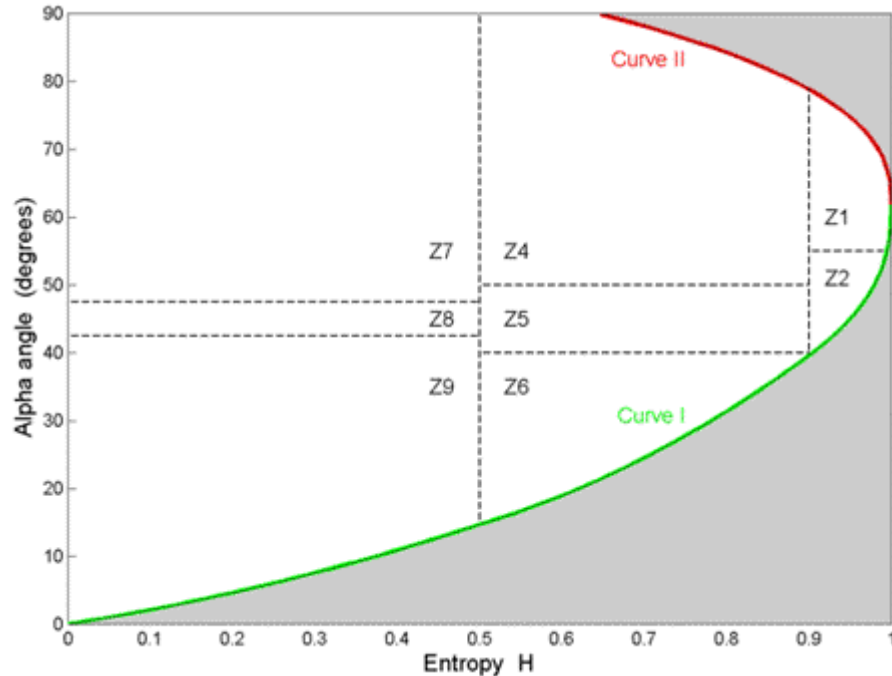


Figure 8.3: The H/α plane showing the model-based classes and their partitioning. A description of the classes (Z1 - Z9) is given in Section 8.3.2.

The bounds in Figure 8.3 (Curve I and Curve II) show that when the entropy is high, the ability to classify different scattering mechanisms is very limited. An initial partition into nine classes (eight usable) has been suggested by Cloude and Pottier, and is shown in Figure 8.3. Classes are chosen based on general properties of the scattering mechanism and do not depend on a particular data set. This allows an unsupervised classification based on the physical properties of the signal. The class interpretations suggested by Cloude and Pottier are as follows :

- Class Z1:** Double bounce scattering in a high entropy environment;
- Class Z2:** Multiple scattering in a high entropy environment (e.g., forest canopy);
- Class Z3:** Surface scattering in a high entropy environment (not a feasible region in the H / α space);
- Class Z4:** Medium entropy multiple scattering;

- Class Z5:** Medium entropy vegetation (dipole) scattering;
- Class Z6:** Medium entropy surface scattering;
- Class Z7:** Low entropy multiple scattering (double or even bounce scattering);
- Class Z8:** Low entropy dipole scattering (strongly correlated mechanisms with a large imbalance in amplitude between HH and VV);
- Class Z9:** Low entropy surface scattering (e.g., Bragg scatter [8.2] and rough surfaces).

It is important to note, however, that the boundaries are somewhat arbitrary and depend upon the radar calibration, the measurement noise floor and the variance of the parameter estimates. Nevertheless, this classification method is linked to physical scattering properties, making it independent of training data sets. The number of classes needed as well as the usability of the method depends upon the application.

The third variable of polarimetric anisotropy¹ has been used to distinguish different types of surface scattering. The H/A plane representation for surface scattering is given in Figure 8.4, where the shaded region is not feasible. The line delineating the feasible region can be calculated using a diagonal coherency matrix with small lower eigenvalues λ_2 and λ_3 , with λ_3 varying from 0 to λ_2 [8.2].

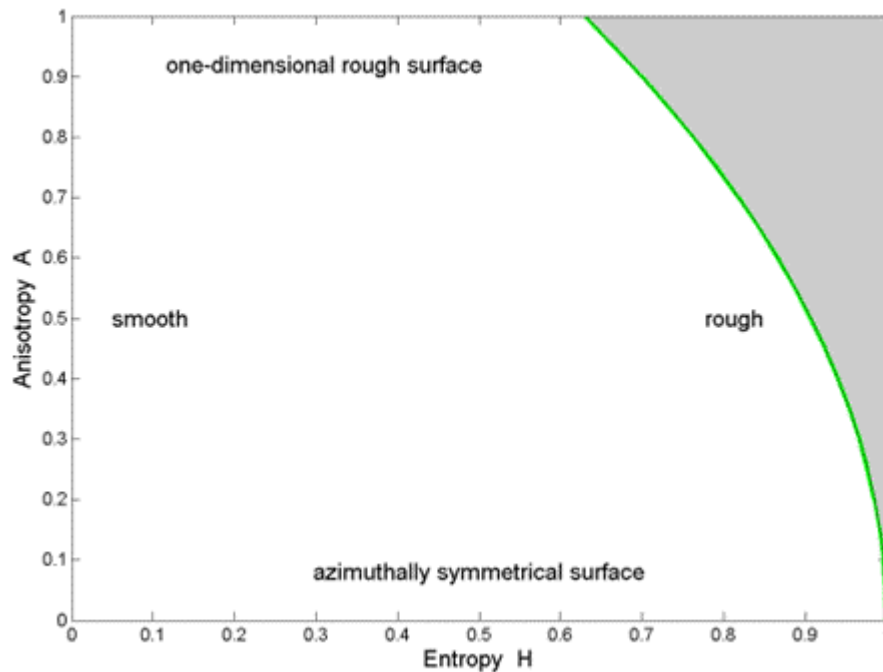


Figure 8.4: Types of surface scattering in the Entropy/Anisotropy plane.

¹ A parameter derived in the Cloude eigenvector decomposition of the coherency matrix to describe the importance of the secondary scattering mechanisms [8.2].

The introduction of the anisotropy to the feature set yields a third parameter that can be used in the classification. One approach is simply to divide the space into two H/α planes using the green plane shown in the 3-D space of Figure 8.4, one side for $A \leq 0.5$ the other side for $A > 0.5$. This introduces 16 classes if the H/α planes are divided according to Figure 8.3. Note that the upper limit of H is restricted when $A > 0$, as shown in Figure 8.4.

The $H/A/\alpha$ classification space, given in Figure 8.4, now provides additional ability to distinguish between different scattering mechanisms. For example, high entropy and low anisotropy ($\lambda_2 \approx \lambda_3$) correspond to random scattering whereas high entropy and high anisotropy ($\lambda_2 \gg \lambda_3$) indicate the existence of two scattering mechanisms with equal probability.

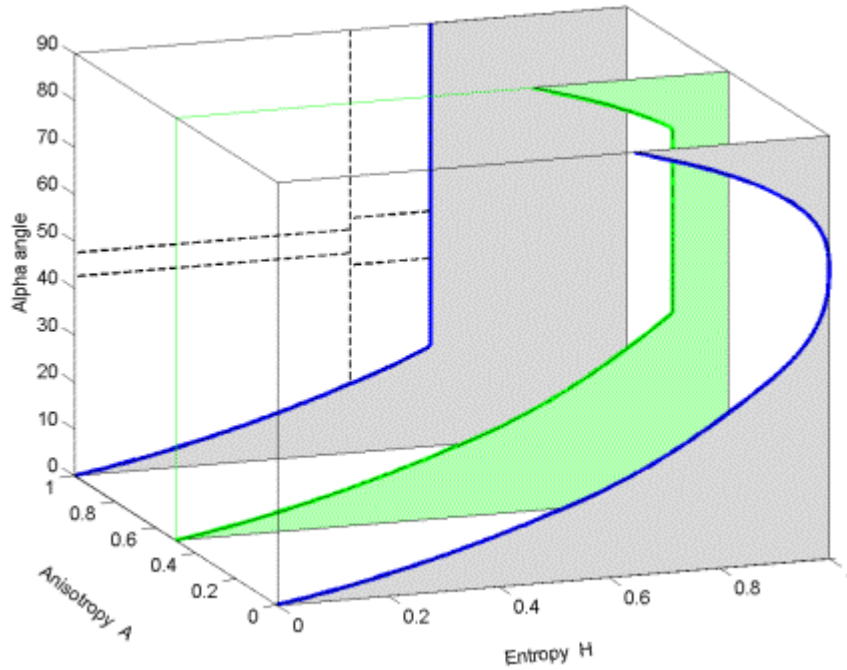


Figure 8.5: Illustration of how an $A = 0.5$ plane (green) creates 16 classes from the original 8 H/α classes shown in Figure 8.3. This gives 16 regions in the Entropy/Anisotropy/ α space for use in an unsupervised classifier.

The three parameters H , A and α are based on eigenvectors and eigenvalues of a local estimate of the 3 x 3 Hermitian coherency matrix ("Hermitian" means a square matrix A such that $A=A^{*T}$, i.e., it has real eigenvalues). The basis invariance of the target decomposition makes these three parameters roll invariant, i.e. the parameters are independent of a rotation of the target about the radar line of sight. It also means that the parameters can be computed independently of the polarization basis. Estimation of the three parameters H , A and α allows a classification of the scene according to the type of scattering process within the sample (H , A) and the corresponding physical scattering mechanism (α). The data needs to be averaged in

order to allow an estimation of H , A and α (without averaging, the coherency matrix has rank 1), which has the benefit of reducing speckle noise.

8.4 CLASSIFICATION ACCURACY ASSESSMENT

The final step in any classification procedure is an accuracy assessment. It is an integral part of any classification project, which explains and completes the results of the employed classifier. The most common way to represent the classification accuracy of remote sensed data is in the form of an error matrix (also called confusion matrix). This is a square array of numbers set out in rows and columns which expresses the number of sample units (pixels) assigned to a particular category. Usually, the columns represent the reference data while the rows indicate the classification generated from the acquired data. The confusion matrix is a very effective way to represent accuracy as much as the accuracies of each category are plainly described along with both the error of inclusion (commission errors) and error of exclusion (omission errors) present in the classification [8.4], [8.5].

The error matrix can then be used as a starting point for deriving a series of descriptive and analytical statistical parameters. Perhaps the simplest descriptive statistic is the overall accuracy (OA), which is computed by dividing the total number of correctly assigned pixels (i.e., the sum of the diagonal of the confusion matrix), by the total number of pixels in the error matrix. In addition, accuracies of individual categories can be computed in a similar manner. Traditionally, the total number of "correct pixels" in a category is divided by the total number of pixels of that category as derived from the reference data (i.e. the column total). This accuracy measurement indicates the probabilities of a reference pixel being correctly classified and is really a measure of the omission error: it is often called producer's accuracy (Pr. Acc.) because the producer of the classification is interested in how well a certain area is classified. On the other hand, if the total number of "correct pixels" in a category is divided by the total number of pixels classified in that category, then this result gives a measure of the commissions. This measure, called user's accuracy (Us.Acc.), is indicative of the probability that a pixel in the image actually represents that category on the ground.

8.5 CONTEXTUAL CLASSIFICATION

8.5.1 THE CONCEPT OF SPATIAL CONTEXT

So far we treated classifiers which label a pixel just according to its spectral properties, without taking into consideration any neighbouring pixels. In any real image, adjacent pixels are strongly correlated, both because imaging sensors acquire significant portions of energy from adjacent pixels and because ground cover types generally occur over a region that is large as compared with the size of a pixel. In an agricultural area, for example, if a particular image pixel represents wheat it is highly likely that its neighbouring pixels will also be wheat. This knowledge of neighbourhood relationships is a rich source of information. In addition, the inclusion of context is important not only because it exploits spatial information, but also because sensitivity to the correct context of a pixel can improve a thematic map by helping to remote individual pixel labelling errors that might result from noisy data, or unusual classifier performance. Classification methods that take into account the labelling of neighbours in order to determine the most appropriate class for a pixel are said to be context sensitive, or simply contextual classifiers. They attempt to develop a thematic map that is consistent both spectrally and spatially.

The degree to which adjacent pixels are strongly correlated depends on the spatial resolution of the sensor and on the scale of natural regions on the Earth's surface. For the same sensor, therefore, adjacent pixels over a urban region may not show a strong correlation, while adjacent pixels over an agricultural region are strongly correlated.

8.5.2 CONTEXT CLASSIFICATION BY IMAGE PRE-PROCESSING

It is often useful before classification to process the image data in order to modify or enhance its spatial properties and so it is possible to exploit the spatial context. Specifically, the application of simple averaging filters (possibly with edge preserving thresholds) can be used to impose a degree of homogeneity among the brightness values of adjacent pixel thereby increasing the chance that neighbouring pixels may be given the same label. An alternative is to generate a separate channel of data that associates spatial properties with pixels. For example, a texture channel could be added and classification can be carried out (using a suitable algorithm) on the combined multispectral and texture channels.

One of the most useful spatial pre-processing techniques is that used in the ECHO classification methodology [8.6]. In ECHO (Extraction and Classification of Homogeneous Objects) regions of similar spectral properties are grown before classification is performed. Several region growing techniques are available. The simplest of these ones consists of aggregating pixels into small regions by comparing their brightnesses in each channel and then aggregating the small regions into bigger regions in a similar manner. When this is done, ECHO classifies the regions as single objects and only resorts to standard maximum likelihood classification when it has to treat individual pixels that could not be put into regions.

8.5.3 POST CLASSIFICATION FILTERING

After having generated a thematic map, using a simple point classifier, some degree of spatial context can be developed by logically filtering the map. For example, if the map is examined

in 3×3 windows, a label at the centre of the window might be changed to the label most represented in the window.

8.5.4 PROBABILISTIC LABEL RELAXATION

Using the process of label relaxation it is possible to develop spatial consistency in a classified image product. It allows the spatial properties of a region to be carried into the classification process in a logically consistent way. The process starts by assuming that a classification, based on spectral data alone, has already been carried out. There is available therefore, for each pixel, a set of probabilities that describe the chance that the pixel belongs to each of the possible ground cover classes under consideration. This set of probabilities could be computed from (8.2) and (8.3) if maximum likelihood classification has been used first. If another classification method had been employed, then some other assignment processes will be required. Let the set of probabilities for a pixel m be represented by

$$p_m(\omega_i) \quad \text{for } i = 1, \dots, K \quad (8.18)$$

where K is the total number of classes: $p_m(\omega_i)$ is the probability that ω_i is the correct class for the pixel m . Obviously

$$\sum_i p_m(\omega_i) = 1 \quad (8.19)$$

We suppose now that a neighbourhood is defined surrounding pixel m . This can be of any size and, in principle, should be large enough to ensure that all the pixels considered to have any spatial correlation with m are included. For high resolution imagery this is not practical and simple neighbourhoods are often adopted such as in Figure 8.6.

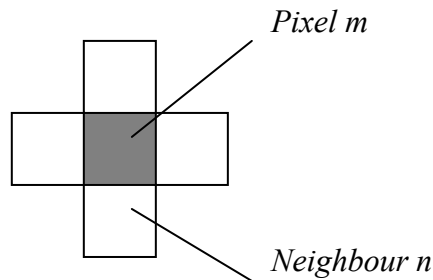


Figure 8.6: Definition of a simple neighbourhood around pixel m , pixel n is one of the neighbours of pixel m .

Supposing to find a neighbourhood function $Q_m(\omega_i)$ which allows the pixels in the prescribe neighbourhood to influence the possible classification of pixel m the (8.18) expresses this influence:

$$p'_m(\omega_i) = \frac{p_m(\omega_i)Q_m(\omega_i)}{\sum_i p_m(\omega_i)Q_m(\omega_i)} \quad (8.20)$$

The (8.20) has been obtained by multiplying the label probabilities in (8.18) by the $Q_m(\omega_i)$ and by normalizing new values. This modification is made to the set of label probabilities for all pixels by moving over the image from its top left hand to bottom right hand corners. The neighbourhood function $Q_m(\omega_i)$ depends on the label probabilities of the neighbouring pixels, so that if all the pixel probabilities are modified in the manner just described then the neighbours for any given pixel have also been altered. Consequently (8.20) should be applied again to give newer estimates of the label probabilities. Indeed, it is necessary to apply (8.20) until the $p'_m(\omega_i)$ have stabilised, i.e. until they do not change significantly in successive iteration. In addition, $p'_m(\omega_i)$ represents the correct set of label probabilities for the pixel, having taken into account both the spectral data (in the initial determination of label probabilities) and the spatial context (through the neighbourhood functions). Since the process is iterative, (8.20) is usually written as an explicit formula:

$$p^{k+1}_m(\omega_i) = \frac{p^k_m(\omega_i)Q^k_m(\omega_i)}{\sum_i p^k_m(\omega_i)Q^k_m(\omega_i)} \quad (8.21)$$

where k is the iteration counter. Depending on the size of the image and on its spatial complexity, the number of iterations required to the label probabilities to powerful may be quite large. However, most changes in the label probabilities occur in the first few iterations. Consider now just one of the neighbours of pixel m in Figure 8.6, called pixel n . We can suppose there is a measure of compatibility of the current labelling of pixel m and its neighbouring pixel n . Particularly, let $r_{mn}(\omega_i, \omega_j)$ describe numerically how compatible it is to have pixel m classified as ω_i and a neighbouring pixel n classified as ω_j . This measure will be high if the adjoining pixels are both labelled in the same class otherwise it is low. These “compatibility coefficients” can be defined in several ways. An intuitively definition is based on conditional probabilities. Thus, the compatibility measure $p_{mn}(\omega_i|\omega_j)$ is the probability

that ω_i is the correct label for pixel m if ω_j is the correct label on pixel n . Specifically, the probability of the joint event that pixel m is labelled as ω_i and pixel n is labelled as ω_j is $p_{mn}(\omega_i|\omega_j) p_n(\omega_j)$. As probabilities for all possible labels on pixel n are available, the total evidence from pixel n in favour of ω_i being the correct class for pixel m will be the sum from all pixel n 's labelling possibilities

$$\sum_j p_{mn}(\omega_i|\omega_j) p_n(\omega_j) \quad (8.22)$$

Considering the full neighbourhood of the pixel m all the neighbours contributions are simply added, via the use of *neighbour weights* d_n that recognise that some neighbours may be more influential than other ones. Thus, at the k^{th} iteration, the local neighbourhood support for pixel m being classified as ω_i is:

$$Q_m^k(\omega_i) = \sum_n d_n \sum_j p_{mn}(\omega_i|\omega_j) p_n^k(\omega_j) \quad (8.23)$$

This is the definition of the neighbourhood function. There are several methods to determine values for compatibility coefficients $p_{mn}(\omega_i|\omega_j)$. For example, a spatial model for the region can be available or another approach is to compute values for the compatibility coefficients from ground truth pixels, although the ground truth needs to be in the form of training regions that contain heterogeneous and spatially representative cover types. At the completion of the relaxation process, or at any intervening stage, each of the pixels can be classified according to the highest label probability. Particularly, the process can be allowed to converge to a natural completion at which further iteration leads no changes in the label probabilities for all pixels. In this process there are, however, two difficulties:

- Up to several hundred iteration may be involved leading to a computationally very intensive post classification step;
- The relaxation process improves the classification results in the first few iterations. If the process is not terminated, the thematic map, after a lot of iterations, can be worse than before the application of the techniques.

So in order to avoid these problems a control mechanism is needed. For example, it can be useful to stop after just a few iterations or to check the labelling errors remaining at each

iteration, so that the process can be terminated when the labelling error is minimum. In addition, the relaxation labelling approach is essentially heuristic and does not involve any statistical model for the joint distribution of the context in the image. Therefore, convergence although usually achieved on an experimental basis, cannot be proved, which further confirms the possibility irregular asymptotic behaviour of the method.

9 MRF-ICM ALGORITHM

9.1 GIBBS RANDOM FIELDS

Gibbs Random Fields (GRF) represent a global models for the labels in a classification map Ω , specifying the probability function in terms of energy functions through the following formula:

$$P(\Omega) = \frac{1}{Z} \exp\left[\frac{-U_T(\Omega)}{T}\right] \quad (9.1)$$

where:

- Ω is the configuration of all the labels in the class map;
- T is a parameters named “temperature”;
- $U_T(\Omega)$ is a function of the label configuration named “energy”;
- Z is a normalizing constant named “partition function”.

Such notations and terminology date back to statistical mechanics where a formally identical probabilistic model raises from the study of a canonical thermodynamical systems [9.3].

A GRF can be used to define a global description of image trough its properties. However, note that in a such model the description involved the joint probability distribution of all image labels, whose estimation is a computationally intractable task. Hence, analytical simplifications of such a model are required to make it useful to image-processing purposes.

9.2 MARKOV RANDOM FIELDS

A random field is a Markov Random Fields (MRF) with respect to a given neighbourhood system (see Section 8.5.4) if its probability distribution exhibits the following properties:

- **Positivity**

For each possible classification map Ω of the image, $P(\Omega) > 0$

- **Markov Property**

$P(\omega | \Omega') = P(\omega | \Omega_N)$, where ω is the label of a given image pixel Q , Ω' is the configuration of all the other labels of the image (i.e., $\Omega' = \Omega - \{\omega\}$) and Ω_N is the configuration of the labels of the neighbours of the pixel Q .

Differently from GRFS, MRFS are explicitly defined on a local basis, formalized through the use of a neighbourhood system. The specific properties of such system are explicitly investigated in the next section.

9.2.1 NOTION OF NEIGHBOURHOOD

The notion of neighbourhood is a key-point for MRF models. With this definition it is possible to state a relation between GRFs (global models) and MRFs (local models). For this purpose, it is necessary to introduce the concept of “*clique*”. A “*clique*” is a set of pixels in which every couple of pixels mutually neighbours. For a second order neighbourhood, the possible types of “*clique*” are as in Figure 9.1.

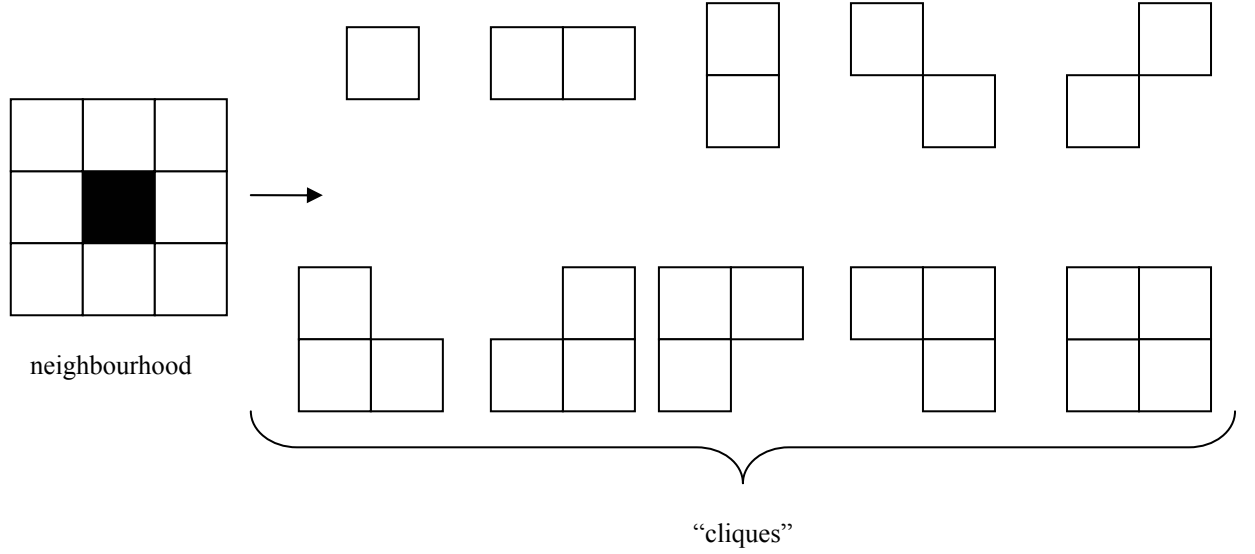


Figure 9.1: Types of “clique” for a neighbourhood of second order of the pixel highlighted in black.

A potential function $V_Q(\Omega)$ on a “clique” q define the energy contribution associated to the “clique” q , for all the each possible “clique” in the neighbourhood system. The total energy function is simply the sum of potential functions:

$$U_T(\Omega) = \sum_{q \in Q} V_q(\Omega) \quad (9.2)$$

where Q is a collection of all cliques in the neighbourhood system.

9.3 EQUIVALENCE MRF-GRF

If the Markov property is satisfied for a random field in an image, it is possible to pass from a global model (Gibbs Random Field) to a local model (Markov Random Field) which is reduced to the neighbourhood (context) of the pixel where the great part of correlation is concentrate. The MRF model leads to a simpler and practise formulation of the problems. These properties are summarized by the Hammersley-Clifford theorem [9.4]:

“Considering a neighbourhood system S , Ω is on MRF with respect to S if and only if $P(\Omega)$ is a Gibbs distribution with energy function formulated by (9.2), where Q is the collection of cliques associated to S .”

This theorem provides a direct connection between the local (thus, computationally tractable) MRF model and the highly general energy-based GRF models.

9.4 THE OPTIMIZATION PROBLEM

In order to minimize the probability of error classification of a image, it is necessary to identify the label map which maximizes the “*a posteriori*” probability (MAP criterion) [9.5]:

$$\text{Max}_{\Omega} \{P(\Omega|I)\} \quad (9.3)$$

where I is the configuration of all the pixel vectors in the image.

From the point of view of MRFs, finding the MAP estimate of Ω is equivalent to find the solution with minimum energy. For an $S_1 \times S_2$ image, the estimation of MAP requires the minimization of $S_1 \times S_2$ variables; this represents a huge problem from the point of view of computation but this complexity is strongly reduced by the Markov property. There are several algorithms which permit to address this classification task [9.]. In this chapter, we focus on the ICM (Iterated Conditional Mode) optimization algorithm which has been adopted in the present thesis, since it is known from the literature as an effective trade-off between optimization of the class accuracy and the minimization of all overall execution time.

9.5 ITERATED CONDITIONAL MODE (ICM)

The ICM algorithm is a well-known optimization algorithm for MRF-based classification. Although this algorithm converges to local, minimum point of energy represents an efficient and simple alternative from the point of view of computation. With an iterative scheme based on processing at the pixel level, ICM permits to simplify the conditioning on the entire observed image I , by considering the current estimation about the image labels. For each pixel, ICM introduce the following equation of proportionality:

$$P(\omega | I, \Omega') \propto P(x | \omega) \cdot P(\omega | \Omega_N) \quad (9.4)$$

where x is the feature vector of the pixel Q under analysis, Ω' and Ω_N are referred to the current estimation of the class labels, respectively, for the other pixels of the image and for the pixels included in a predefined neighbourhood system.

9.6 PROBLEM FORMULATION

In order to explain the adopted MRF-ICM algorithm, some preliminary consideration are required: here, I will be the multifrequency and polarimetric image to be classified and $\Omega = \{\omega_1, \omega_2, \dots, \omega_N\}$ is the configuration of possible class labels for the entire area of study.

The MAP classification Ω of the entire image is derived from:

$$\text{Max}_{\Omega} \{P(\Omega | I)\} \quad (9.5)$$

But, under appropriate assumptions, the ICM method permits to reduce the problem in (9.5) to [9.6]:

$$\text{Max}_{\omega_i} \{P(x|\omega_i) \cdot P(\omega_i|C_{sp})\} \quad (9.6)$$

where C_{sp} (spatial context) represents the current estimation of the class labels in the spatial neighbourhood. Considering the model based on MRFs, the maximization of (9.6) is equivalent to the minimization of an energy function:

$$\text{Min}_{\omega_i} \{U_T(\omega_i, x, C_{sp})\} \quad (9.7)$$

where U_T represents the total energy associated to the spectral and contextual spatial information of the pixel under analysis. In addition, ICM assumes the separability between spectral and contextual information. In this way, it is possible to split the energy function in two different kinds of energy as follows.

$$U_T(\omega_i, x, C_{sp}) = U_X(\omega_i, x) + U_{SP}(\omega_i, C_{sp}) \quad (9.8)$$

where $U_X(\omega_i, x)$ and $U_{SP}(\omega_i, C_{sp})$ are the spectral and spatial contributions respectively.

There are, therefore, two kinds of energy which are considered separately. In this way, it is simpler to define the overall energy functions. Instead of using the conditional probability density $P(x|\omega_i)$, we use the ratio between the “a posteriori” probability and the “a priori” probability about the classes to define the spectral energy of the pixel:

$$U_X(\omega_i, x) = -\beta_x \log[P(\omega_i|x) / P(\omega_i)] \quad (9.9)$$

as this quantity is equal to $[-\log p(x|\omega_i)]$ up to an additive constant.

The other energy is connected to spatial context and it depends on the number of neighbouring pixels we consider:

$$U_{SP}(\omega_i, C_{sp}) = -\beta_{sp} \sum_{\omega_k \in C_{sp}} I(\omega_i, \omega_k) \quad (9.10)$$

where $I(*)$ represents the “Indicator Function” which counts the number of times that ω_i is presented in C_{sp} :

$$I(\omega_i, \omega_k) = \begin{cases} 1, & \text{if } \omega_k = \omega_i, \\ 0, & \text{otherwise} \end{cases} \quad (9.11)$$

The parameters β_x and β_{sp} represent local and spatial weight parameters and they can be set manually or automatically. In addition, it is interesting to remember that no specific constraints are given on the choice of the density function $p(x|\omega_i)$.

9.7 AUTOMATIC PARAMETERS SETTING

In order to automate the choice of the β parameters which are used in the energy functions we use the Ho-Kashyap method [9.1]. We have seen that the energy function can be seen as a sum of two contributions which are considered independently. In addition, every of these functions have a parameter (β) which has to set in order to obtain a good classification. This is a particular case of more general linear energy model:

$$U_T(\omega_i, x, C_{sp}) = \sum_{l=1}^L \lambda_l \varepsilon_l(\omega_i, x, C_{sp}), \quad (9.12)$$

where $\varepsilon_l(*)$ is the l -th contribution and the parameter λ_l (which correspond to the β parameters that we want to obtain automatically) plays the role of the weight of the energy contribution $\varepsilon_l(*)$ ($l = 1, 2, \dots, L$). In our case the number of parameter is 2 (so $L = 2$) because we splitted the energy function in two different types of energy. We denote by S^0 the given label vector at the initialization of ICM method. When the ICM algorithm is started, a given pixel is classified correctly by the minimum-energy rule if:

$$U_T(s^*, x, C_{sp}) \leq U_T(\omega_i, x, C_{sp}) \quad \text{for} \quad \omega_i \neq s^* \quad (9.13)$$

or equivalently:

$$\sum_{l=1}^L \lambda_l [\varepsilon_l(\omega_i, x, C_{sp}) - \varepsilon_l(s^*, x, C_{sp})] \geq 0 \quad \text{for} \quad \omega_i \neq s^* \quad (9.14)$$

where s^* represents the true label of the generic pixel obtained from the training-set. It is simple to note that the inequalities in (9.14) are linear respect to λ , i.e. the correct classification of a training pixel is expressed ad a set of $(M-1)$, where M is the number of

training classes) linear inequalities with respect to the model parameters. More formally, collecting the energy differences contained in (9.14) in a single L -dimensional column vector:

$$\varepsilon_l = \text{col}[\varepsilon_l(\omega_i, x, C_{sp}) - \varepsilon_l(s^*, x, C_{sp}) : l = 1, 2, \dots, L] \quad (9.15)$$

The pixel is classified correctly by ICM if $\varepsilon_i^T \lambda \geq 0$ for all the class labels $\omega_i \neq s^*$. Denoting by E the matrix obtained by juxtaposing all the row vectors ε_i^T ($\omega_i \neq s^*$), we conclude that ICM classifies the whole set correctly if and only if:

$$E\lambda \geq 0 \quad (9.16)$$

Operatively the matrix E includes all the coefficients (i.e., the energy differences) of the linear inequalities (9.14): hence, E has L columns and has a row for each inequality. In order to compute the solution of (9.16), in the ICM algorithm, the Ho-Kashyap method was adopted, which is completely automatic [9.2]. This approach jointly optimizes λ according to the following quadratic problem:

$$\begin{aligned} & \min_{l, b} \|El - b\|^2, \\ & l \in \mathbb{R}^L, b \in \mathbb{R}^L \\ & b_r > 0 \quad \text{for } r = 1, 2, \dots, R \end{aligned} \quad (9.17)$$

which is solved by alternating iteratively an MSE (Minimum Square Error) step to update λ and a gradient-like to update the vector b , named “margin vector”[9.2].

The proposed approach to the MRF parameter setting problem allows exploiting the theoretical results known about the Ho-Kashyap method in the context of linear discriminant functions. Specifically, one can prove that, the Ho-Kashyap algorithm converges to a solution of (9.16) in a finite number τ of iterations (i.e., we obtain $E\lambda^\tau = b^\tau \geq 0$) [9.2] if the matrix inequality (9.16) has solution. Instead, if the inequality (9.16) has no solution, it is possible to prove that each component of $E\lambda^t - b^t$ (t indicates the iterations) either vanishes for $t \rightarrow +\infty$ or takes on non-positive values, while the $\|E\lambda^t - b^t\|$ converges to a positive limit, bounded away from zero.

9.8 THE OVERALL CONTEXTUAL CLASSIFIER

Now, it is possible to describe all the steps of the ICM algorithm which has been used in this thesis. First of all, it is necessary to say that MRF-ICM is a method which requires a training set and an appropriate PDF to describe the class probability. In our program this probability is Gaussian or Log-normal. As shown in the following chapter, if the input image of MRF-ICM is a transformed image without several transformations the probability model changes for each transformation. For instance, if we use the image with a LOG transformation as an input to the classifier, then the probability model is not Gaussian but LOG-Normal. The ICM algorithm is summarized in the following steps:

- **not-contextual step:** generate an initial not-contextual classification map by applying a given supervised non-contextual (in our case, Bayesian with MAP criterion) classifier;
- **energy difference step:** compute the energy-difference matrix E according to the adopted MRF model and to the label vector S^0 ;
- **Ho-Kashyap step:** compute an optimal parameter vector λ^* by running the Ho-Kashyap procedure (applied to the matrix E) up to convergence;
- **ICM step:** generate a contextual classification map by running ICM up to convergence.

A complete flow-chart of the method is shown in Figure 9.2.

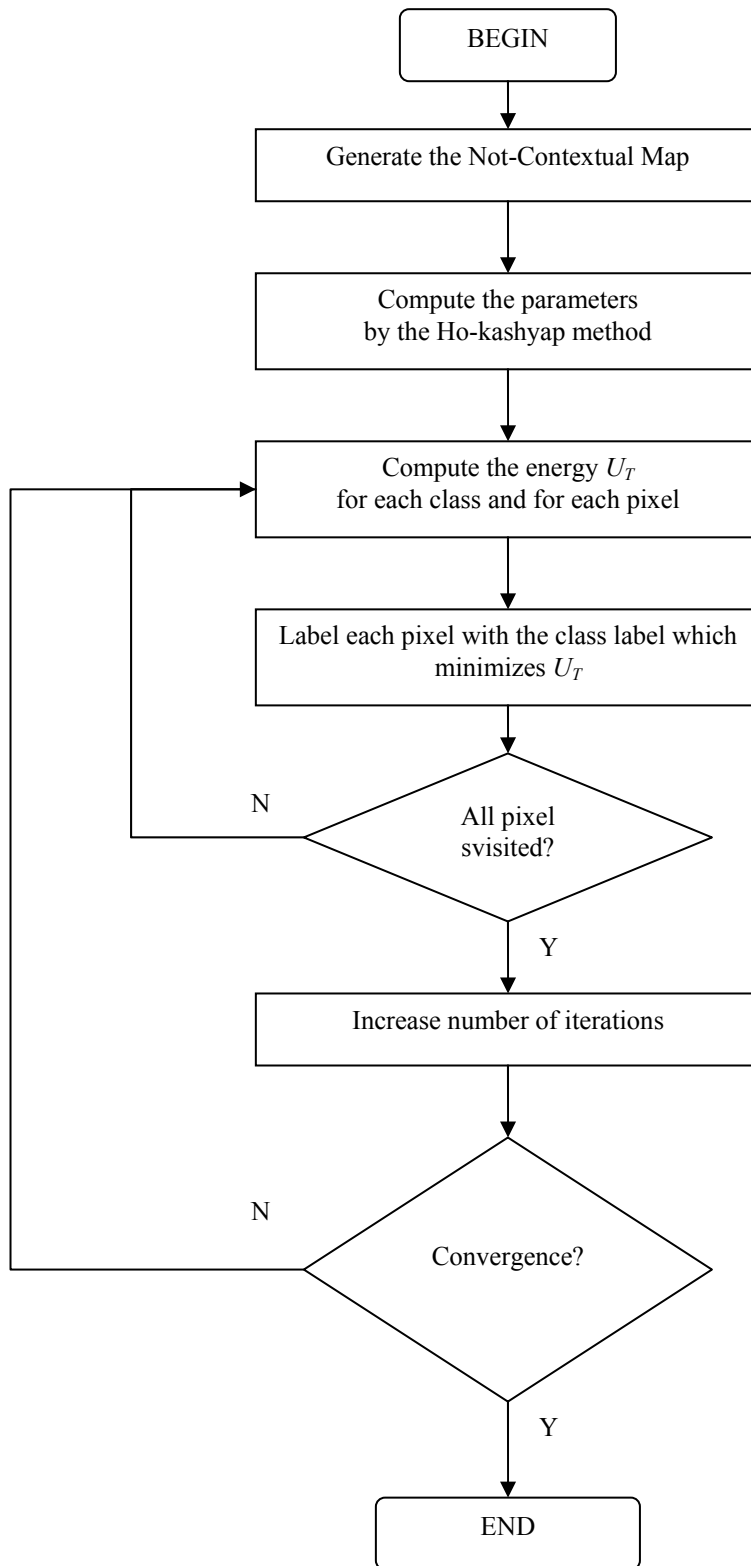


Figure 9.2: Flowchart of the adopted MRF-ICM algorithm.

10 FEATURE SPACE TRANSFORMATION

10.1 ORTHOGONAL TRANSFORM

In this section we present several orthogonal transformations employed for classifying POLSAR data. Specifically, if T is an orthogonal square transformation matrix, following relation must be valid the:

$$T^T T = I \quad (10.1)$$

where T^T is the transpose matrix and I is the unitary matrix. The relation above shows that the inverse transformation T^{-1} is equal to T^T .

Consider now the vector x which represents a given input discrete signal:

$$x^T = [x(0), x(1), \dots, x(N-1)] \quad (10.2)$$

Through the transformation T we obtain:

$$X = Tx \quad (10.3)$$

where

$$X^T = [X(0), X(1), \dots, X(N-1)] \quad (10.4)$$

is the new vector in the transformed space.

Indeed, the following relations are valid:

$$x^T x = \|x\|^2 \quad \text{and} \quad X^T X = \|X\|^2 \quad (10.5)$$

where $\| \cdot \|$ denotes the Euclidean norm operator.

And, plugging (10.1) into (10.3), we obtain:

$$\|x\|^2 = \|X\|^2 \quad (10.6)$$

This relation shows that the orthogonal transformations save the norm, thus the signal power is not modified by such transformations. The orthogonal transformations are defined through the orthogonal vectors $\{\varphi_i\}$ such that:

$$T^T = [\varphi_0 \varphi_1 \dots \varphi_{N-1}] \quad (10.7)$$

The orthogonality condition (10.1) can thus be expressed as:

$$\varphi_i^T \varphi_j = \delta_{ij} \quad (10.8)$$

In addition, considering (10.3) and (10.7) the following relations are obtained:

$$x = T^T X = [\varphi_0 \varphi_1 \dots \varphi_{N-1}] X = X(0)\varphi_0 + X(1)\varphi_1 + \dots + X(N-1)\varphi_{N-1} = \sum_{i=0}^{N-1} X(i)\varphi_i \quad (10.9)$$

which expands x on the basis $\{\varphi_i\}_{i=0}^{N-1}$.

10.2 KARHUNEN-LOEVE (KL) TRANSFORM

The discrete *Karhunen-Loeve* (KL) transform (also called the Principal Component transform) is a linear orthonormal transform which optimally decorrelates image data, provided that the underlying distribution of the data is a zero-mean stationary Gaussian random process [10.1]. If the second order statistics of the source random process are known, eigenvectors can be determined and used as the basis vectors for the KL expansion. The optimality of the KL transform is reflected by the fact that, for a given total number of bits, the representation error is minimized in the mean-square sense. This transform also has the highest energy compaction as compared to the other deterministic transforms, such as *Discrete Cosine Transform* (DTC), *Fourier Transform* or *Walsh-Hadamard Transform*.

Specifically, the *Karhunen-Loeve* (KL) transform permits to convert a distribution with arbitrary covariance matrix Σ to one with diagonal covariance. Actually, the aim of KL is to generate d de-correlated¹ features $y_1, y_2 \dots y_d$ starting from d generic features $x_1, x_2 \dots x_d$. In the

¹ The features are de-corellated in the sense of having a diagonal *covariance matrix*.

Gaussian case, the result of features are still independent. In order to obtain this transformation, we collect the eigenvectors $e_1, e_2, \dots, e_d$ of Σ as a row juxtaposition, thus obtaining the following linear transformation:

$$A = \begin{bmatrix} e_1^t \\ e_2^t \\ \vdots \\ e_d^t \end{bmatrix}, \quad y = Ax \quad (10.10)$$

The covariance matrix in the transform space is diagonal:

$$A = \begin{bmatrix} e_1^t \\ e_2^t \\ \vdots \\ e_d^t \end{bmatrix}, \quad y = Ax \quad (10.11)$$

$$\text{cov}\{y\} = \begin{bmatrix} \lambda_1 & 0 & 0 & 0 \\ 0 & \lambda_2 & 0 & 0 \\ 0 & 0 & \vdots & 0 \\ 0 & 0 & 0 & \lambda_d \end{bmatrix} \quad (10.12)$$

when λ_i is the eigenvalue corresponding of the eigenvector e_i ($i = 1, 2, \dots, d$).

The Karhunen-Loeve (KL) transform creates, through the diagonalization of Σ , d de-correlated features $y_1, y_2, \dots, y_d$ whose variances coincide with the eigenvalues of Σ :

$$\text{var}[y_i] = \lambda_i, \quad i=1,2,\dots,d. \quad (10.13)$$

There are not fast algorithms for the computation of the *Karhunen-Loeve (KL)* transform. Thus, the *Karhunen-Loeve (KL)* transform is seldom implemented, but sub-optimally transformations are usually preferred. However, it is a very useful reference. Among the orthogonal transformations, the other used ones are the *Discrete Cosine transform*, the *Discrete Fourier transform* and the *Walsh-Hadamard transform*.

10.3 DISCRETE COSINE TRANSFORM - DCT

10.3.1 DEFINITION

The *Discrete Cosine Transform (DCT)* of a sequence $x(n)$ of length N , is a sequence $X(k)$, of length N , defined by:

$$X(k) = \sqrt{\frac{2}{N}} \beta(k) \sum_{n=0}^{N-1} x(n) \cos\left(\pi \frac{k(2n+1)}{2N}\right) \quad (10.14)$$

with

$$\beta(k) = \sqrt{\frac{1}{2}} \quad \text{for} \quad k = 0 \quad (10.15)$$

$$\beta(k) = 1 \quad \text{for} \quad 0 < k \leq N-1 \quad (10.16)$$

while the *Inverse Cosine Transform (I.D.C.T.)* of a sequence $X(K)$ of length N is the following sequence $x(n)$, of length N :

$$x(n) = \sqrt{\frac{2}{N}} \sum_{k=0}^{N-1} \beta(k) X(k) \cos\left(\pi \frac{k(2n+1)}{2N}\right) \quad (10.17)$$

10.3.2 DCT AND IDCT AS DEVELOPMENT OF THE SEQUENCE ON ORTHOGONAL BASE

Defining the vectors:

$$x(n) = \sqrt{\frac{2}{N}} \sum_{k=0}^{N-1} \beta(k) X(k) \cos\left(\pi \frac{k(2n+1)}{2N}\right) \quad (10.18)$$

$$\begin{aligned} \underline{X} &= [X(0), X(1), \dots, X(N-1)]^T \\ \underline{x} &= [x(0), x(1), \dots, x(N-1)]^T \end{aligned} \quad (10.19)$$

and the matrices C and D of dimension $N \times N$, whose entries $\{c_{nk}\}$ and $\{d_{nk}\}$ are given by ($n, k = 0, 1, \dots, N-1$):

$$\begin{aligned} \{c_{nk}\} &= \sqrt{\frac{2}{N}} \beta(k) \cos\left(\pi \frac{k(2n+1)}{2N}\right) \\ \{d_{kn}\} &= \sqrt{\frac{2}{N}} \beta(k) \cos\left(\pi \frac{k(2n+1)}{2N}\right) \end{aligned} \quad (10.20)$$

the transformations DCT and $IDCT$ can thus be expressed by the linear operators C and $D = C^{-1} = C^T$ as:

$$\underline{X} = C \underline{x} \quad \underline{x} = D \underline{X} = C^{-1} \underline{X} \quad (10.21)$$

As the vectors which compose the matrices C and D are orthogonal and of unitary norm, the correspondent operators are simple rotations of coordinates in $\leq^n$. For these transformations all the geometric properties of the orthonormal transformations are valid.

10.3.3 RELATION WITH THE DISCRETE FOURIER TRANSFORM - DFT

DFT (Discrete Fourier Transform) is the expansion of a signal on a base of cosines and sines, while *DCT* involves a base of only cosines. Indeed, it is possible to develop a real signal in only cosines if its Fourier Transform is real with even symmetry. Any signal can be expressed in series of only cosines if the domain $\{0, 1, \dots, N-1\}$ of the signal is replicated through a symmetry around the ordinate axis (see Figure 10.1).

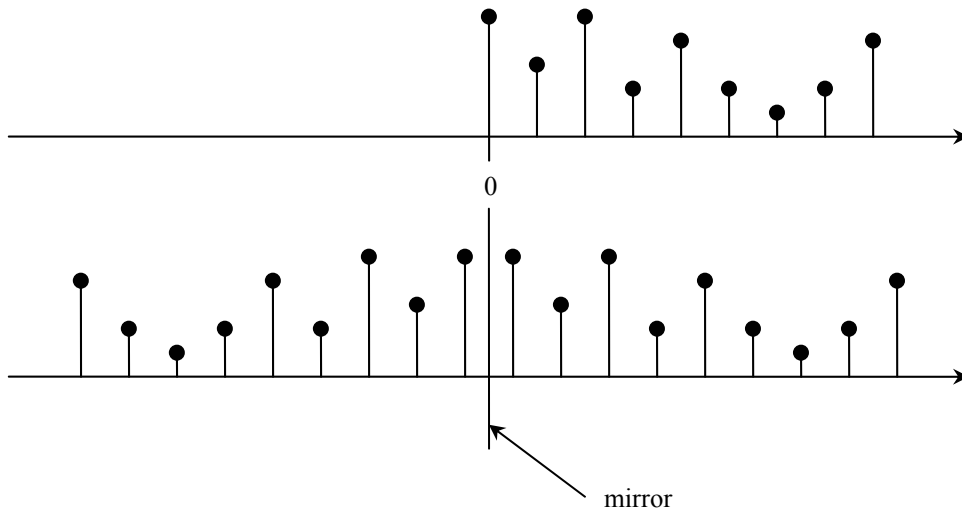


Figure 10.1: Examples of replication of a discrete signal defined in $\{0, 1, \dots, 8\}$ by symmetry with respect to the ordinate axis.

After this step, the number of samples is doubled as compared to the original one. In addition, the original samples are translated of a half duration. From these observations, the expression of *DCT* can be derived in term of *DFT* and the one of *IDCT* in term of *IDFT*. Actually, defined the sequence $x_2(n)$ obtained after the symmetry operation above 0, as shown in Figure 10.2.

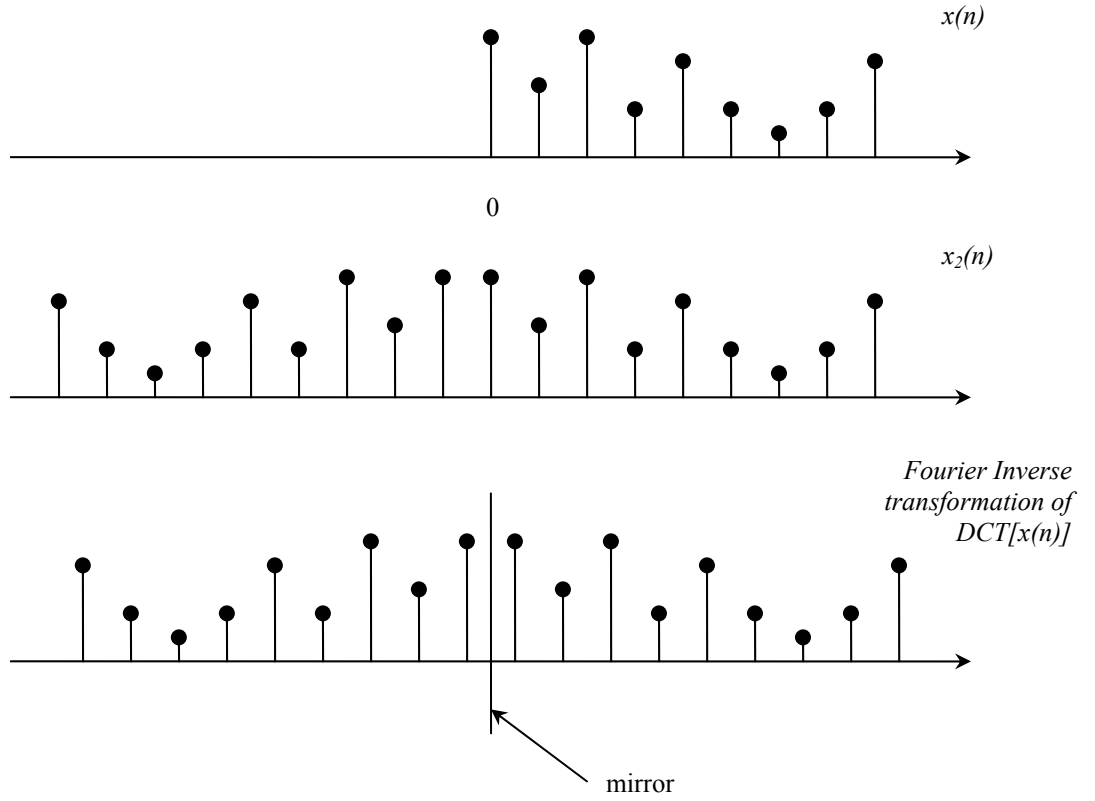


Figure 10.2: Fourier Inverse transformation of $DCT[x(n)]$ is equal to $x_2(n)$.

$DCT \{X(k)\}$ of the sequence of N samples coincides numerically with $DFT \{X_2(k)\}$ on $2N$ samples of the sequence $x_2(n)$, multiplied by a factor of delay corresponding to half sampling period of the sequence $x(n)$, that is [10.2]:

$$X(k) = e^{-j\pi k/(2N)} X_2(k) \quad (10.22)$$

This enables the fast calculation of the DCT of N samples through FFT (*Fast Fourier Transforms*) algorithms on $2N$ points. Similarly, if we compute the $2N$ samples of the transformations $DFT \{X_2(k)\}$ as follows:

$$X_2(k) = \begin{cases} X(0) & \text{for } k = 0 \\ X(k) e^{j\pi k/(2N)} & \text{for } 1 \leq k \leq N-1 \\ 0 & \text{for } k = N \\ -X(2N-k) e^{j\pi k/(2N)} & \text{for } N+1 \leq k \leq 2N-1 \end{cases} \quad (10.23)$$

and we compute $IDFT \{X_2(k)\}$ on $2N$ points of $X(K)$, we obtain the same sequence $x_2(n)$ centered in 0. Thus, it turns out that [10.1]:

$$x(n) = x_2(n) \text{ for } 0 < n \leq N-1, \quad (10.24)$$

which enables the fast calculation of $IDCT$ through inverse FFT ($IFFT$) on $2N$ samples. From a compression viewpoint, the validity of the representation though DCT consists in the capability of representing in a compact form (using fewer bits) the coefficients of the expansion.

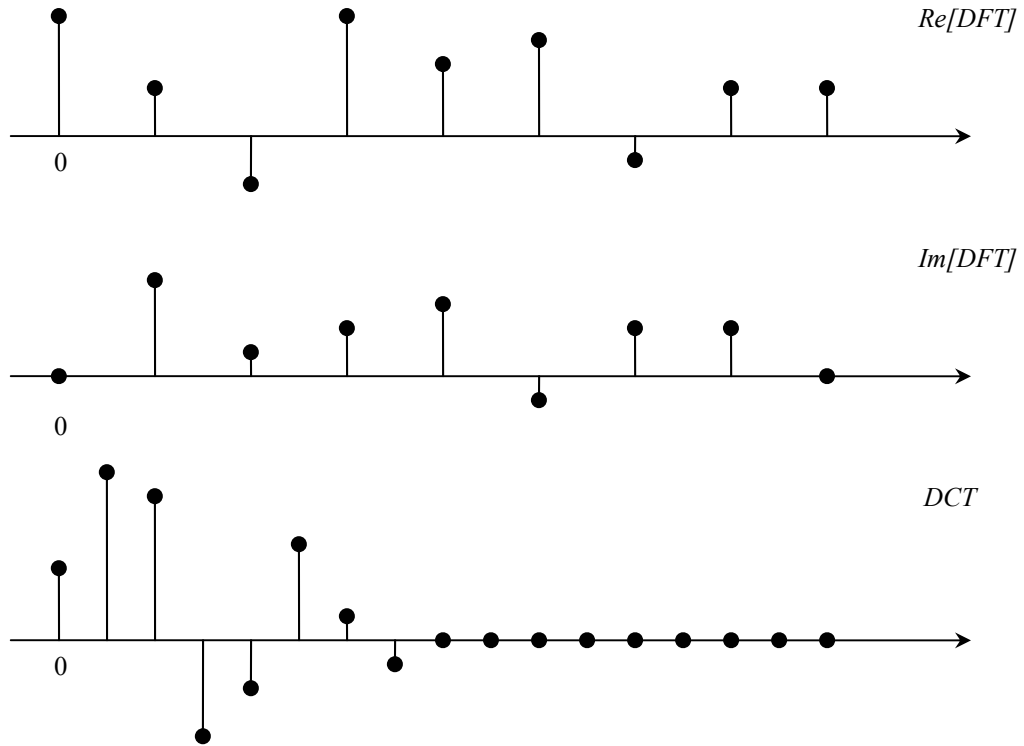


Figure 10.3: Coefficients for DFT and DCT. The DCT coefficients energy is more concentrated in the low frequencies.

In other words, although the total energy of the coefficients of the expansion (among the different bases which can be used for representing the signals) must be equal for orthonormal transformations, the mean energy of the coefficients DCT often is not homogenous, the entity of the first coefficients (low frequencies) being major and the entity of the last ones (high frequency) being minor. This property enables to quantize many of the DCT coefficients and

it is possible to neglect them during the transmission (cost in bit is very low) if they are very small. If, for example, we decide to transmit only half or a quarter of the coefficients, the error of rounding down would not be relevant for *DCT*, while it would be relevant for *DFT*.

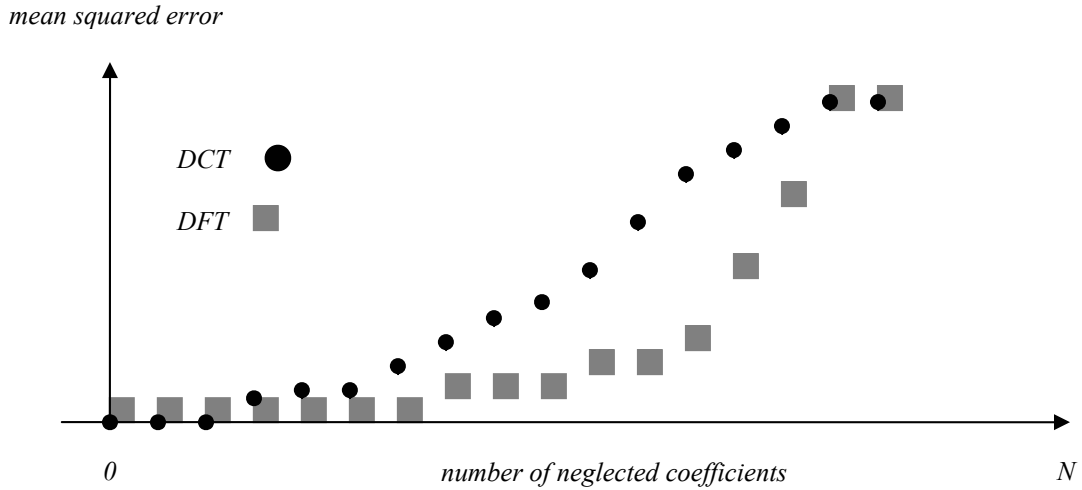


Figure 10.4: Mean squared error of DFT and DCT with different number of coefficients.

We can note how the energy of the error is identical for the 0 and N coefficients, omitted in both the case of use of *DFT* and *DCT*. Both are complete bases for the representation of signals or images. The efficiency of representation is different if we only consider a sub-set of expansion coefficients. Actually, many *DCT* coefficients can be quantized to 0 and thus neglected in the transmission, using a lower numbers of bits. The property of compaction of the energy of *DCT* is used in usual transmission methods or in image processing (standard *JPEG* for fixed and photographic images and *MPEG* for video-sequences).

10.4 DISCRETE FOURIER TRANSFORM - DFT

The *Discrete Fourier Transform (D.F.T.)* is a Fourier Transform operator defined for a discrete complex-valued sequence. Consider a complex sequence $x(n)$ with N samples of the form

$$\{x(0), x(1), x(2), \dots, x(N-1)\} \quad (10.25)$$

where $x(n)$ is a complex number

$$x(n) = x_{real}(n) + jx_{imag}(n), \quad n = 0, 1, \dots, N-1 \quad (10.26)$$

Assume that the sequence outside the range $\{0, 1, \dots, N-1\}$ is extended N -periodically, that is, $x(n) = x(n+N)$ for all $n \in \mathbb{Z}$. The *DFT* of this series is denoted $X(k)$, it has also N samples. The forward transform is defined as [10.2]:

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-jk2\pi n/N} \quad \text{for } k=0 \dots N-1 \quad (10.27)$$

The inverse transform is defined as [10.2]:

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{jk2\pi n/N} \quad \text{for } n=0 \dots N-1 \quad (10.28)$$

In general, the transform into the frequency domain will be a complex valued function, that is, with well-defined modulus and phase:

$$\text{magnitude} = |X(K)| = \sqrt{X_{real}^2 + X_{imag}^2} \quad (10.29)$$

$$\text{phase} = \tan^{-1} \left(\frac{X_{imag}}{X_{real}} \right) \quad (10.30)$$

10.5 WALSH'S FUNCTIONS AND WALSH HADAMARD TRANSFORMATIONS

10.5.1 WALSH'S FUNCTIONS

In many applications it is very useful to represent functions or sequences through sums of members of a system of simple functions, which are easy to define and to generate. Among these ones orthogonal suitable functions can approximate a generic function of the time with an arbitrary accuracy degree.

There are systems of orthogonal functions which (although they have not properties of the harmonic functions useful in the field of linear time invariant systems) are favorable in other applications, for example in the field of digital technologies. In this context Walsh's functions are the most important. Walsh's functions are an example of orthogonal functions. They are an ordered system of rectangular waves and, although they only taken on two values (typically $+1$ and -1), they have many properties similar to the co-sinusoidal system. They are defined on a bounded interval $[0, T]$, known as "*base of times*". As for the sinusoidal functions, two values are required for the complete definition of the generic Walsh's function: the value of the time variable t (usually normalized on the amplitude of the base of the times in t/T) and a number n of order, related to the concept of frequency in analogy with the harmonic functions. So the generic Walsh function is written: $WAL(n,t)$.

10.5.2 ORDERING OF WALSH'S FUNCTIONS

There are two ways of ordering Walsh's functions, which find applications in several fields:

- Ordering in sequence: put the functions in increasing order of zero crossing. We refer to this ordering with the notation $WAL(n,t)$

- Natural ordering : it presents analytic advantages as compare to the precious one. It is obtained by the Hadamard matrices (see below). We refer to it through the notation $PAL(n,t)$.

The natural ordering is used in image transmission for its computation of efficiency. The ordering in sequence is preferred in communication fields and in signal processing, for example, spectral analysis and filtering. Different algorithms which compute the transformation supply a set of coefficients ordered in bit-reversed order with respect to the sequences PAL or WAL :

	0	1	2	3	4	5	6	7
binary equivalent	000	001	010	011	100	101	110	111
inverse order	000	100	010	110	001	101	011	111
in decimal number	0	4	2	6	1	5	3	7

Table 10.1: Growing and inverse order of numbers in equivalent binaries.

10.5.3 DERIVATION OF WALSH'S FUNCTIONS

The approaches can be employed to obtain *Walsh's* functions:

- **Derivation of *Walsh's* functions by *Hadamard's* matrices**

Hadamard's matrices are quadratic matrices whose coefficients take on values $+1$ and -1 and whose rows (and column) are orthogonal among them. They are in normal form if they present the first row and the first column made of " $+1$ ".

$$H_2 = \begin{bmatrix} +1 & +1 \\ +1 & -1 \end{bmatrix} \quad H_4 = \begin{bmatrix} H_2 & H_2 \\ H_2 & -H_2 \end{bmatrix} \quad H_{2N} = \begin{bmatrix} H_N & H_N \\ H_N & -H_N \end{bmatrix} \quad (10.31)$$

($N = 2, 3, \dots$)

The Hadamard's matrix of the lowest order is the one of order two. Matrices of higher order, which are powers of 2, can be obtained by the recursive relation shown in (10.31). Each row of the Hadamard's matrix of order N defines a Walsh function. The order of this derivation is that one of a series of Walsh functions with positive phase (the functions are all in phase in the sense that they start from the value +1) and disposed in a natural bit-reversed ordering. The use of Hadamard matrices is the simplest way to compute the *WH* transform.

- **Derivation by Boolean synthesis**

For this derivation we can use the expression found by Gibbs that describes Walsh's discrete function, $WAL(i, t)$, through the product [10.3]:

$$WAL(i, t) = -1^{\sum_{k=1}^{\infty} (i_k + i_{k+1}) t_k} \quad (10.32)$$

From expressions like this one fast algorithms are obtained for the computation of *WH* transform [10.3].

10.5.4 WALSH-HADAMARD (WH) TRANSFORM

We consider a discrete series of N terms obtained by sampling a continuous functions in N equally spaced points in the opened interval. In order to make to corresponding properties of continuous and discrete systems, we must take N like power of 2, that is $N = 2^p$ ($p \in \infty$). We can so write a pair of discrete and finished transforms as:

$$X_n = \frac{1}{N} \sum_{i=0}^{N-1} x_i \cdot WAL(n, i) \quad x_i = \sum_{n=0}^{N-1} X_n \cdot WAL(n, i) \quad (10.33)$$

These equations are the direct and inverse transforms and they can be computed through a multiplication between matrices. Actually, if x is the N -dimensional vector which represents the input sequence, X the N -dimensional vector which represents the transformation and H is the *Hadamard matrix* of order N (with $N = 2^p$), then:

$$NX = Hx \quad (10.34)$$

while the inverse transform is expressed by:

$$x = HX \quad (10.35)$$

As the matrices for the WH transform are symmetric the direct and inverse transformations have an identical expression, except the normalization factor $1/N$. In addition, we can note that the ordering of the coefficients of the expansion in Walsh's series which are in the vector X is natural bit-reversed.

11 ILLUSTRATION AND STUDY OF RESULTS

This chapter shows the experiments which have been done by using the implemented algorithms. Before explaining the results of our studies it is necessary we describe the programs we used and their operations. Particularly, the main program which we implemented in C++ is based on the method MRF-ICM (Markov Random Fields - Iterated Conditional Mode) that is to say on a supervised classification technique. Indeed, two versions of the MRF-ICM method have been used: MRF-ICM which works using a 3 by 3 sliding window and that one which works with a 5 by 5 sliding window. Therefore two kinds of results are obtained through these algorithms. As some different input parameters are necessary to start the program, which is a supervised classifier, it is useful to describe them shortly. They are shown in Table 11.1.

-i [input image]	-d [training-set image]
-t [test image]	-m [init-map image]
-w [width of input image]	-h [height of input image]
-icm [number of iterations of ICM mode]	-hok [max number of iterations of the HoKashyap-method]
-n [number of bands of input image]	-? Help

Table 11.1: List of input commands of the MRF-ICM program.

It is easy to see that the program needs a training image for the classification and a test image to compute the confusion matrix in order to assess the quality of the class results. In order to improve the classification we used also the IDL language¹ (Interactive Data Language) to implement the algorithms which permit the transformations of the images.

All these ones will be described in the following. The other experiments regard the unsupervised classification and specially the K-means algorithm and the entropy-alpha

¹ IDL is an interpretative programming language. Information about it can be found on <http://www.rsinc.com>.

classification of a POL-SAR image. For these experiments it is not possible to extract a confusion matrix in order to discuss the results. The results are only analyzed from the point of view of a comparison between the K-means results about POL-SAR image and the one by the entropy – alpha parameters. Therefore, there are two parts in this chapter: one for supervised classification and the other one for unsupervised classification.

11.1 SUPERVISED CLASSIFICATION OF SAR POLARIMETRIC DATA

The experimentation in this context was finalized to test the use of methods of supervised classification of SAR polarimetric data. Indeed, these kinds of images are very noisy due to the phenomenon of speckle. In addition the noise is like a multiplicative component and not like an additive component. This explains the reason why it is difficult to classify this kind of images. Particularly, the SAR image used for tests presents another important problem regarding the number of bits per pixel which are used to store the image. Originally, our image was stored with 16 bits for pixel with a range of gray-level values from 0 to $2^{16}-1$. Then, it was saved with 8 bits for pixel and this has determined a distribution in the image statistics. Particularly, the problem consists in having a peak in the histogram of this image at the level 255, causing an increase in the image granularity. The original image is shown in Figure 11.1.

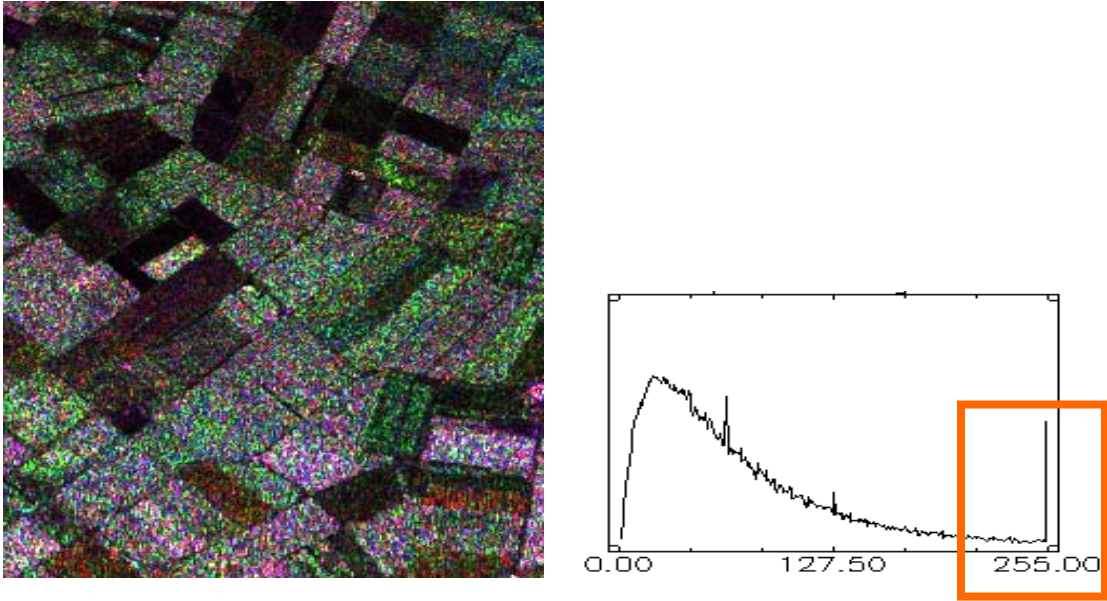


Figure 11.1: Original image with the relative histogram.

Another important problem which renders this type of classification very difficult is that the classes, in which the image is divided, are very similar. The training-set, in our specific case, identifies 5 different classes, all representatives of soil vegetative covers. Particularly, the classes represent: carrots, potatoes, bare soil, stubble, and sugar (Figure 11.2). The area under examination (Figure 11.1) is actually an agricultural zone around the city of Feltwell (United Kingdom) and the classes under exam are thus all vegetative soil covers. The sensor taken into consideration is the air-transported polarimetric NASA/JPL sensor, which acquires three distinct polarizations (HH , HV and VV) about three distinct frequencies (L , C and P), generating 9 channels.

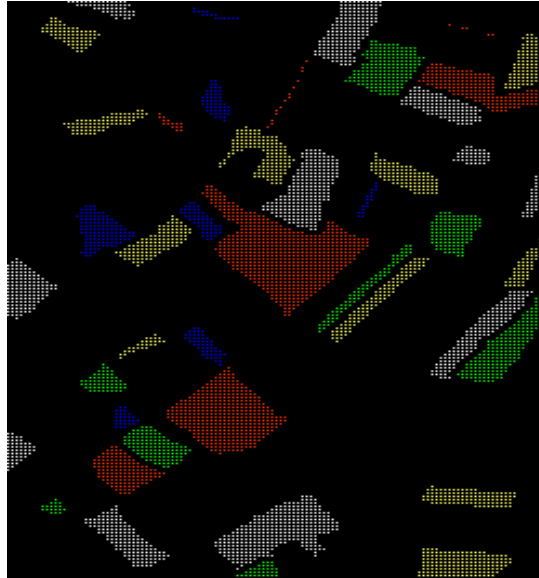


Figure 11.2: Training Map for the “Feltwell” data set.. To show the classes in the map each classes was assigned a different colour. The pixels in black are not in the training set.

SUGAR BETTS	
STUBBLE	
BARE SOIL	
POTATOES	
CARROTS	
OTHER	

Table 11.2 : Set of classes of the training-set and relative colour in the map.

Hence the pixels, pertaining to different classes, are characterizes by feature vectors frequently very similar. The feature vector of a pixel (indicated with $\underline{x}$) describes the polarimetric measures acquired by N image components in correspondence of the pixel. $\underline{x}$ is, therefore, an array of N elements and every element of this vector represents the gray value of every channel relative to that pixel. Like the test-map, the training set was used again. In this way we have two different data in the same image.

11.1.1 CLASSIFICATION WITH MRF-ICM

The program used for supervised classification tests is, as said above, written in C++ and it requires some parameters to carry out the processing. Among all the parameters, it is better to focus on those ones which regard the image and the classification. First of all, the size of the image is 250 per 350, with 9 channels² so the height is 350 (*-h 350* in the command line of the program), weight is 250 (*-w 250*) and the number of bands is 9 (*-n 9*). The most important parameters for the classification are the number of iterations of ICM (Iterated Conditional Mode) (*-icm* in the program) and the maximum number of iterations of the HoKashyap-method (*-hok*). For every test which we did, we defined 50 as the number of ICM iterations (*-icm 50*) and 100 as the maximum number of HoKashyap iterations (*-hok 100*). The *HoKashyap*-method is used to compute the coefficients which are necessary to evaluate the energy for every pixel [11.1].

After the short description of the classifier, it is now the moment to discuss the tests and the used methods. In order to obtain a good classification it is mandatory to determine the suitable PDF (probably density function). The most suitable distribution, derived from empiric measures [11.1], is the “Log-Normal” distribution. This distribution is related to the Gaussian distribution, i.e., a random variable $\underline{x}$ is Log-Normal if and only if its logarithm ($\ln \underline{x}$) is a Gaussian with the same parameters [11.2]. An analogous modelling (in terms of average vector and covariance matrix) adapts itself to operate with a greater number of random variables with jointly Log-Normal distribution. For this reason, the first test regards the classification with MRF-ICM, with 3 by 3 and 5 by 5 sliding window. As the distribution of pixels follows a Log-Normal PDF, a LOG transformation of the image has been made and then the classification with a Gaussian MRF-ICM has been performed. MRF-ICM uses a MAP criterion to compute the energy of every pixel. The MAP criterion uses, in its turn, the Gaussian PDF to compute $p(x|\omega_i)$ which represents the probability of finding a pixel belonging to a given class ω_i in a position x of the multispectral space. So, the results of these classifications with a 3 by 3 and a 5 by 5 sliding window are summarized in a confusion

² The image of study is divided in 9 channels derived from different frequencies and polarization.

matrix and in the parameter OA³. In the Table 11.3 these results are reassumed (ω_1 represents carrots, ω_2 potatoes, ω_3 sugar beets, ω_4 bare soil and ω_5 stubble).

NOT-CONTEXTUAL

TEST	MAP					CLASS ACCURACY
	ω_1	ω_2	ω_3	ω_4	ω_5	
ω_1	498	78	13	277	101	0.514995
ω_2	20	653	5	35	171	0.738688
ω_3	104	66	128	241	16	0.230631
ω_4	202	16	219	870	64	0.634573
ω_5	132	461	10	101	1339	0.655409
OA =						0.599313

CONTEXTUAL MRF-ICM (3 BY 3)

TEST	MAP					CLASS ACCURACY
	ω_1	ω_2	ω_3	ω_4	ω_5	
ω_1	676	36	10	206	39	0.699069
ω_2	7	774	4	34	65	0.875566
ω_3	122	77	169	187	0	0.304505
ω_4	113	7	350	854	47	0.622903
ω_5	48	155	6	65	1769	0.865884
OA =						0.728866

Table 11.3: Confusion Matrix of Non-Contextual and Contextual-MRF-ICM with a 3 by 3 sliding window.

The confusion matrix exhibit an OA equal to about 60% (0,599313) in the Not-Contextual case with a good class accuracies in every classes, particular in the case of the class ω_3 which has an accuracy around 23%. This value is very low and determines a mistaken classification of the pixels of this class. For this reason the value of OA is quite low, as well. In order to improve this value there is the contextual MRF-ICM. Particularly, the confusion matrix of this method with a 3 by 3 sliding window shows that the OA is increased from 60% to 73% with good class accuracies⁴ for the classes ω_2 and ω_5 . Also the CA (Class Accuracy) of the class ω_3 is increased but unfortunately it is still low (30,45%).

³ OA: Overall Accuracy, it is described in the chapter 8.

⁴ Class Accuracy (CA) is described in the chapter 8.

CONTEXTUAL MRF-ICM (5 BY 5)						
TEST	MAP					
	ω_1	ω_2	ω_3	ω_4	ω_5	CLASS ACCURACY
ω_1	684	32	1	207	43	0.707342
ω_2	5	794	0	35	50	0.89819
ω_3	123	77	154	201	0	0.277477
ω_4	49	7	66	1218	31	0.888403
ω_5	23	86	0	58	1876	0.918257
OA =						0.812027

Table 11.4: Confusion Matrix of the Contextual-MRF-ICM classifier with 5 by 5 sliding window.

Better results are obtained from the classification with a 5 by 5 sliding window. We can see that OA is improved and it is about 81% i.e., a 16% of increase in OA is achieved by the MRF-based approach. Unfortunately, there is the same problem about the class accuracy of ω_3 . Indeed, the CA of this class is diminished from the value 30,45% of the 3 by 3 case to 27,75% for this classification. The classification with a 5 by 5 window improves OA but it does not increase the CA of the famous class.

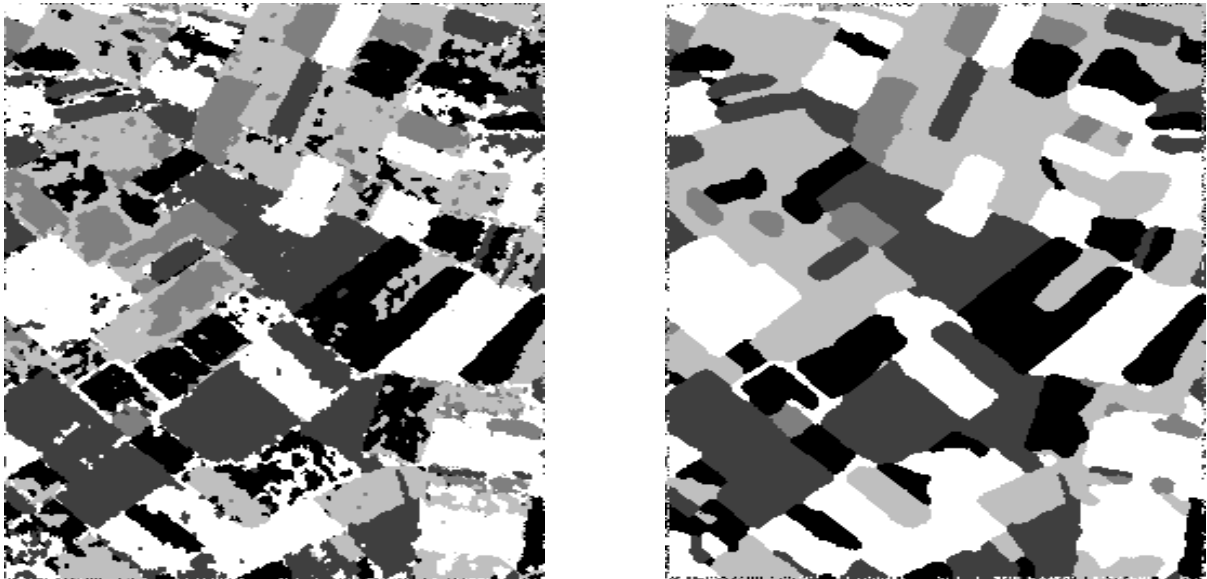


Figure 11.3: Results of the classification with the MRF-ICM program with a 3 by 3 sliding window (left) and a 5 by 5 sliding window (right).

As it is possible to see in Figure 11.3, the result at the left side presents a granular aspect due to the already cited phenomenon of speckle which increases the difference between

neighbouring pixels. In the second result (right), in homogenous points are nearly absent which confirms the quantitative results represented by the Confusion Matrix.

Now, we describe shortly the results of the image classification without the LOG transformation in order to emphasize that the PDF model chosen for the image is correct. Actually, the results of these tests are not good and are worse than the previous results in the two cases above described. As above, such results are summarized by a confusion matrix of all the two cases (3 by 3 and 5 by 5):

CONTEXTUAL MRF-ICM (3 BY 3)						
TEST	MAP					CLASS ACCURACY
	ω_1	ω_2	ω_3	ω_4	ω_5	
ω_1	399	31	50	273	61	0.490172
ω_2	15	1145	137	59	55	0.811481
ω_3	2	8	42	272	17	0.123167
ω_4	180	12	35	656	187	0.613084
ω_5	104	74	54	113	1143	0.768145
					OA =	0.660617

CONTEXTUAL MRF-ICM (5 BY 5)						
TEST	MAP					CLASS ACCURACY
	ω_1	ω_2	ω_3	ω_4	ω_5	
ω_1	409	32	1	305	67	0.502457
ω_2	1	1190	110	63	47	0.843373
ω_3	0	4	23	290	24	0.0674487
ω_4	203	5	4	601	257	0.561682
ω_5	16	64	2	102	1304	0.876344
					OA =	0.688329

Table 11.5: Confusion Matrices of the Contextual-MRF-ICM classifier, with a 3 by 3 and a 5 by 5 sliding windows.

These two contextual classifications explain that the Log-Normal PDF is a better distribution than the Gaussian one to characterize the random variable $\underline{x}$. Actually, the results from the classification of image without the transform have an OA of 66% (3 by 3) and 69% (5 by 5), respectively. Therefore, these values are worse than the results of image classification with LOG transformation.

The results obtained from classification are comparable with the experiments of the pasts. Actually, there are some classification of SAR images and, in particular, there is an experiments which was made to allow the interpretation of the “neural networks” for the

classification of multisensor remote-sensing image [11.4]. It is useful to compare the results of these classifications with those ones obtained from our tests. The results show that the best overall accuracy of classification is about 74%. In our experiments the best classifications presents, respectively, an OA of 72% for the case 3 by 3 sliding window and an OA of 81% for the case 5 by 5 sliding window. We consider the tests with LOG transformation. This shows that the Log-normal probability density function is a correct statistical model and, in one case, there is an increase of the overall accuracy. Obviously, it is necessary to consider the fact that they are two different procedures and that the attention of our work is focused in classification of SAR images but the attention of the other work is focused on neural networks as a method for classification of SAR and optical images [11.4].

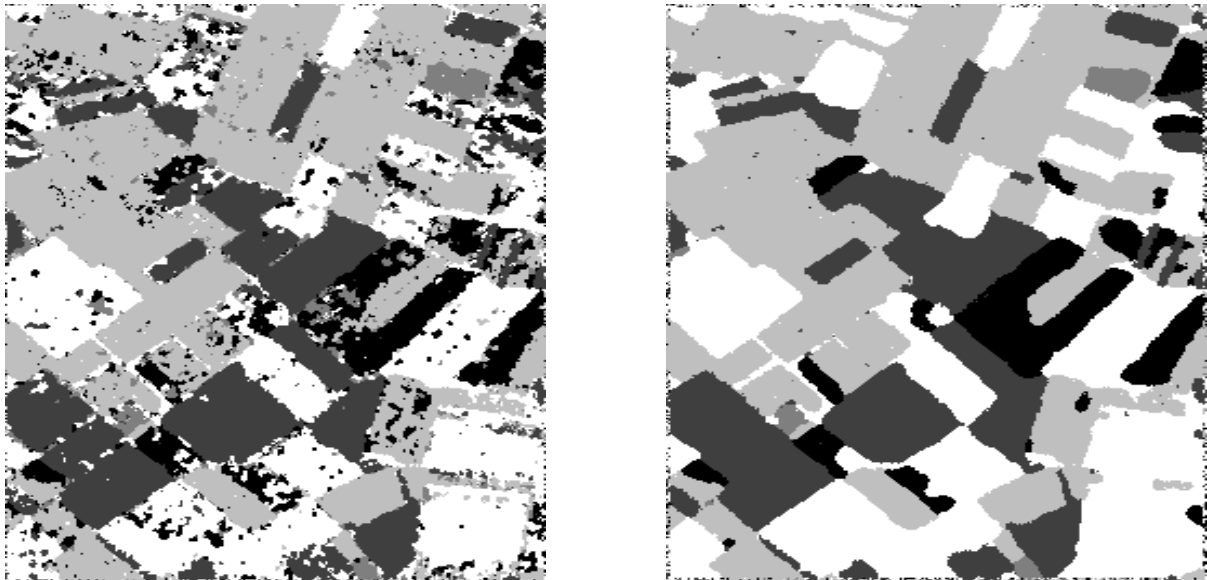


Figure 11.4: Classification with MRF-ICM of the image without the LOG transform (3 by 3 on the left side, 5 by 5 on the right side).

In addition, it is obvious that the last two classifications are not good because the CA of ω_3 is still lower than in the previous cases (12,31% for 3 by 3 and nearly 6,74% for 5 by 5). There is an important aspect to consider with regard to the problem of the CA of ω_3 . All these poor results could depend on the number of pixels of this class presented in the training-set and on the fact that this class represents a little portion of the image and, thus, it is easy to confuse a class with another. This is the case of ω_3 . In order to improve the classification analyzing the confusion matrix and the overall accuracy it is necessary to do transformations which permit to obtain different images from the original ones and to gain some advantages from the point

of view of classification or time-execution of processing of image. Considering the problem of the image to classify, in the next paragraphs the usual classification is described with different transformations of the input image. These ones are transformations which change the image in different ways. Indeed, for our tests, we used several orthogonal transformations like DCT (Discrete Cosine Transform), DFT (Discrete Fourier Transform), WH (Walsh – Hadamard Transform) and KL (Karhunen Loeve Transform). In addition, we tested despeckling algorithm based on GMRFs by Marc Walessa [11.3]. The purpose of the next pages will be to explain and compare the transformations and the classifications of these ones. The results of the first classification of the image with LOG transformations will be used as a reference in order to improve the classification and to find a better transformation.

11.1.2 TRANSFORMATIONS OF THE IMAGE

To implement the transformations which were described in the previous paragraph, it was useful the use of the IDL language. It is a programming language which is useful when work with images and vectors. Every transformation was developed through this algorithm. As the image has the problems above described, nearly all the transformations were done after the LOG transformation. In this way, there is not the peak in the maximum value of the image histogram, but it appears a new problem arises deriving from the range of gray values which are reduced. Actually, the image is stored with 8 bits for pixel, so the maximum value is 256, thus if we apply the LOG transformation we obtain a max number which is lower than 256. However, as we can see in the next sections, these transformations after the LOG mapping give better results as compared to the case without LOG transformation.

11.1.2.1 DCT AND CLASSIFICATION

The DCT is one of the most used orthogonal transformations. Actually, the KL transform is the best theoretical transformation but it requires computing its covariance matrix. DCT is an approximation of the KL transform. Fortunately, as we use Markov Random Fields the results of DCT are like the ones with KL. There are some advantages to do this processing. First of all, the energy of the image is concentrated in a few bands and this is good for the classification because we only have to extract a few transformed features.

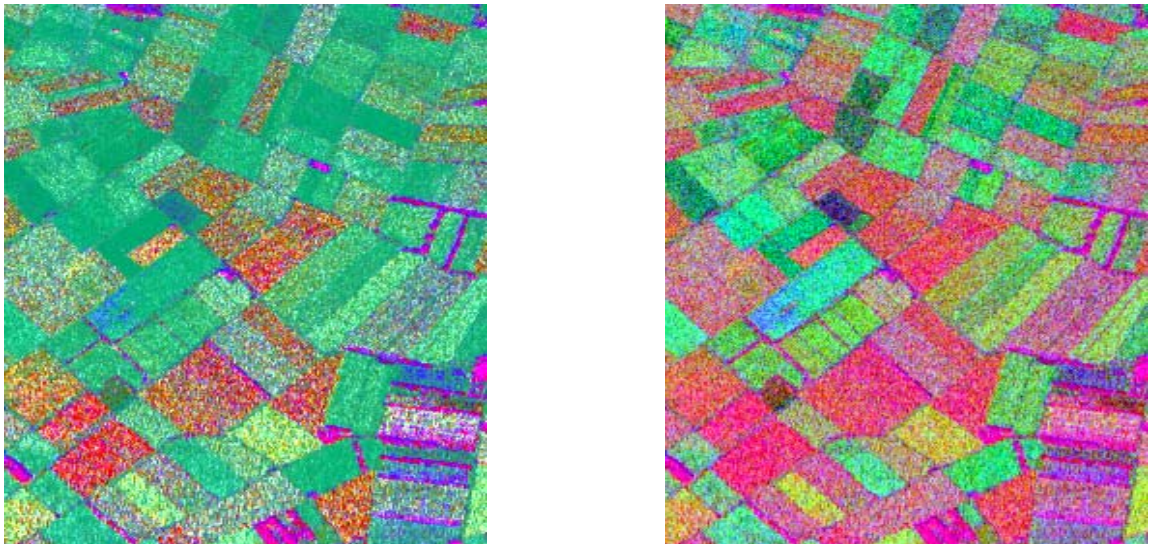


Figure 11.5: DCT Image on the left side and LOG+DCT image on the right side. The shown images were derived from the first 3 channels of the two images in an RGB false colour composition.

The LOG transformation is necessary and the results confirm this conclusion. Actually, if we consider the classification often DCT only it is obvious to note that the accuracy is the same as the classification of the original image.

CONTEXTUAL MRF-ICM
(3 BY 3)

TEST	MAP ω_1	ω_2	ω_3	ω_4	ω_5	CLASS ACCURACY
ω_1	399	31	50	273	61	0.490172
ω_2	15	1145	137	59	55	0.811481
ω_3	2	8	42	272	17	0.123167
ω_4	180	12	35	656	187	0.613084
ω_5	104	74	54	113	1143	0.768145
OA =						0.660617

Table 11.6: Confusion Matrix of the Contextual-MRF-ICM classifier with a 3 by 3 sliding window, after applying the DCT.

Actually, the confusion matrix is exactly the same as in Table 11.5. Hence the OA is the same and for this reason the confusion matrix of the 5 by 5 test is not shown. It is not necessary to discuss the output classification of these tests because they are identical to the previous tests. Therefore, we explain the tests with the LOG transformation whose results is shown in Figure 11.5. The results of this classification are nearly the same as the reference classification discussed in the previous paragraph. It is better to consider OA but they are comparable and they result, after all, in the same classification.

CONTEXTUAL MRF-ICM
(3 BY 3)

TEST	MAP ω_1	ω_2	ω_3	ω_4	ω_5	CLASS ACCURACY
ω_1	676	36	10	206	39	0.699069
ω_2	7	774	4	34	65	0.875566
ω_3	122	77	169	187	0	0.304505
ω_4	113	7	350	854	47	0.622903
ω_5	48	155	6	65	1769	0.865884
OA =						0.728866

CONTEXTUAL MRF-ICM
(5 BY 5)

TEST	MAP ω_1	ω_2	ω_3	ω_4	ω_5	CLASS ACCURACY
ω_1	684	32	1	207	43	0.707342
ω_2	5	794	0	35	50	0.89819
ω_3	123	77	154	201	0	0.277477
ω_4	49	7	66	1218	31	0.888403
ω_5	23	86	0	58	1876	0.918257
OA =						0.812027

Table 11.7: Confusion Matrix of Contextual MRF-ICM with a 3 by 3 and a 5 by 5 sliding window with the LOG+DCT transformation.

Unfortunately there is the usual problem with the class accuracy of ω_3 . As the results for this class are nearly the same in these two tests, the classification with the 5 by 5 sliding window is better. Obviously, the problem remains but the OA is greater.

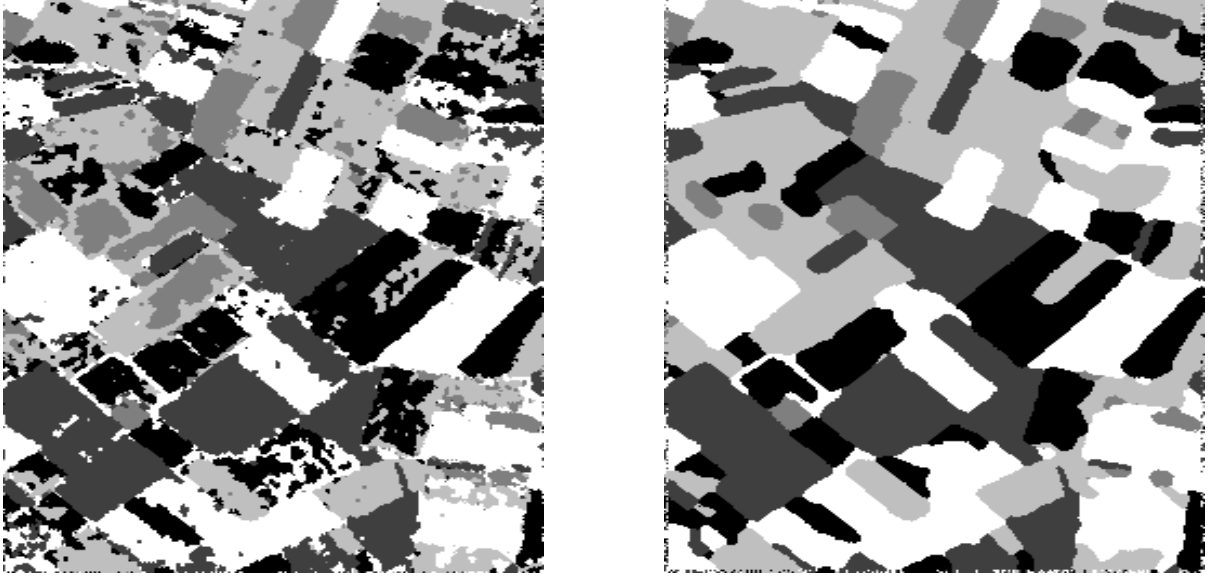


Figure 11.6: Classification with MRF-ICM of image with the LOG+DCT transform (3 by 3 on the left side, 5 by 5 on the right side).

An important advantage of DCT is for compression purpose. Actually, if we cut out a part of the energy (for example: 90%), we obtain few transformed bands. Before discussing the results of classification with this approach, it is useful to discuss shortly about the energy of the image and the extraction of the principal components. The computation of the energy is obtained from the function:

$$E_i = \sum_{j=0}^{N-1} |x_{ji}|^2 \quad (11.1)$$

where E_i is the energy of i -th band of the image, x_j is the j -th pixel in the i -th band and N is the number of pixels. In this way, it is possible to compute the energy for every band and to extract only the bands which have the highest energy. Actually, for the tests carried out the 90% of the energy. The image with the LOG–DCT transformation after this extraction has only three bands and this is a great advantage from the point of view of compression. The data show a curious thing: the classifications with this reduction has good results.

CONTEXTUAL MRF-ICM**(3 BY 3)**

TEST	MAP					CLASS ACCURACY
	ω_1	ω_2	ω_3	ω_4	ω_5	
ω_1	617	36	1	244	69	0.638056
ω_2	7	788	1	41	47	0.891403
ω_3	115	76	22	341	1	0.0396396
ω_4	87	12	19	1215	38	0.886214
ω_5	61	227	3	49	1703	0.833578
OA =						0.746564

CONTEXTUAL MRF-ICM**(5 BY 5)**

TEST	MAP					CLASS ACCURACY
	ω_1	ω_2	ω_3	ω_4	ω_5	
ω_1	568	38	0	281	80	0.587384
ω_2	18	817	0	28	21	0.924208
ω_3	123	77	0	355	0	0
ω_4	43	8	5	1280	35	0.933625
ω_5	28	113	0	59	1843	0.902105
OA =						0.77457

Table 11.8: Confusion Matrix of the Contextual-MRF-ICM classifier with a 3 by 3 and a 5 by 5 sliding window after saving only 90% of energy of image with LOG+DCT transformation.

The confusion matrix shows that for the test with the 3 by 3 sliding window the OA is better but the class accuracy of ω_3 is really poor. Quite worse is the test with a 5 by 5 sliding window where the class accuracy for ω_3 is exactly 0. This means that every pixel of the test-set which belongs to class ω_3 in the classification is labelled erroneously. For this reason the test with the 5 by 5 sliding window is not acceptable. Instead, the test with the 3 by 3 sliding window is good for the OA, but the class accuracy of ω_3 is not acceptable. These results demonstrate that it is very difficult to improve the classification with transformations without making worse the accuracy of the aforesaid class. However, there is an advantage from this classification viewpoint and it regards the compression of the image. Actually, from the extraction of the 90% of the energy from the image, the image has only 3 bands which permits a classification through the program with a shorter execution time. The execution time of ICM is about 2-3 seconds for the image with known dimensions (height: 350, weight: 250). As the number of iterations is fixed as an input datum and the execution is the same at every iteration, it is possible to determine the execution time of the classification. So, the time is about 18-27 seconds with 9 channels while it is 6-9 seconds with extraction of the energy. In this case, it is not so different because 30 seconds are not too much. But, if you work with larger images, this aspect becomes relevant. Actually, the compromise between the

classification accuracy and the execution time could be acceptable. Fortunately, this is not our case because the execution time is very short in every case. It is necessary to remember that computing the execution time, HoKashyap-method is not considered because is a not-deterministic task and depends on the size of the image as well as on the specific data. Therefore, the number of bands is not so determinant.

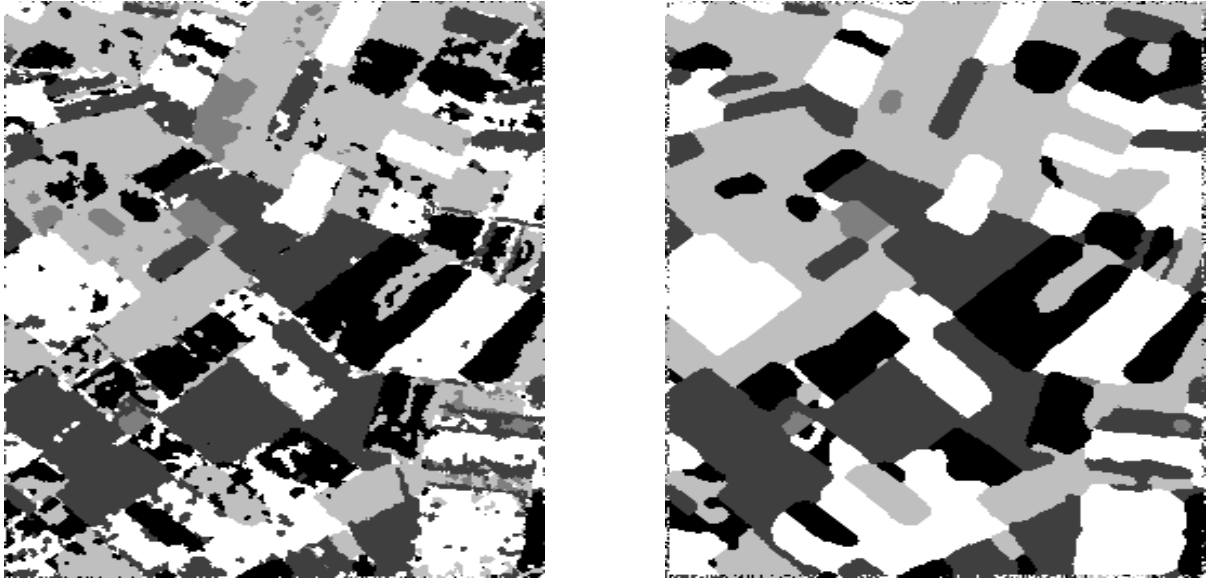


Figure 11.7: Classification with MRF-ICM retaining only the DCT features with the 90% of energy of image after a LOG+DCT transform (3 by 3 on the left side, 5 by 5 on the right side).

11.1.2.2 DFT AND CLASSIFICATION

The previous section showed that the DCT is an effective transformation and it is possible to use it for classification purposes. Instead, the DFT proves not to be very effective. The results of the classification of the LOG-DFT transform are not so satisfactory. Also with this transformation we did two different tests: one with DFT of LOG image and another one with the extraction of 90% of energy. The DFT in our tests is the most effective transform from the point of view of compression. Actually, with the extraction of 90% of image energy, the obtained image has only two bands and this one is very good.

CONTEXTUAL MRF-ICM**(3 BY 3)**

TEST	MAP					CLASS ACCURACY
	ω_1	ω_2	ω_3	ω_4	ω_5	
ω_1	675	37	9	208	38	0.698035
ω_2	8	773	5	32	66	0.874434
ω_3	122	77	171	185	0	0.308108
ω_4	112	7	356	849	47	0.619256
ω_5	52	154	6	59	1772	0.867352
OA =						0.728522

CONTEXTUAL MRF-ICM**(5 BY 5)**

TEST	MAP					CLASS ACCURACY
	ω_1	ω_2	ω_3	ω_4	ω_5	
ω_1	671	32	1	220	43	0.693899
ω_2	5	794	0	35	50	0.89819
ω_3	123	77	129	226	0	0.232432
ω_4	51	7	73	1208	32	0.881109
ω_5	22	86	0	60	1875	0.917768
OA =						0.803608

Table 11.9: Confusion Matrix of the Contextual MRF-ICM with a 3 by 3 and a 5 by 5 sliding window after with the LOG+DFT transformation.

The confusion matrix for a 3 by 3 sliding window is similar to the one of DCT in the previous section, thus, the conclusions are the same. For the test with a 5 by 5 sliding window, the OA is a bit lower than that one for corresponding classification with DCT. In order to show that DFT is not so effective as DCT it is necessary to discuss the results from the classification of image with only 90% of energy (2 channels). Unfortunately, the classification is not good and there is a decrease in the OA and in the class accuracy. Actually, these results are shown (Table 11.10) in the confusion matrix and in every case the accuracy is low.

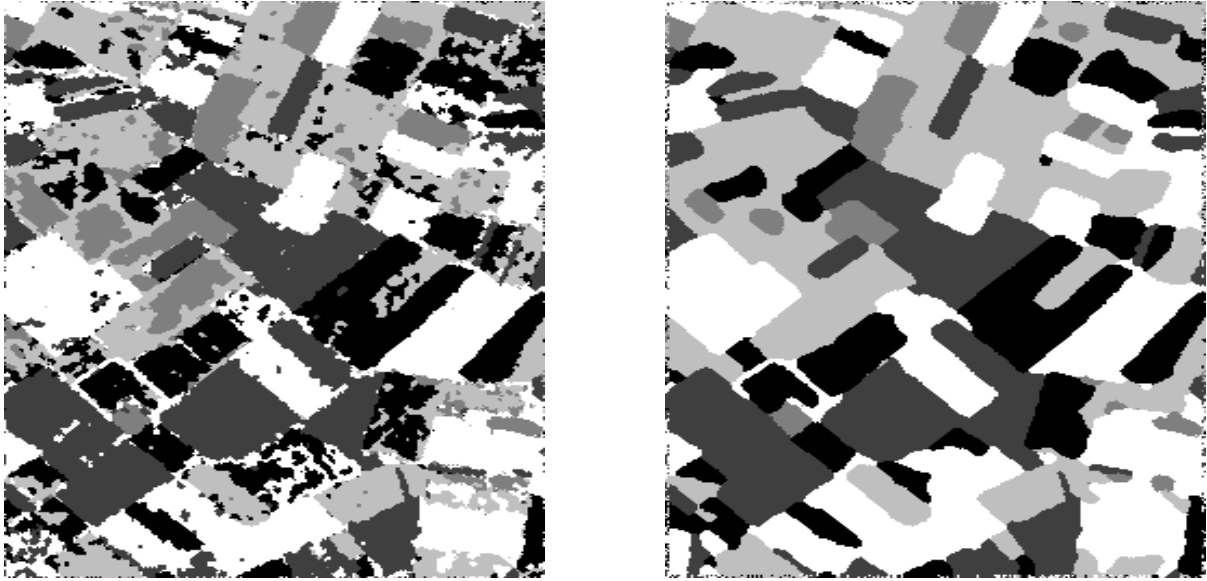


Figure 11.8: Results of classification with MRF-ICM program with a 3 by 3 sliding window (left) and a 5 by 5 sliding window (right) of image with the LOG+DFT transformation.

This is due to the fact that also in these tests a low class accuracy is obtained (for ω_3 and for ω_1) and for this reason these classifications confirm that the DFT classification is worse than with DCT.

**CONTEXTUAL MRF-ICM
(3 BY 3)**

TEST	MAP					CLASS ACCURACY
	ω_1	ω_2	ω_3	ω_4	ω_5	
ω_1	367	39	1	440	120	0.379524
ω_2	17	782	0	32	53	0.884615
ω_3	17	77	0	461	0	0
ω_4	130	15	15	1179	32	0.859956
ω_5	30	210	2	52	1749	0.856094
OA =						0.700515

**CONTEXTUAL MRF-ICM
(5 BY 5)**

TEST	MAP					CLASS ACCURACY
	ω_1	ω_2	ω_3	ω_4	ω_5	
ω_1	293	40	0	464	170	0.302999
ω_2	12	807	0	32	33	0.912896
ω_3	4	77	0	474	0	0
ω_4	60	9	1	1271	30	0.927061
ω_5	16	102	0	57	1868	0.914342
OA =						0.728351

Table 11.10: Confusion Matrix of the Contextual MRF-ICM with a 3 by 3 and a 5 by 5 sliding window of 90% of energy of image with the LOG+DFT transformation.

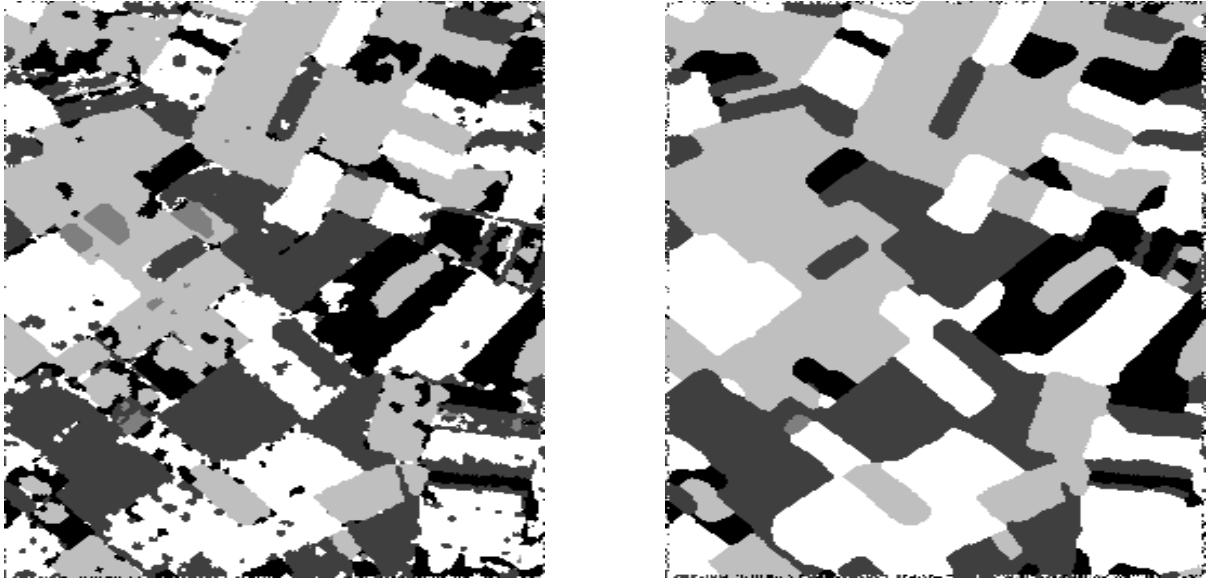


Figure 11.9: Results of classification with MRF-ICM program with 3 by 3 sliding window (left) and 5 by 5 sliding window (right) of image with LOG+DFT transformation.

We note that the poor results given by DFT may be interpreted as due to a mismatch between the Gaussian model used by MRF-ICM here and the data statistics. DFT generates, in fact, a complex-valued transformed set of features, which has been mapped onto a real-valued set of features, by computing its modulus. However, the non-linearity of the modulus operator does not allow the Gaussianity of the log-transformed data to be saved. On the contrary, DCT provides by itself a set of real-valued linearly-transformed features which makes the Gaussian-conditional assumption quite realistic.

11.1.2.3 KL TRANSFORMATION AND CLASSIFICATION

Another transformation which we used for the tests is KL (Karhunen Loeve). The tests are only focused on the classification with a 3 by 3 sliding window. The test with a 5 by 5 sliding window was not presented because are not very interesting. Actually, preliminary experiments show that the results are very lower than the correspondent results with DFT or DCT. The reason for these results, is that to obtain good results with this transformation a great de-correlation between pixels is necessary. Finally, KL is lower than DCT from the point of view of computation. Actually, it requires the eigenvectors which are extracted from a covariance

matrix to do the transformation. In this way the processing time is longer than for the previous transformation.

CONTEXTUAL MRF-ICM

(3 BY 3)

TEST	MAP					CLASS ACCURACY
	ω_1	ω_2	ω_3	ω_4	ω_5	
ω_1	399	31	50	273	61	0.490172
ω_2	15	1145	137	59	55	0.811481
ω_3	2	8	42	272	17	0.123167
ω_4	180	12	35	656	187	0.613084
ω_5	104	74	54	113	1143	0.768145
OA =						0.660617

Table 11.11: Confusion Matrix of the Contextual MRF-ICM with a 3 by 3 sliding window of image with the KL transformation.

The reason for this poor behaviour of KL may be that this transform adopts a data-specific basis, given by the covariance eigenvectors. However, the class-conditional normality for the log-data implies that the overall data distribution is a Gaussian mixture. Therefore the KL procedure can be strongly affected by the inaccuracy of modelling the whole data set as a single Gaussian mode.

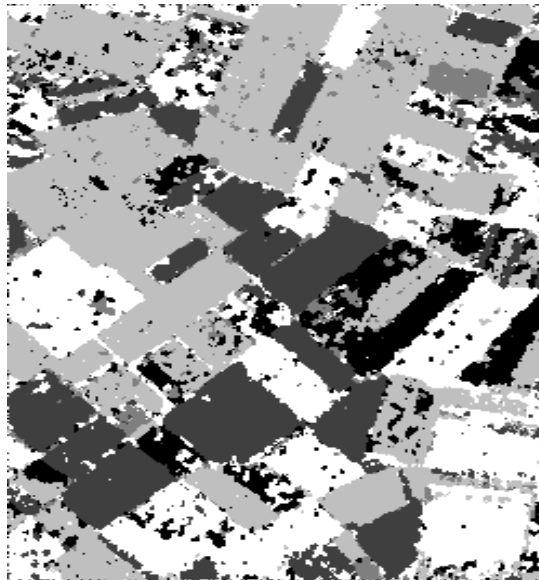


Figure 11.10: Classification with MRF-ICM with a 3 by 3 sliding window of image with the KL transform.

11.1.2.4 WH TRANSFORMATION AND CLASSIFICATION

This section shows the results which are obtained with WH (Walsh-Hadamard). Actually, this transformation proves similar to KL from the point of view of the results. We tested it with a 3 by 3 sliding window as in the case of KL.

CONTEXTUAL MRF-ICM

(3 BY 3)

TEST	MAP					CLASS ACCURACY
	ω_1	ω_2	Ω_3	ω_4	ω_5	
ω_1	399	31	50	273	61	0.490172
ω_2	15	1145	137	59	55	0.811481
ω_3	2	8	42	272	17	0.123167
ω_4	180	12	35	656	187	0.613084
ω_5	104	74	54	113	1143	0.768145
OA =						0.660617

Table 11.12: Confusion Matrix of the Contextual MRF-ICM with a 3 by 3 sliding window of the image with the WH transformation.

Therefore, WH does not seem to be very appropriate to the classification of polarimetric SAR data, at least with regard to the considered data set and classification scheme. An advantage of this transformation is that it is fast to compute because it has coefficients which do not require computing trigonometric function like DCT or DFT.

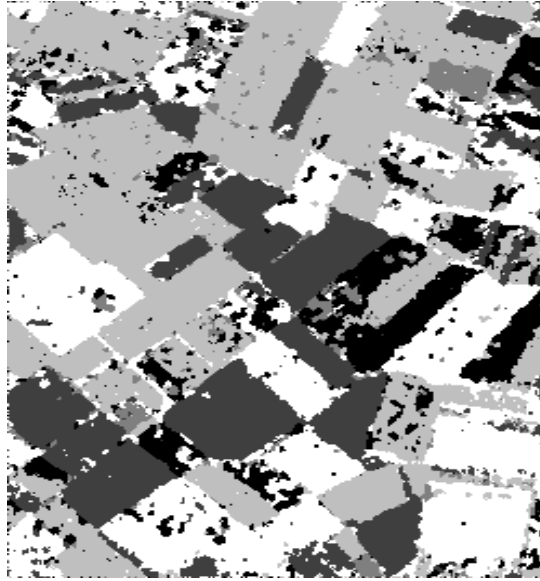


Figure 11.11: Classification with MRF-ICM with a 3 by 3 sliding window of the image with the WH transform.

11.1.2.5 DESPECKLING OF IMAGE AND CLASSIFICATION

To conclude the discussions about supervised classification we introduce some experiments involving also filtering. First of all, an important transformation which we use is a despeckling based on GMRF [11.1]. Other used filters are LPF (Low Pass Filter) and the Median filter. Not all such filters permit a good classification results.

Speckle is a phenomenon which rends difficult to improve the classification. For this reason, many algorithms were developed in order to reduce this problem and to obtain a good classification. As the image has the usual problem with the peak in the maximum value (255), it was difficult to apply the despeckling filter to the image. Actually, the results are poor considering the fact that an image without speckle was used previously.

CONTEXTUAL MRF-ICM

(3 BY 3)

TEST	MAP ω_1	ω_2	ω_3	ω_4	ω_5	CLASS ACCURACY
ω_1	458	28	178	82	68	0.562654
ω_2	10	1171	149	28	53	0.829908
ω_3	6	27	173	122	13	0.507331
ω_4	264	19	229	373	185	0.348598
ω_5	106	58	53	78	1193	0.801747
OA =						0.657299

Table 11.13: Confusion Matrix of the Contextual MRF-ICM with a 3 by 3 sliding window of the image with the DESPECKLE filter.

The confusion matrix shows that OA is not increased but is lower than that one of the original image. In addition, the problem of class accuracy is switched from ω_3 to ω_4 . Hence, this is not an accurate classification and for this reason it was necessary to find another solution to improve the accuracy of classification. Hence, to resolve the problem of image histogram we use the LPF before despeckling.

$$\frac{1}{5} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix} \quad (11.2)$$

This is the mask was used for the LPF. Indeed, this filter permits to execute the multilook⁵ technique in order to reduce the phenomenon of the speckle. Unfortunately, increasing the number of multiple observations, the spatial resolution of the image decreases. In this way, we try to reduce the peak of the image histogram in order to do a good despeckling of the image. Unfortunately, the results are not so good thus the problem still persists.

⁵ Multilook is a processing technique for SAR images which utilizes more observations of the same area dividing the antenna beam in more sectors to do an average in order to reduce the random variations of gray values caused by speckle.

CONTEXTUAL MRF-ICM
(3 BY 3)

TEST	MAP	ω_1	ω_2	ω_3	ω_4	ω_5	CLASS ACCURACY
ω_1		636	26	194	86	25	0.657704
ω_2		29	792	11	7	45	0.895928
ω_3		45	77	419	14	0	0.754955
ω_4		133	7	820	314	97	0.22903
ω_5		178	131	50	118	1566	0.76652
						OA =	0.640378

Table 11.14: Confusion Matrix of the Contextual MRF-ICM with a 3 by 3 sliding window of the image with the LPF+DESPECKLE transformation.

In order to improve the classification and to obtain a satisfactory test we tested other different processings before despeckling. The most interesting results considering the confusion matrix are gained by the following steps:

- *LOG transformation;*
- *Median Filters with 11 by 11 sliding window;*
- *Despeckling elaboration.*

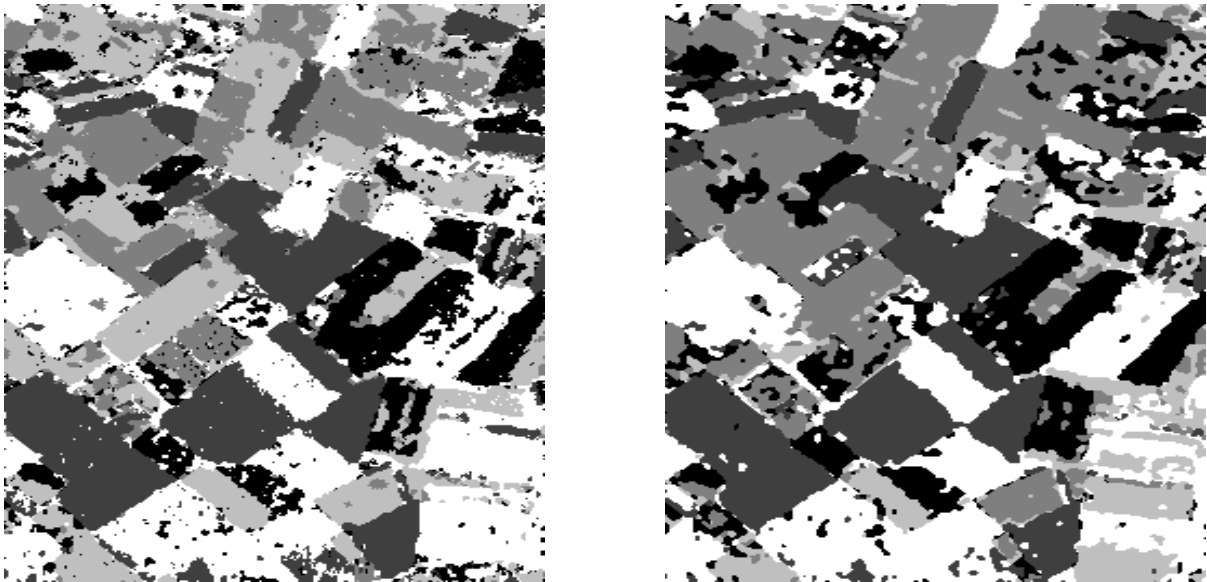


Figure 11.12: Results of classification with MRF-ICM with a 3 by 3 sliding window of the image with the DESPECKLE filter (left) and with LPF+DESPECKLE elaboration (right).

The results are better than with other filters but it is useful to remember that the confusion matrix is an estimate of the accuracy but it is computed only on the pixels which are labelled

on the training-set. In addition, it is necessary to consider that in the test-set the classified pixels are few and hence it could happen that a bad classification can be mistaken as a good classification.

CONTEXTUAL MRF-ICM

(3 BY 3)

TEST	MAP ω_1	ω_2	ω_3	ω_4	ω_5	CLASS ACCURACY
ω_1	625	29	65	227	21	0.646329
ω_2	34	712	50	23	65	0.80543
ω_3	79	77	159	240	0	0.286486
ω_4	76	2	206	1029	58	0.750547
ω_5	19	90	9	130	1795	0.87861
OA =						0.742268

Table 11.15: Confusion Matrix of the Contextual MRF-ICM with a 3 by 3 sliding window of the image with the LOG+MEDIAN+DESPECKLE transformation.

The confusion matrix shows OA is good but we have the usual problem of class accuracy on ω_3 . This classification is, thus, comparable with that one of the LOG-transformation. Other tests have been done but the results are not so satisfactory and for this reason are not detailed here. In addition, the described tests are concentrated on the experiments with a 3 by 3 sliding window. The tests with a 5 by 5 sliding window are not described because they are not so interesting from the point of view of the results. Actually, the classifications do not present advantages with these kinds of elaborations.



Figure 11.13: Classification with MRF-ICM with a 3 by 3 sliding window of the image with the LOG+MEDIAN+DESPECKLE transformation.

11.1.3 CONCLUSION ABOUT SUPERVISED CLASSIFICATION TESTS

Finally, the most important results which are gained from these tests are summarised in Table 11.16.

IMAGE TRANSFORM	OA (3x3: MRF-ICM window)
LOG	72,86
LOG + DCT	72,88
LOG + MEDIAN11x11 + DES	74,22
LOG + DCT +90E (3 bands)	74,65

Table 11.16: List of the best results and of the relative OAs for MRF-ICM with a 3 by 3 sliding window.

In order to define a better classification OA is not only sufficient. A method to do this evaluation is to compute the standard deviation (St.Dev.) of all the respectively confusion matrices. In this way, if this value is great for a classification, it means that at least one class accuracy is not good.

IMAGE TRANSFORM	Average of Class Accuracy (3x3: ICM-MRF window)	St.Dev. of Class Accuracy (3x3: ICM-MRF window)
LOG	0,673	2,486
LOG + MEDIAN11x11 + DES	0,673	2,484
LOG + DCT	0,673	2,486
LOG + DCT +90E (3 bands)	0,657	2,684

Table 11.17: List of the best result and the relative statistic data.

From Table 11.17 it is possible to see how the LOG and LOG+DCT have exactly the same results. The second transformation in the table has a low standard deviation. and a good average of accuracy. Finally, they are quite good classifications even if there is the usual problem with class accuracy which persists in every classification.

11.2 UNSUPERVISED CLASSIFICATION

This part of the chapter shows the results of the tests about unsupervised classification. In order to do these tests we used the ENVI software which is useful to do the unsupervised classification and to show the images. In addition to this software, we implemented other algorithms in order to obtain the entropy-alpha coefficients from the image. These algorithms are developed with IDL. To explain the results, we have to consider which methodology we used to do an unsupervised classification.

Among all of the possible methods we use the K-Means algorithm. This program computes the classification without training data and it needs the number of clusters in which we want to classify the image. To label a pixel this algorithms use the minimum Euclidean distance. The image used for these tests has 4 channels of different polarizations (HH , VV , HV , and VH) and with size of 2990 by 4140 pixels. It is acquired from DLR (German Aerospace Center) and it represents a place near Oberpfaffenhofen (Germany). This image is larger than that one used for the supervised classification. As we have not the test-set, it is not possible to compute a confusion matrix, hence we only write a description of the classification with different number of clusters as input. After this classification we did a transformation of the image from 4 channels to 3 channels in order to compute the aforesaid entropy and alpha coefficients. Therefore, this image is used as an input to K-Means.



Figure 11.14: Input image of unsupervised classification. Also this image is shown in the RGB mode. The used bands are the HH , VV and HV polarization.

11.2.1 K-MEANS CLASSIFICATION OF THE IMAGE

To analyze the classification with “K-Means” it is necessary to choose the number of clusters in which the image is divided. To do this, we try classifications with different numbers of clusters as input to this program. To compare the different results, we need to define a coefficient which describes the classification. Hence, for every classification, we extracted some parameters in order to do an accurate analysis. As “K-Means” groups the pixels in clusters, it is possible to evaluate the covariance matrix for every cluster and then to compute the sum of these ones for all the clusters. In this way, we have a parameter which indicates the quality of the classification. Obviously, if we increase the number of clusters as an input to classifier it is normal to expect that the sum of class covariances diminishes more and more slowly and stabilises. The experiments will show better what has been said above. In our tests we did 9 classifications of the image with different numbers of clusters (from 2 to 10 clusters). The other input parameters of K-Means are the maximum number of iterations and the percent change threshold. For us, the chosen values are, respectively, 100 for the maximum number of iterations and 0.01% for the change threshold. After the description of the algorithm and of the used methodologies, it is, finally, possible to explain the tests.

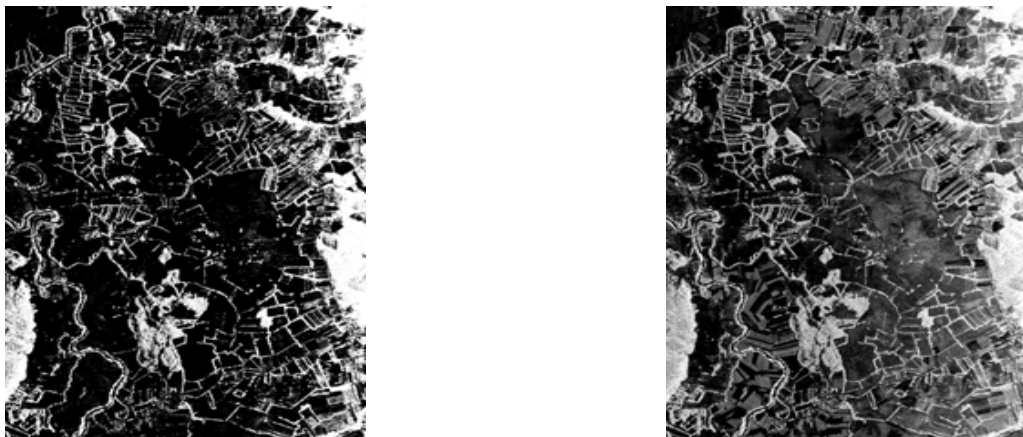


Figure 11.15: Results of K-Means classification with 2 clusters (left) and 4 clusters (right).

Without statistical data it is very difficult to define the best classification. Obviously the one in Figure 11.15 with 2 clusters is not good but it is only shown for illustration purposes.

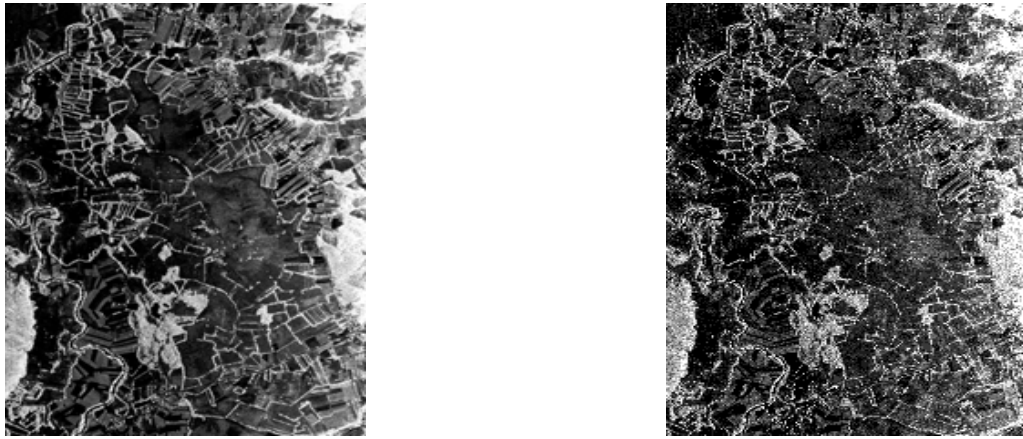


Figure 11.16: Results of K-Means classification with 5 clusters (left) and 7 clusters (right).

Thus, we computed the covariance for every cluster in each classification in this way: we compute the covariance matrix for the cluster i ; then we calculated the determinant of these matrices which are summed at the end. The resulting sum is named here “overall variance” of the clustering solution. All this it is summarized in Table 11.18.

Cluster 1 – Covariance Matrix			
1190,28	744,51	428,58	437,53
744,51	1221,02	379,57	387,93
428,58	379,57	654,13	639,02
437,53	387,93	639,02	659,80
Determinant: 1,52E+17			
Cluster 2 – Covariance Matrix			
1190,28	744,51	428,58	437,53
744,51	1221,02	379,57	387,93
428,58	379,57	654,13	639,02
437,53	387,93	639,02	659,80
Determinant: 1,48E+19			
Overall variance: 149E+19			

Table 11.18: Statistic data obtained from K-Means with 2 clusters.

In Table 11.18 we show all the statistical data for the classification obtained with 2 clusters. The overall variance is the value which will be used to compare the classification with other ones. We note that the values of overall variances follow the exponential and for this reason it is necessary to do a compromise between the number of clusters and overall variance.

Number of clusters	Overall variances
2,00	1,49E+19
3,00	1,14E+19
4,00	8,46E+18
5,00	6,76E+18
6,00	6,71E+18
7,00	5,78E+18
8,00	5,64E+18
9,00	5,04E+18
10,00	4,31E+18

Table 11.19: List of statistic data obtained from K-Means classification with several clusters.

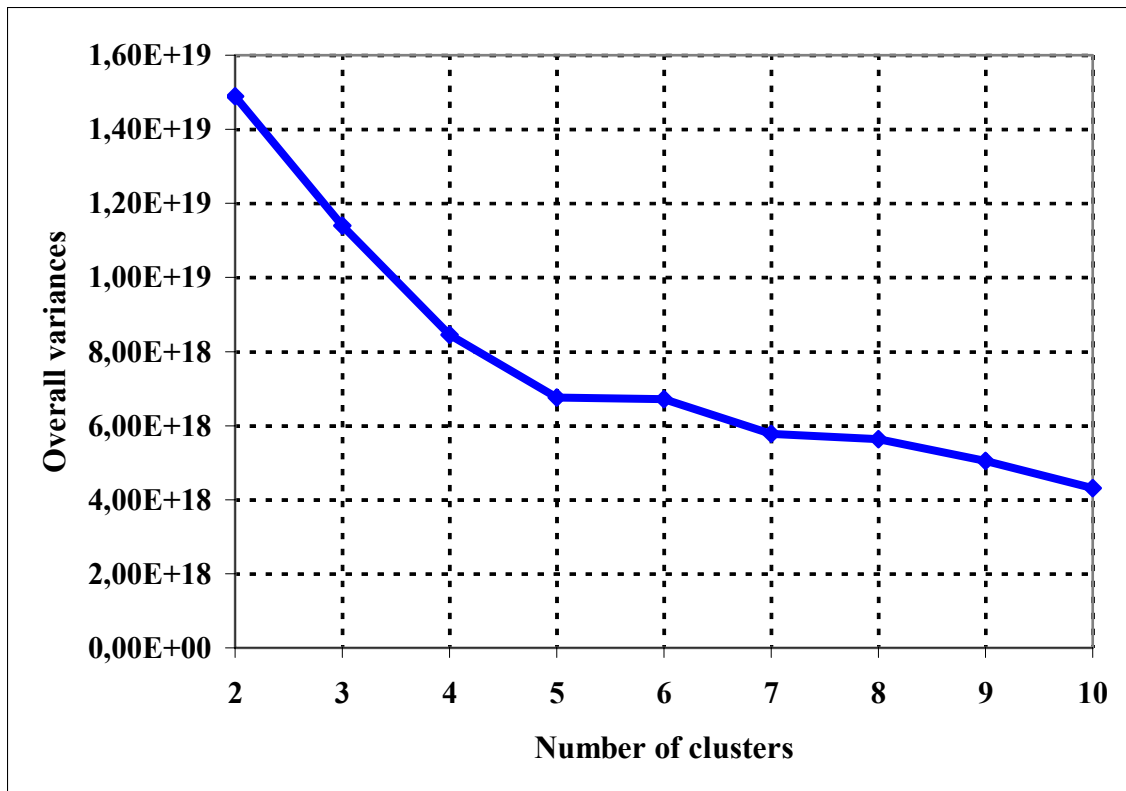


Figure 11.17: Behavior of overall variances at a function of the number of clusters in the K-Means classification.

In order to obtain the best classification and to choose the correct number of clusters, it is necessary to consider how many different clusters could be present in the image and the plot in Figure 11.17. From this plot, we observe that the curve present a point of partial stabilization in correspondence of the number of clusters 5. The best compromise is so when we choose 5 clusters which is a correct number in relation to what is present in the image and to the value of the overall variance. For these reasons, the chosen classification for this test is that one with 5 clusters as it is shown in Figure 11.16. It is necessary to remember that these

classifications label every pixel and they leave no unclassified pixels. In addition, the best classification from the tests will be used to compare the classification with the entropy-alpha parameters.

11.2.2 ENTROPY–ALPHA PARAMETERS AND CLASSIFICATION WITH K-MEANS

This last experiment which we executed shows the K-Means classification from another point of view. Before executing K-Means a transformation of the image has been done. This transformation changes the image from 4 channels (HH , VV , HV and VH) to 3 channels ($HH+VV$, $HH-VV$ and $2VH$) considering that the polarizations HV and VH are very similar or almost equal. After this operation, the image has 3 bands and it is possible to evaluate the entropy and alpha parameters for every pixel in order to make another image with 2 bands. To compute these parameters it is necessary to process for each pixel the coherence matrix from which it is possible to gain these parameters.

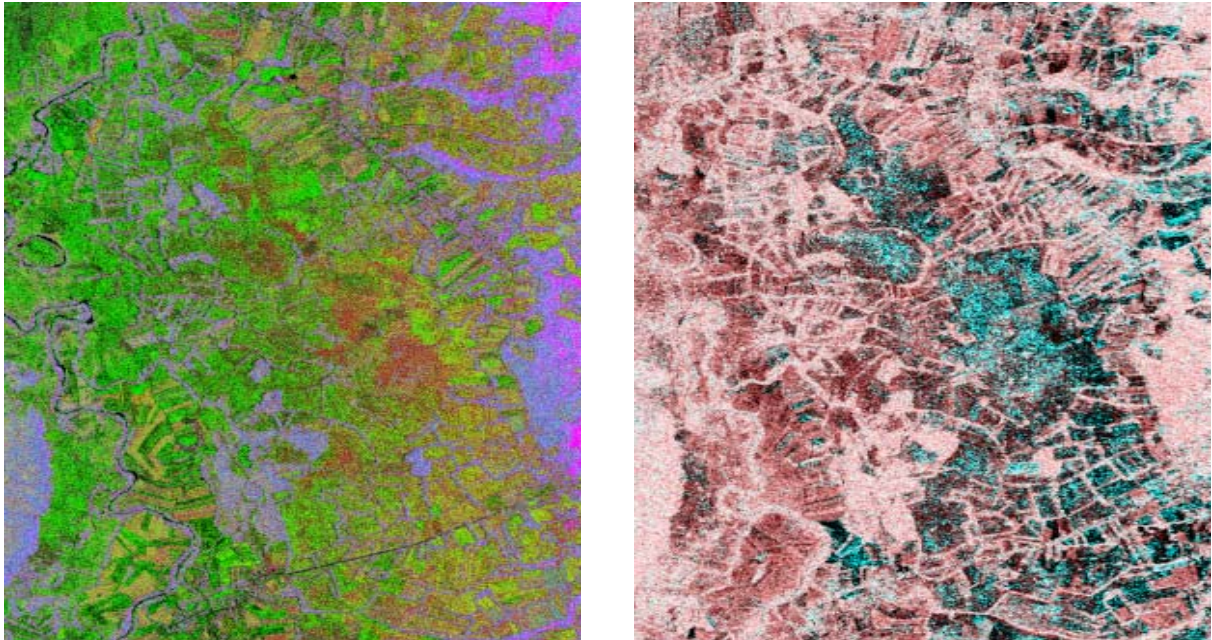


Figure 11.18: Input image with 3 channels ($HH+VV$, $HH-VV$, $2HV$) in RGB mode (left) and image with 2 channels (entropy and alpha) in the right (R = entropy, G = alpha and B = alpha).

Entropy and alpha have been evaluated, therefore, from this matrix and now it is possible to use the K-Means classifier at the same way as in the previous tests. Also in this case we did different experiments to find the correct number of clusters. Without describing again step by step the same procedures, we focus on the results and we compare this classification with the one in the previous sections.

Number of clusters	Overall variances
2,00	8722455,00
3,00	6932461,20
4,00	5976794,00
5,00	5476096,10
6,00	5051407,10
7,00	4634269,40
8,00	4232431,40
9,00	3652888,20
10,00	2950530,50

Table 11.20: List of statistical data obtained from K-Means with several clusters.

Computing the overall variances for every clustering result, it is easy to see that in this case the values are lower than in the previous experiments. This means that the labelled pixels for each cluster are closer to each other than before. First of all, it is necessary to choose how many clusters may be present in this case.

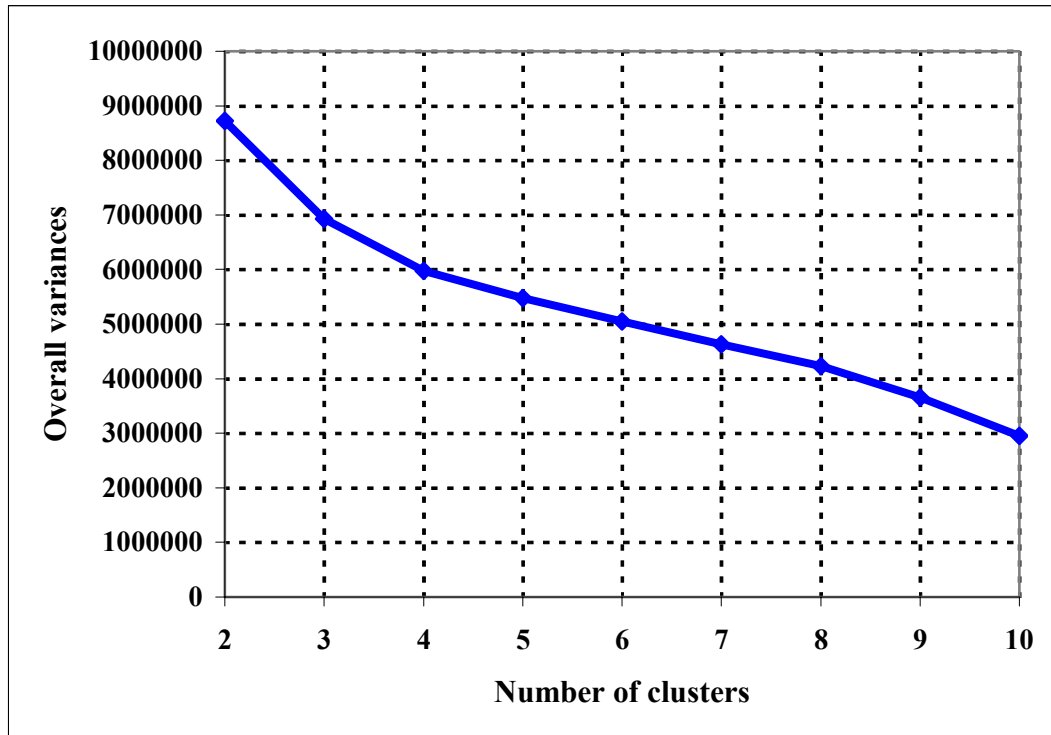


Figure 11.19: Behavior of covariances as a function of the number of the clusters in the K-Means classification.

In this case the most suitable number of clusters, considering the statistical data and the structures present in the image (it is possible to see at a first sight 5 or 6 different type of clusters) oscillates from 4 to 5. Both of them are good for the classification and thus they are used as the results of this test. Actually, there are not more differences between these two classifications even if the one with 5 clusters is preferable because some different parts are not confused as the case with 4 clusters.

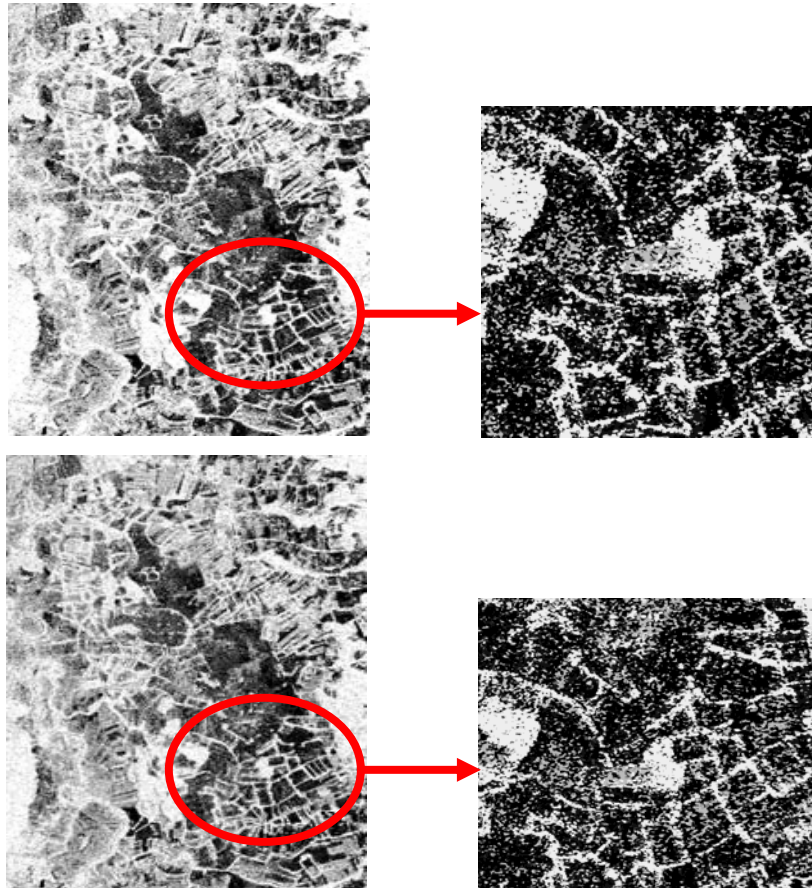


Figure 11.20: Results of the K-Means classification with 4 clusters (up) and 5 clusters (down).

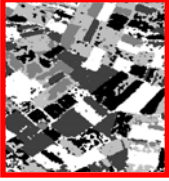
11.2.3 CONCLUSIONS ABOUT UNSUPERVISED CLASSIFICATION

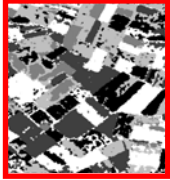
Summing up the results of the unsupervised classification experiments, we analysed briefly differences between the K-Means classification of the original image with 4 bands and the one with the entropy and alpha parameters. First of all, the values of overall variances are better in the second case because a lower value of variance indicates that the pixels are more gathered around the mean values of the clusters. As it is possible to see from the Table 11.21, the classification with the latter method contains more details which supports an overall better quality of classification map. These considerations are only described in a qualitative way because a the test-set is not available for this data-set, in order to do an accurate analysis by confusion matrix and the overall accuracy. Therefore, without these additional information, we can only perform an analysis with the statistical data which can be extracted directly from the clustering result.

11.3 CONCLUSIONS ABOUT ALL TESTS

In this chapter, two different kinds of classification have been described: supervised and unsupervised with different experiments and tests. In the supervised classification we used different transformations in order to improve the overall accuracy and the class accuracies. Among all of these ones, the *LOG+DCT* is the best combination of transformations together with *LOG + MEDIAN FILTER (11 by 11) + DESPECKLE* and *LOG* transformations which permit a good classification to be obtained even if it remains the problem of a low class accuracy for class ω_3 . This is due to the strong overlapping between the 5 agricultural classes in the SAR feature space. Instead, in the unsupervised classification, the work of the K-Means classifier was described together with the entropy – alpha transformation. As this is a classification without a training-set, a test-set was not available to compare the results and to

compute quality measures as in the supervised classification. The operation of the classifier and the transformations has been described, from a more qualitative and visually-oriented viewpoint. In particular, this analysis has confirmed the effectiveness of the use of the H/α parameters which seem to allow a better identification of the details to be obtained in the clustering map.

LOG Transformation		OA	CA
 Classification with window 3 by 3		★ ★ ★	★ ★ ★ ★
	 Classification with window 5 by 5		

LOG+DCT		OA	CA
 Classification with window 3 by 3		★ ★ ★	★ ★ ★

LOG+MEDIANO+DESPECKLE		OA	CA
 Classification with window 3 by 3		★ ★ ★ ★	★ ★ ★

LOG+DCT+Estrazione Energia (90%)		OA	CA
	Classification with window 3 by 3	★ ★ ★ ★	★ ★
		★ ★ ★ ★	★ ★

Table 11.21: Best results of supervised classification with MRF-ICM algorithm.

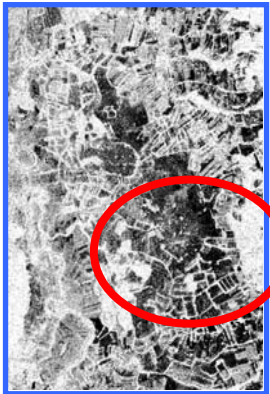
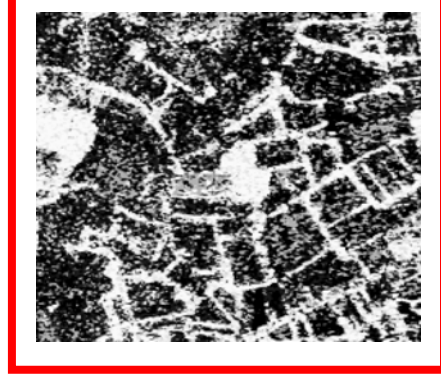
Immagine POLSAR with 4 parameters ★ ★		
Parameters H/α ★ ★ ★		

Table 11.22: Best results of unsupervised classification with K-Means algorithm.

12 CONCLUSIONS

In the present thesis the problem of the classification of polarimetric and/or multifrequency SAR data has been addressed. This classification task involves significant difficulties, due to the strong intrinsic presence of noise in radar imagery, related to the scattering phenomenon of speckle. Specifically, the classification of SAR data has been tackled by working both in a supervised and in an unsupervised contexts.

To classify the images we adopted two algorithms: Markov Random Field with Iterated Conditional Mode (MRF-ICM) for supervised classification and K-Means for unsupervised classification.

MRF-ICM was applied for its effectiveness in using the contextual information in the classification. In particular, a multivariate generalization of the Log-Normal distribution has been employed to model the statistics of the classes in the polarimetric SAR data, and has been plugged in a standard MRF Potts model. In addition, the adopted MRF-based classifier has been endowed with recently proposed parameter-optimization technique, which makes the method fully automatic, with no need for the user's intervention.

K-Means is a simple clustering algorithm. It was chosen due to its simplicity and short computation time and was combined with the extraction of different parameters (H/α) which represent the image in other ways. The H/α features are SAR specific transformed features introduced in the context of the analysis of vegetated terrains by polarimetric data.

These techniques provided satisfying results when tested on two real PolSAR images. The MRF approach proved effective in exploiting in the classification context and gave very high accuracy improvements, as compared to the non-contextual initialisation. The good results are obtained, in particular, using a multivariate Log-Normal model which supports a good accuracy of this empirical distribution for POLSAR data. Results have been obtained also through linear transformations (DCT – Discrete Cosine Transform, DFT – Discrete Fourier Transform, WH – Walsh Hadamard and KL – Karhunen Loeve) which try to decrease the correlation between pixels. Specifically, an improvement in the classification accuracy has been granted by the DCT transform, which also provided additional information about the possibility to reduce the dimensionality of the

polarimetric feature space employed for classification, without significantly affecting the quality of the resulting classification maps.

A different analysis has been carried out for the unsupervised classification. Actually, due to the lack of training-data it is not possible to assess the classification results in a fully qualitative way. On the other hand, interesting information can be drawn from a visual (qualitative) analysis of the resulting clustering maps. In particular, the clustering results based on the use of the H/α transformed features turned out to distinguish better the structures and the details in the scene, or compared with the results obtained by using only the SAR amplitude and or intensity features.

Interesting developments of this preliminary experimental investigation would be, for instance, a joint use of the polarimetric channels and of the H/α transformed features for classification purposes, or the use of supervised linear features transforms, which explicitly optimizes the class separation.

BIBLIOGRAPHY

CHAPTER 2

- [2.1] H.D. Lüke, "Korrelationssignale. Korrelationsfolgen und Korrelationsarrays in Nachrichten- und Informationstechnik, Meßtechnik und Optik", Springer Verlag, Berlin, 1992.
- [2.2] H.D. Lüke, "Signalübertragung. Grundlagen der digitalen und analogen Nachrichtenübertragungssysteme", Springer Verlag, Berlin, 1990.
- [2.3] R. M. Barnes, "Antenna polarization calibration using in-scene reflectors", Lincoln Lab., Lexington, MA, Proj. Rep.TT-65, Sept.1986.
- [2.4] R.L. Jordan, B.L. Huneycutt and M. Werner: "The SIR-C/X-SAR Synthetic Aperture Radar System", Proceedings of the IEEE, vol.79, No.6, pp.827-838, 1991.
- [2.5] J.R. Bennett, I.G. Cumming and R.A. Deane: "The Digital Processing of SEASAT Synthetic Aperture Radar Data", Proceedings of the IEEE International Radar Conference, pp.168-174, 1980.
- [2.6] Oliver, Chris; Quegan and Shaun: "Understanding Synthetic Aperture Radar Images", SciTech Publishing, 2004

CHAPTER 3

- [3.1] C. Wiley: "Pulsed Doppler Radar Method and Means", US Patent No.3.196.436, 1954.
- [3.2] L.J. Cutrona, E.N. Leith, C.J. Palermo and L.J. Porcello: "Optical Data Processing and Filtering Systems", IRE Transactions on Information Theory, vol.IT-6, pp.386-400, 1960.
- [3.3] J.R. Bennett, I.G. Cumming and R.A. Deane: "The Digital Processing of SEASAT Synthetic Aperture Radar Data", Proceedings of the IEEE International Radar Conference, pp.168-174, 1980.
- [3.4] C. Wu: "A Digital Fast Correlation Approach to Produce SAESAT SAR Imagery", Proceedings of the IEEE International Radar Conference, pp.153-160, 1980.
- [3.5] F.T. Ulaby, R.K. Moore and A.K. Fung: "Microwave Remote Sensing Volume I", Addison-Wesley, Reading, MA, 1981.
- [3.6] F.T. Ulaby, R.K. Moore and A.K. Fung: "Microwave Remote Sensing Volume II", Addison-Wesley, Reading, MA, 1982.
- [3.7] L.C. Graham: "Synthetic Interferometric Radar for Topographic Mapping", Proceedings of the IEEE, vol.62, pp.763-768, 1974.
- [3.8] H.A. Zebker and R.M. Goldstein: "Topographic Mapping from Interferometric Synthetic Aperture Radar Observations", Journal of Geophysical Research, vol.91, pp.4993-4999, 1986.
- [3.9] A. Gabriel and R.M. Goldstein: "Crossed Orbit Interferometry: theory and Experimental Results from SIR-B", International Journal of Remote Sensing, vol.9, No.5, pp.857-872, 1988.
- [3.10] A.L. Gray and P.J. Farris-Manning: "Repeat-Pass Interferometry with Airborne Synthetic Aperture Radar", IEEE Transactions on Geoscience and Remote Sensing, vol.31, No.1, pp.180-191, 1993.
- [3.11] "Space at the Service of Our Environment", Proceedings of the first ERS-1 symposium, vol.1-2, Cannes, France, 1992.

- [3.12] "Space at the Service of Our Environment", Proceedings of the third ERS-1 symposium, vol.1-3, Florence, Italy, 1997.
- [3.13] R.L. Jordan, B.L. Huneycutt and M. Werner: "The SIR-C/X-SAR Synthetic Aperture Radar System", Proceedings of the IEEE, vol.79, No.6, pp.827-838, 1991.
- [3.14] J. Way and E.A. Smith: "The Evolution of Synthetic Aperture Radar Systems and their Progression to the EOS SAR", IEEE Transactions on Geoscience and Remote Sensing, vol.29, No.6, pp.962-985, 1991.
- [3.15] M. Werner: "Shuttle Radar Topography Mission (SRTM) / Mission Overview", Proceedings of EUSAR '2000, pp.209-212, Munich, Germany, 2000.
- [3.16] E.L. Christensen, N. Skou, J. Dall, K.W. Woelders, J.H. Jorgensen, J. Granholm and S.N. Madsen: "EMISAR: An Absolutely Calibrated Polarimetric L- and C-band SAR", IEEE Transactions on Geoscience and Remote Sensing, vol.36, No.6, pp.1852-1865, November 1998.
- [3.17] F. Holecz, J. Moreira, P. Pasquali, S. Voigt, E. Meier and D. Nüesch: "Height Model Generation, Automatic Geocoding and Mosaicing Using Airborne AeS-1 InSAR Data", Proceedings of IGARSS' 97, pp.1929-1931, Singapore, 1997.
- [3.18] S.N. Madsen, J. Martin and H.A. Zebker: "Analysis and Evaluation of the NASA/JPL TOPSAR Across-Track Interferometric System", IEEE Transactions on Geoscience and Remote Sensing, vol.33, No.2, pp.383-391, 1995.
- [3.19] R. Scheiber: "Single-Pass Interferometry with the E-SAR System of DLR", Proceedings of EUSAR '98, pp.47-50, Friedrichshafen, Germany, 1998.
- [3.20] A. Ferretti, C. Prati and F. Rocca: "Nonlinear Subsidence Rate Estimation Using Permanent Scatterers in Differential SAR Interferometry", IEEE Transactions on Geoscience and Remote Sensing, vol.38, No.5, pp.2202-2212, September 2000.
- [3.21] W.M. Boerner, H. Mott, E. Lüneburg, C. Livingstone, B. Brisco, R.J. Brown and J.S. Paterson: "Polarimetry in Radar Remote Sensing: Basic and Applied Concepts", Manual of Remote Sensing vol.2, Chapter 5, John Wiley & Sons, New York, 1998.
- [3.22] D. Massonet, P. Briole and A. Arnaud: "Deflation of Mount Etna Monitored by Spaceborne Radar Interferometry", Nature, vol.375, pp.567-570, 1995.
- [3.23] D. Massonet, M. Rossi, C. Carmona, F. Adragna, G. Pelzer, K. Feigl and T. Rabaute: "The Displacement Field of the Landers Earthquake mapped by Radar Interferometry", Nature, vol.364, pp.138-142, 1993.

- [3.24] H. Rott, C. Mayer and A. Siegel: "On the Operational Potential of SAR Interferometry for Monitoring Mass Movements in Alpine Areas", Proceedings of EUSAR' 2000, pp.43-46, Munich, Germany, 2000.
- [3.25] X.Wu, K.H. Thiel and P. Hartl: "Estimating Ice Changes by SAR Interferometry ", Proceedings of the 3rd International Airborne Remote Sensing Conference and Exhibition, pp.110-117, Copenhagen, Denmark, 1997
- [3.26] J. Askne, P.B. Dammert, L.M. Ulander and G. Smith: "C-band Repeat-Pass Interferometric SAR Observations of the Forest", IEEE Transactions on Geoscience and Remote Sensing, vol.35, No.1, pp.25-35, 1997.
- [3.27] J.O. Hagberg, L.M. Ulander, and J. Askne: "Repeat-Pass Interferometry over Forested Terrain", IEEE Transactions on Geoscience and Remote Sensing, vol.33, No.2, pp.331-340, 1995.
- [3.28] K.P. Papathanassiou, A.Reigber and M. Coltelli: "On the Interferometric Coherence: A Multifrequency and Multitemporal Analysis", Proceedings of the Fringe '96 workshop, pp.319-330, Zurich, Switzerland, 1996.
- [3.29] R.N. Treuhaft, S.N. Madsen, M. Moghaddam, and J.J. van Zyl: "Vegetation Characteristics and Underlying Topography from Interferometric Data", Radio Science, vol.31, pp.1449-1495, 1996.
- [3.30] R.N. Treuhaft and P.R. Siqueira: "The Vertical Structure of Vegetated Land Surfaces from Interferometric and Polarimetric Radar", Radio Science, vol.35, pp.141-177, 2000.
- [3.31] G. Sinclair: "Modification of the Radar Range Equation for Arbitrary Targets and Arbitrary Polarization", Report 302-19, Antenna Laboratory, The Ohio State University Research Foundation, 1948.
- [3.32] G. Sinclair: "The Transmission and Reception of Elliptically Polarized Waves", Proceedings of the IRE, vol.38, No.2, pp.148-151, 1950.
- [3.33] J.R. Huynen: "Phenomenological Theory of Radar Targets", PhD. thesis, University of Technology, Delft, Netherlands, 1970.
- [3.34] W.M. Boerner and M.B. El-Arini: "Polarisation Dependence in Electromagnetic Inverse Problems", IEEE Transactions on Antenna and Propagation, vol.29, No.2, pp.262-271, 1981.
- [3.35] S.R. Cloude and E. Pottier: "En Entropy Based Classification Scheme for Land Applications of Polarimetric SAR", IEEE Transactions on Geoscience and Remote Sensing, vol.35, No.1, pp.68-78, 1997.

- [3.36] J.S. Lee, M.R. Grunes and R. Kwok "Classification of Multi-look Polarimetric SAR Imagery Based on Complex Wishart Distribution", *International Journal of Remote Sensing*, vol.15, No.11, pp.2299-2311, 1994.
- [3.37] E. Rignot and R. Chellappa: "Segmentation of Polarimetric Synthetic Aperture Radar Data", *IEEE Transactions on Image Processing*, vol.1, pp.281-300, 1992.
- [3.38] S.R. Cloude and E. Pottier: "A Review of Target Decomposition Theorems in Radar Polarimetry", *IEEE Transactions on Geoscience and Remote Sensing*, vol.34, No.2, pp.498-518, 1996.
- [3.39] S.R. Cloude, I. Hajnsek and K.P. Papathanassiou: "Eigenvector Methods for the Extraction of Surface Parameters in Polarimetric SAR", *Proceedings of CEOS SAR Workshop 1999*, CNES, Toulouse, France, 26-29 October 1999.
- [3.40] J.S. Lee, M.R. Grunes, T. Ainsworth, D. Lu, D.L. Schuler and S.R. Cloude: "Unsupervised Classification Using Polarimetric Decomposition and Complex Wishart Distribution", *Proceedings of IGARSS'98*, pp.2178-2180, Seattle, USA, 1998.
- [3.41] E. Pottier and J.S. Lee: "Unsupervised Classification Scheme of POLSAR Images Based on the Complex Wishart Distribution and the 'H/A/ α ' Polarimetric Decomposition Theorem", *Proceedings EUSAR' 2000*, pp.265-268, Munich, Germany, 2000.
- [3.42] S.R. Cloude and E. Pottier: "A Review of Target Decomposition Theorems in Radar Polarimetry", *IEEE Transactions on Geoscience and Remote Sensing*, vol.34, No.2, pp.498-518, 1996.
- [3.43] I. Hajnsek, S.R. Cloude, J.S. Lee and E. Pottier: "Inversion of Surface Parameters from Polarimetric SAR Data", *Proceedings of IGARRSS' 2000*, pp.1095-1097, Honolulu, Hawaii, 2000.
- [3.44] K.P. Papathanassiou: "Polarimetric SAR Interferometry", PhD. thesis, Technische Universität Graz, Austria, February 1999.
- [3.45] S.R. Cloude and K.P. Papathanassiou: "Polarimetric SAR Interferometry", *IEEE Transactions on Geoscience and Remote Sensing*, vol.36, No.5, pp.1551-1565, 1998.
- [3.46] K.P. Papathanassiou, S.R. Cloude and A. Reigber: "Estimation of Vegetation Parameters using Polarimetric SAR Interferometry Part I and II", *Proceedings of CEOS SAR Workshop 1999*, CNES, Toulouse, France, 26-29 October 1999

- [3.47] K.P. Papathanassiou, S.R. Cloude and A. Reigber: "Single and Multibaseline Polarimetric SAR Interferometry over Forested Terrain", Proceedings of EUSAR'2000, pp.123-126, Munich, Germany, 2000.
- [3.48] D.L. Evans, T.G. Farr, J.J. van Zyl and H. Zebker: "Imaging Radar Polarimetry: Analysis Tools and Applications", IEEE Transactions on Geoscience and Remote Sensing, vol.26, No.5, pp.774-789, 1988.
- [3.49] G.F. Lemoine, D.H. Hoekman and A.J. Sieber: "Polarimetric Contrast Classification of Agricultural Fields Using MAESTRO-1 AIR-SAR Data", International Journal of Remote Sensing, vol.15, No.14, pp.2851-2869, 1994.
- [3.50] Y. Oh, K. Sarabandi and F.T. Ulaby: "An Empirical Model and an Inversion Technique for Radar Scattering from Bare Soil Surfaces", IEEE Transactions on Geoscience and Remote Sensing, vol.30, No.2, pp.370-381, 1992.
- [3.51] E. Rignot, J. Way, C. Williams and L. Viereck: "Radar Estimates of Aboveground Biomass in Boreal Forest Using SAR Imagery", IEEE Transactions on Geoscience and Remote Sensing, vol.32, No.5, pp.1117-1123, 1994.
- [3.52] D.L. Schuler, J.S. Lee, and K.W. Hoppel: "Polarimetric SAR Image Signatures of the Ocean and Gulf Stream Features", IEEE Transactions on Geoscience and Remote Sensing, vol.31, No.6, pp.1210-1221, 1993.

CHAPTER 4

- [4.1] J. A. Stratton, "Electromagnetic theory ", McGraw-Hill Book Company, New York, USA, 1941.
- [4.2] J. D. Jackson, "Classical electrodynamics", John Wiley and Sons, New York, USA, 1975
- [4.3] J. A. Kong, "Electromagnetic wave theory", John Wiley and Sons, New York, 1986

- [4.4] E.M. Kennaugh, "Effects of the type of polarization on echo characteristics", Technical Report 381-1 to 394-24, Antenna Laboratory, The Ohio State University Research Foundation, Columbus, Ohio, USA, 1949-1954.
- [4.5] "IEEE standard definitions of the terms for antennas", IEEE Transactions on Antennas and Propagation, vol.AP-31 (6): pages 1-26, 1983.
- [4.6] H. A. Zebker, J. J. Van Zyl and D. N. Held, "Imaging radar polarimetry from wave synthesis ", J. Geophys. Res., vol 92, pp.683-701, Jan. 1987.
- [4.7] R. M. Barnes, "Antenna polarization calibration using in-scene reflectors", Lincoln Lab., Lexington, MA, Proj. Rep.TT-65, Sept.1986.
- [4.8] Y. Oh, K. Sarabandi and F.T. Ulaby: "An Empirical Model and an Inversion Technique for Radar Scattering from Bare Soil Surfaces", IEEE Transactions on Geoscience and Remote Sensing, vol.30, No.2, pp.370-381, 1992.

CHAPTER 5

- [5.1] S.R. Cloude, "Symmetry, zero correlations and target decomposition theorems", in Proc. 3rd Int. Workshop on Radar Polarimetry (JIPR'95), IRESTE, University of Nantes, Mar.1995, pp.58-68.
- [5.2] Ngheim, S. H. Yueh, R. Kwok and F. K. Li, "Symmetry properties in polarimetric remote sensing", Radio Sci., vol.27.no.5, pp.693-711, Oct.1992.

CHAPTER 7

- [7.1] R. T. Cov, "Probability, frequency and reasonable expectation", Amer. J Phys, vol 17, pp.1-13, 1946.

- [7.2] E. T. Jaynes, "Information theory and statistical mechanics", J. Phys. Rev., vol.106, pp.620-630, 1957.
- [7.3] R. E. Blahut, Principles and Practice of information Theory. New York Addison-Wesley, 1987.
- [7.4] C. P. Robert, "Springer texts in statistics", in the Bayesian Choice, Berlin, Germany: Springer-Verlag, 1996.
- [7.5] Mihai Datcu, Klaus Seidel, and Marc Walessa, "Spatial Information Retrieval from Remote-Sensing Images—Part I: Information Theoretical Perspective", IEEE Transactions on Geosciences and Remote Sensing, Vol.36, No.5, September 1998.

CHAPTER 8

- [8.1] J. A. Richards, "Remote sensing digital image analysis", Springer-Verlag, Berlin, Germany, 1994.
- [8.2] S. R. Cloude and E. Pottier, "An entropy based classification scheme for land application of polarimetric SAR", IEEE Transactions on Geoscience and Remote Sensing, vol.35 (1): pages 68-78, January 1997.
- [8.3] A. Papoulis, "Probability, random variables and stochastic processes", McGraw-Hill Book Company, New York, USA, 1965.
- [8.4] R. G. Congalton, "A review of assessing the accuracy of classification of remotely sensed data", Remote Sensing of Environment, vol.37: pages 35-46, 1991.
- [8.5] S. V. Stehman, "Selecting and interpreting measures of thematic classification accuracy", Remote Sensing of Environment, vol.62: pages 77-89, 1997.
- [8.6] V. Alberga, "Comparison of polarimetric methods in image classification and SAR interferometry applications".

CHAPTER 9

- [9.1] G. Moser, J. Zerubia and S.B. Serpico, "SAR amplitude probability density function estimation based on a generalized Gaussian Model", IEEE Transactions on Image Processing, in print.
- [9.2] J.T. Tou and R.C. Gonzalez, "Pattern Recognition Principles", Addison-Wesley Publ. Comp., MA, 1974
- [9.3] Zemansky, Mark, "Heat And Thermodynamics - Second Edition" McGraw-Hill, New York, 1943
- [9.4] O.Duda, Peter E.Hart, David G.Stork, "Pattern Classification, Second Edition", Richard
- [9.5] J. Besag, "On the statistical analysis of dirty pictures," *Journal of the Royal Statistical Society, Series B*, Vol.48, No.3, pp.259–302, 1986.

CHAPTER 10

- [10.1] Rosenfeld and A. Kak, "Digital Image Processing", Academic Press, Orlando, FL, 1982.
- [10.2] Jain and A.Kak. "Fundamentals of digital image processing", Prentice-Hall, Englewood Cliffs, 1989, NJ.
- [10.3] A.Hedayat and W.Wallis, "Hadamard Matrices and their Applications. The Annals of Statistics.", 1978.

CHAPTER 11

- [11.1] J. Richards and X. Jia. Remote sensing digital image analysis. 3rd Edition, Springer-Verlag, Berlin, 1999.
- [11.2] G. Moser and S.B. Serpico, "MFR Model parameter estimation for contextual supervised classification of remote-sensing images", Proceedings of IGARSS 2005, in print.
- [11.3] G. Moser, J. Zerubia and S. B. Serpico, "SAR amplitude probability density function estimation based on a generalized Gaussian Model", IEEE Transactions on Image Processing, in print.
- [11.4] S.B. Serpico and F. Roli, "Classification of Multisensor Remote Sensing Images by Structured Neural Networks", IEEE Transactions on Geoscience and Remote Sensing, Vol.33, No.3, May 1995.